

Question ID cbe9966a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	

ID: cbe9966a

$$0.10x + 0.20y = 0.17(x + y)$$

The equation gives a volume x , in gallons, of a **10%** solution that could be mixed with a volume y , in gallons, of a **20%** solution to produce a **17%** solution. According to this equation, what volume, in gallons, of the **20%** solution could be mixed with **90.0** gallons of the **10%** solution to produce a **17%** solution? (Assume that the volume of the mixture is the sum of the volumes of the two solutions before they were mixed.)

ID: cbe9966a Answer

Correct Answer: 210

Rationale

The correct answer is **210**. It's given that the equation $0.10x + 0.20y = 0.17(x + y)$ represents mixing a **10%** solution and a **20%** solution to obtain a **17%** solution. It's also given that x represents the volume, in gallons, of the **10%** solution. If **90.0** gallons of the **10%** solution is used, it follows that $x = 90.0$. Substituting **90.0** for x in the given equation yields $0.10(90.0) + 0.20y = 0.17(90.0 + y)$, or $9 + 0.20y = 15.3 + 0.17y$. Subtracting $0.17y$ from both sides of this equation yields $9 + 0.03y = 15.3$. Subtracting **9** from both sides of this equation yields $0.03y = 6.3$. Dividing both sides of this equation by **0.03** yields $y = 210$. Therefore, the volume of the **20%** solution that could be mixed with **90.0** gallons of the **10%** solution to produce a **17%** solution is **210** gallons.

Question Difficulty: Hard

Question ID ceb93469

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: ceb93469

Sam has a flag with an area of **110 square inches (in^2)**. Sam plans to enlarge the flag by sewing on pieces of colored nylon, where each piece would increase the flag's area by **64 in^2** . Which equation gives the flag's total area **y** , in square inches, after Sam sews on **x** pieces of nylon?

- A. **$y = 64x$**
- B. **$y = 64 + 110x$**
- C. **$y = 110 + 64x$**
- D. **$y = 174x$**

ID: ceb93469 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that Sam has a flag with an area of **110 square inches (in^2)** and that each piece of colored nylon Sam sews on increases the flag's area by **64 in^2** . If Sam sews on **x** pieces of nylon and each piece increases the area by **64 in^2** , it follows that the expression **64 x** represents the total increase in area, in square inches, from sewing on **x** pieces of nylon. Adding this increase in area, **64 x in^2** , to the original area of the flag, **110 in^2** , gives the total area of the flag. Therefore, the equation that gives the flag's total area **y** , in square inches, after Sam sews on **x** pieces of nylon is **$y = 110 + 64x$** .

Choice A is incorrect. This equation represents the total area, in square inches, of **x** pieces of colored nylon.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 0e8b6c57

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: 0e8b6c57

Students and chaperones from a middle school are going on a field trip to a museum. Admission to the museum will cost **\$8.50** for each student and **\$12** for each chaperone. It will cost a total of **\$730** for admission to the museum for all **83** people on the field trip. How many students are going on the field trip?

- A. **31**
- B. **70**
- C. **76**
- D. **86**

ID: 0e8b6c57 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that admission to the museum will cost **\$8.50** for each student and **\$12** for each chaperone. It's also given that it will cost a total of **\$730** for admission to the museum for all **83** people on the field trip. Let x represent the number of students and y represent the number of chaperones going on the field trip. Since there are a total of **83** people going on the field trip, it follows that $x + y = 83$. Since admission to the museum for each student costs **\$8.50** and each chaperone costs **\$12**, the total cost can be represented by the equation $8.50x + 12y = 730$. The equations $x + y = 83$ and $8.50x + 12y = 730$ form a system of equations. Subtracting x from both sides of the equation $x + y = 83$ yields $y = 83 - x$. Substituting $83 - x$ for y in the equation $8.50x + 12y = 730$ yields $8.50x + 12(83 - x) = 730$, or $-3.50x + 996 = 730$. Subtracting **996** from both sides of this equation yields $-3.50x = -266$. Dividing both sides of this equation by -3.50 yields $x = 76$. Therefore, **76** students are going on the field trip.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 3eebf09c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 3eebf09c

$g(x) = 3(14x - 15)$

What is the y -coordinate of the y -intercept of the graph of $y = g(x) - 2$ in the xy -plane?

- A. -47
- B. -45
- C. -17
- D. -15

ID: 3eebf09c Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that $g(x) = 3(14x - 15)$. Substituting $3(14x - 15)$ for $g(x)$ in the equation $y = g(x) - 2$ yields $y = 3(14x - 15) - 2$. The y -intercept of the graph of a function in the xy -plane is the point where the function crosses the y -axis, which occurs when $x = 0$. Substituting 0 for x in the equation $y = 3(14x - 15) - 2$ yields $y = 3(14(0) - 15) - 2$. It follows that $y = -45 - 2$, or $y = -47$. Therefore, the y -intercept of the graph of $y = g(x) - 2$ in the xy -plane is $(0, -47)$, and the y -coordinate of the y -intercept of the graph is -47 .

Choice B is incorrect. This is the y -coordinate of the y -intercept of the graph of $y = g(x)$ in the xy -plane, not $y = g(x) - 2$.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 978e6cc3

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	

ID: 978e6cc3

Terry has a goal to collect at least **16** items per week for a donation drive. This week, Terry has collected **4** items. If x represents the additional number of items Terry needs to collect this week to meet the goal, which inequality represents this situation?

- A. $4 - x \leq 16$
- B. $4 + x \leq 16$
- C. $4 - x \geq 16$
- D. $4 + x \geq 16$

ID: 978e6cc3 Answer

Correct Answer: D

Rationale

Choice D is correct. It’s given that this week, Terry has collected **4** items. It follows that the expression $4 + x$ represents the total number of items Terry will have collected after collecting x additional items. It’s also given that Terry’s goal is to collect at least **16** items per week. Therefore, the inequality $4 + x \geq 16$ represents this situation.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This inequality represents a situation where the goal is to collect at most, not at least, **16** items per week.

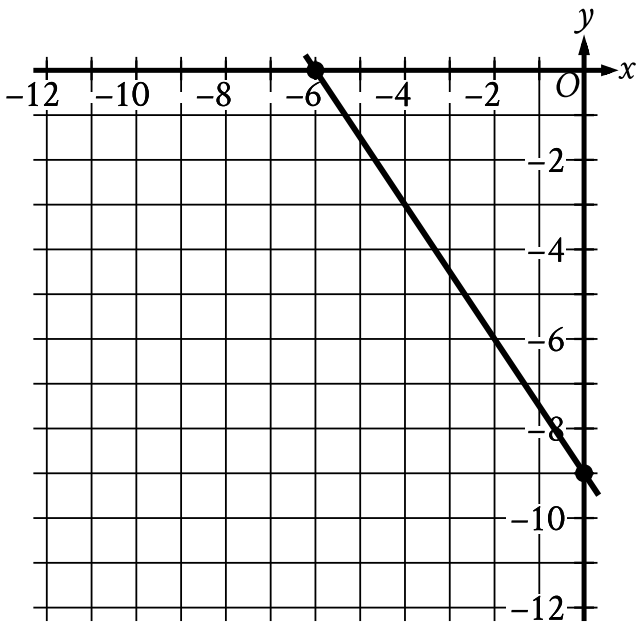
Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 9bf3abe6

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: 9bf3abe6



The graph of line h is shown in the xy -plane. Line k (not shown) is defined by $sx + 48y = t$, where s and t are constants. If line k is graphed in this xy -plane, the result is a graph of two linear equations. This system of two linear equations has no solution. Which of the following is **NOT** a possible value of t ?

- A. -432
- B. -9
- C. 72
- D. 288

ID: 9bf3abe6 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that if line k is graphed in the xy -plane with line h , the result is the graph of a system of two linear equations that has no solution. It follows that line h and line k don't intersect. This means that line k and line h have the same slope and different y -intercepts. The graph shows that the y -intercept of line h is $(0, -9)$. The y -coordinate of the y -intercept of line k is the value of y when $x = 0$ in the equation defining line k . Substituting 0 for x in the equation $sx + 48y = t$ yields $s(0) + 48y = t$, or $48y = t$. Dividing both sides of this equation by 48 yields $y = \frac{t}{48}$. Therefore, the y -coordinate of the y -intercept of line k is $\frac{t}{48}$, so the y -intercept of line k is $(0, \frac{t}{48})$. Since line k and line h have different y -intercepts, $\frac{t}{48}$ can't be equal to -9 . It follows that t can't be equal to $-9(48)$, or -432 . Thus, -432 isn't a possible value of t .

Choice B is incorrect. This isn't a possible value of the y -coordinate of the y -intercept of line k , but it is a possible value of t .

Choice C is incorrect. This is the value of s , not a value of t that isn't possible.

Choice D is incorrect. If $t = 288$, then the y -intercept of line k is $(0, \frac{288}{48})$, or $(0, 6)$, which is different from the y -intercept of line h , $(0, -9)$. Therefore, 288 is a possible value of t .

Question Difficulty: Hard

Question ID 3b102e19

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 3b102e19

$$s(x) = 133 - 4x$$

The function s shown gives the number of spoons, $s(x)$, remaining in a dispenser x minutes after the dispenser had been loaded with spoons. Which statement is the best interpretation of $s(3) = 121$?

- A. The number of spoons remaining in the dispenser decreased by a total of 3 spoons after 121 minutes.
- B. There were 3 spoons remaining in the dispenser 121 minutes after the dispenser had been loaded with spoons.
- C. There were 121 spoons remaining in the dispenser 3 minutes after the dispenser had been loaded with spoons.
- D. The number of spoons remaining in the dispenser decreased by a total of 121 spoons after 3 minutes.

ID: 3b102e19 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that $s(x)$ is the number of spoons remaining in a dispenser x minutes after the dispenser had been loaded with spoons. The expression $s(3)$ represents the value of $s(x)$ when $x = 3$. It follows that $s(3)$ represents the number of spoons remaining in the dispenser 3 minutes after it had been loaded with spoons. Therefore, the best interpretation of the equation $s(3) = 121$ is that there were 121 spoons remaining in the dispenser 3 minutes after it had been loaded with spoons.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect. This would be the best interpretation of $s(121) = 3$, not $s(3) = 121$.

Choice D is incorrect and may result from conceptual errors.

Question Difficulty: Medium

Question ID 7e70df7e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 7e70df7e

The owner of a bakery is giving away free muffins every hour. The equation $f(x) = 108 - 9x$ gives the number of free muffins, $f(x)$, that remain to be given away after x hours. Which of the following is the best interpretation of $f(12) = 0$ in this context?

- A. The owner of the bakery has **12** free muffins to give away.
- B. The owner of the bakery will give away **12** free muffins in **9** hours.
- C. It will take **12** hours to give away all the free muffins.
- D. It will take **108** hours to give away **12** free muffins.

ID: 7e70df7e Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that the equation $f(x) = 108 - 9x$ represents the number of free muffins that remain to be given away after x hours. For the given equation $f(12) = 0$, when the value of x is **12**, the value of $f(x)$ is **0**. It follows that after **12** hours there are **0** free muffins remaining to be given away. Therefore, based on this context, it will take **12** hours to give away all the free muffins.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This represents giving away **12** muffins over **108** hours rather than giving away all **108** muffins in **12** hours.

Question Difficulty: Medium

Question ID e39de699

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: e39de699

$x + y = 20$

One of the two equations in a system of linear equations is given. The system has exactly one solution. Which equation could be the second equation in the system?

- A. $3x + 3y = 60$
- B. $-x - y = 40$
- C. $3x + 3y = 14$
- D. $x - y = 20$

ID: e39de699 Answer

Correct Answer: D

Rationale

Choice D is correct. A system of two linear equations in two variables, x and y , has exactly one solution if the graphs of the lines represented by the equations in the xy -plane have different slopes. In the xy -plane, the slope of a line defined by an equation in the form $Ax + By = C$, where A , B , and C are constants, is $-\frac{A}{B}$. It's given that one equation in the system is $x + y = 20$. It follows that for this equation, $A = 1$ and $B = 1$. Therefore, the value of $-\frac{A}{B}$ is $-\frac{1}{1}$, or -1 . For the equation in choice D, $x - y = 20$, $A = 1$ and $B = -1$. Therefore, the value of $-\frac{A}{B}$ is $-\frac{1}{(-1)}$, or 1 . Since the slopes are different, it follows that the system of $x + y = 20$ and $x - y = 20$ has exactly one solution. Thus, the second equation in the system could be $x - y = 20$.

Choice A is incorrect. The equation $3x + 3y = 60$ can be rewritten as $x + y = 20$. Since $x + y = 20$ is the same as the given equation, the system of $3x + 3y = 60$ and $x + y = 20$ has infinitely many solutions rather than exactly one solution.

Choice B is incorrect. The equation $-x - y = 40$ can be rewritten as $x + y = -40$. The graph of this line has a slope of -1 . Since the graphs of the lines represented by the equations $x + y = -40$ and $x + y = 20$ have the same slope in the xy -plane and the equations aren't equivalent, this system has no solution rather than exactly one solution.

Choice C is incorrect. The equation $3x + 3y = 14$ can be rewritten as $x + y = \frac{14}{3}$. The graph of this line has a slope of -1 . Since the graphs of the lines represented by the equations $x + y = \frac{14}{3}$ and $x + y = 20$ have the same slope in the xy -plane and the equations aren't equivalent, this system has no solution rather than exactly one solution.

Question Difficulty: Hard

Question ID 458fc111

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 458fc111

A yoga studio is offering a deal where the first session is free, the second session is half off the regular price, and the remaining sessions are at the regular price. If the regular price of a session is \$25.40, which function f gives the total cost, in dollars, of purchasing x sessions using this deal, where $x \geq 2$?

- A. $f(x) = 25.40(x - 1) + 12.70$
- B. $f(x) = 25.40(x - 2) + 12.70$
- C. $f(x) = 25.40(x - 1) + 12.70(x - 2)$
- D. $f(x) = 25.40(x - 2) + 12.70(x - 1)$

ID: 458fc111 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that the regular price of a session at a yoga studio is \$25.40. It's also given that the yoga studio is offering a deal where the first session is free, or \$0; that the second session is half off the regular price, or $\$25.40 - \left(\frac{1}{2}\right)(\$25.40)$, which is equal to \$12.70; and that the remaining sessions are at the regular price, \$25.40. Since x represents the total number of sessions and two sessions have already been accounted for, the number of remaining sessions can be represented by $x - 2$. The total cost, in dollars, of purchasing x sessions using this deal can be represented by the function $f(x) = 0 + 12.70 + 25.40(x - 2)$, or $f(x) = 25.40(x - 2) + 12.70$.

Choice A is incorrect. The function represents the total cost of purchasing x sessions when the first session is half off the regular price and the remaining sessions are at the regular price.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 6da32b83

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	

ID: 6da32b83

$$x(r - 9) + 4 = 19x + 27$$

In the given equation, r is a positive integer. If the given equation has exactly one solution, what CANNOT be the value of r ?

- A. 4
- B. 9
- C. 23
- D. 28

ID: 6da32b83 Answer

Correct Answer: D

Rationale

Choice D is correct. In the given equation, $x(r - 9) + 4 = 19x + 27$, r is a positive integer. Distributing on the left-hand side of this equation yields $rx - 9x + 4 = 19x + 27$. Subtracting 4 from both sides of this equation yields $rx - 9x = 19x + 23$. Subtracting $19x$ from both sides of this equation yields $rx - 28x = 23$, which can be written as $(r - 28)x = 23$. For this equation to have exactly one solution, the coefficient of x , $(r - 28)$, cannot be equal to 0. Therefore, 28 CANNOT be the value of r .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID d7a5a66b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: d7a5a66b

x	y
-34	t
-17	$t + 23$
0	$t + 46$

For a linear relationship between x and y , the table gives three values of x and their corresponding values of y , where t is a constant. Which equation represents this relationship?

- A. $y = -2x + t + 23$
- B. $y = 2x + t + 23$
- C. $y = -\frac{23}{17}x + t + 46$
- D. $y = \frac{23}{17}x + t + 46$

ID: d7a5a66b Answer

Correct Answer: D

Rationale

Choice D is correct. It’s given that for a linear relationship between x and y , the table gives three values of x and their corresponding values of y . A linear equation can be written in the form $y = mx + b$, where m is the slope and $(0, b)$ is the y -intercept of the graph of the equation in the xy -plane. The slope of a line that passes through the points (x_1, y_1) and (x_2, y_2) can be calculated using the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. Substituting $(-34, t)$ and $(-17, t + 23)$ for (x_1, y_1) and (x_2, y_2) , respectively, in the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$ yields $m = \frac{(t + 23) - t}{(-17) - (-34)}$, which is equivalent to $m = \frac{t + 23 - t}{-17 + 34}$, or $m = \frac{23}{17}$. Based on the table, the graph of this relationship passes through the point $(0, t + 46)$. It follows that $b = t + 46$. Substituting $\frac{23}{17}$ for m and $t + 46$ for b in the equation $y = mx + b$ yields $y = \frac{23}{17}x + t + 46$. Therefore, the equation $y = \frac{23}{17}x + t + 46$ represents this relationship.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID d7b7acd5

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: d7b7acd5

Ellen bought two types of snack bags for a school event. Each box of chips Ellen bought contained **24** bags and cost **\$11**. Each box of crackers Ellen bought contained **16** bags and cost **\$9**. It cost Ellen **\$497** before tax to buy **968** bags. If x is the number of boxes of chips and y is the number of boxes of crackers that Ellen bought, which of the following systems of equations represents this situation?

- A. $11x + 16y = 968$
 $24x + 9y = 497$
- B. $11x + 9y = 968$
 $24x + 16y = 497$
- C. $11x + 9y = 497$
 $24x + 16y = 968$
- D. $9x + 11y = 497$
 $16x + 24y = 968$

ID: d7b7acd5 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that Ellen bought two types of snack bags: chips and crackers. It's also given that each box of chips contains **24** bags. Thus, if Ellen bought x boxes of chips, the number of bags of chips is $24x$. Additionally, it's given that each box of chips costs **\$11**. Thus, if Ellen bought x boxes of chips, the cost for all the bags of chips is $11x$. It's also given that each box of crackers contains **16** bags. Thus, if Ellen bought y boxes of crackers, the number of bags of crackers is $16y$. Additionally, it's given that each box of crackers costs **\$9**. Thus, if Ellen bought y boxes of crackers, the cost for all the bags of crackers is $9y$. It's given that it cost Ellen **\$497** to buy **968** bags. Since Ellen bought $24x$ bags of chips and $16y$ bags of crackers, the total number of snack bags that Ellen bought can be represented by the equation $24x + 16y = 968$. Additionally, since the cost for all the bags of chips is $11x$ and the cost for all the bags of crackers is $9y$, the total cost for all the snack bags that Ellen bought can be represented by the equation $11x + 9y = 497$. Therefore, the system consisting of the equations $11x + 9y = 497$ and $24x + 16y = 968$ represents this situation.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This system of equations represents a situation where the total cost is **\$968**, not **\$497**, and the total number of bags is **497**, not **968**.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 7dc27752

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: 7dc27752

$y = 7x + 18$

One of the equations in a system of two linear equations is given. The system has no solution. Which equation could be the second equation in the system?

- A. $-21x + y = 54$
- B. $-21x + y = 18$
- C. $-7x + y = 21$
- D. $-7x + y = 18$

ID: 7dc27752 Answer

Correct Answer: C

Rationale

Choice C is correct. A system of two linear equations in two variables, x and y , has no solution if the graphs of the equations in the xy -plane are two distinct and parallel lines, which don't intersect. Two lines in the xy -plane are distinct and parallel if their slopes are the same and the y -coordinates of their y -intercepts are different. A linear equation can be written in slope-intercept form, $y = mx + b$, where m is the slope of the line in the xy -plane and b is the y -coordinate of the y -intercept. Choice C can be written in slope-intercept form by adding $7x$ to both sides of the equation, which gives $y = 7x + 21$. In the equations $y = 7x + 18$ and $y = 7x + 21$, the values of m are each 7 , and the values of b are 18 and 21 , respectively. Since the slopes of these lines are the same and the y -coordinates of the y -intercepts are different, it follows that choice C could be the second equation in the system.

Choice A is incorrect. Adding $21x$ to both sides of this equation gives $y = 21x + 54$. In the equations $y = 7x + 18$ and $y = 21x + 54$, the values of m aren't equal, so the slopes of these lines are different. Therefore, these two lines intersect and the corresponding system of equations has one solution.

Choice B is incorrect. Adding $21x$ to both sides of this equation gives $y = 21x + 18$. In the equations $y = 7x + 18$ and $y = 21x + 18$, the values of m aren't equal, so the slopes of these lines are different. Therefore, these two lines intersect and the corresponding system of equations has one solution.

Choice D is incorrect. Adding $7x$ to both sides of this equation gives $y = 7x + 18$. Since this equation is the same as the given equation, these two lines are not distinct. Therefore, the corresponding system of equations has infinitely many solutions.

Question Difficulty: Hard

Question ID cc5cd4e4

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	

ID: cc5cd4e4

To complete a landscaping project, Adam charges a fee of **\$20.00** for his equipment and **\$9.50** per hour spent working on the project. To complete this same landscaping project, Caroline charges a fee of **\$17.00** for her equipment and **\$10.00** per hour spent working on the project. If x represents the number of hours spent working on the landscaping project, what are all the values of x for which Caroline's total charge is greater than Adam's total charge?

- A. $5 \leq x \leq 6$
- B. $6 \leq x \leq 7$
- C. $x < 5$
- D. $x > 6$

ID: cc5cd4e4 Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that Adam charges a fee of **\$20.00** for his equipment and **\$9.50** per hour spent working on the project. For x hours spent working on the project, it follows that Adam's total charge, in dollars, is $20.00 + 9.50x$. It's also given that Caroline charges a fee of **\$17.00** for her equipment and **\$10.00** per hour spent working on the project. For x hours spent working on the project, it follows that Caroline's total charge, in dollars, is $17.00 + 10.00x$. Therefore, if x represents the number of hours spent working on the landscaping project, then Caroline's total charge is greater than Adam's total charge if $17.00 + 10.00x > 20.00 + 9.50x$, or $17 + 10x > 20 + 9.5x$. Subtracting $9.5x$ from both sides of this inequality yields $17 + 0.5x > 20$. Subtracting **17** from both sides of this inequality yields $0.5x > 3$. Dividing both sides of this inequality by **0.5** yields $x > 6$.

Choice A is incorrect. For example, if $x = 5$, then Caroline's total charge, in dollars, would be $17.00 + 10.00(5)$, or **67**, and Adam's total charge, in dollars, would be $20.00 + 9.50(5)$, or **67.50**. Since Caroline's total charge would be less than Adam's total charge, **5** can't be a possible value of x .

Choice B is incorrect. For example, if $x = 6$, then Caroline's total charge, in dollars, would be $17.00 + 10.00(6)$, or **77**, and Adam's total charge, in dollars, would be $20.00 + 9.50(6)$, or **77**. Since Caroline's total charge would be equal to Adam's total charge, **6** can't be a possible value of x .

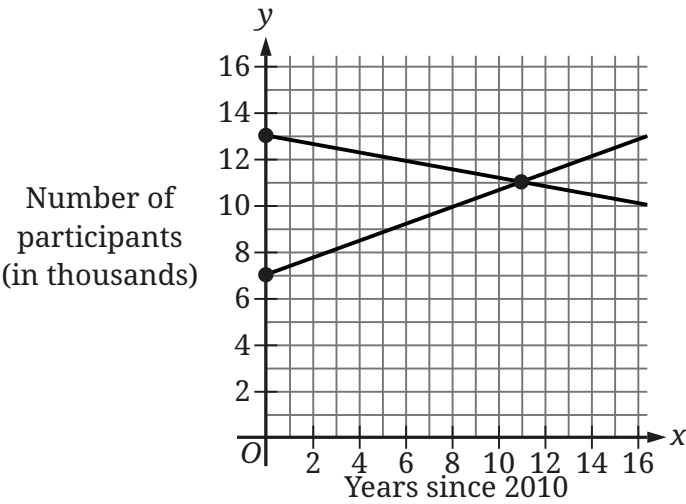
Choice C is incorrect. For example, if $x = 4$, then Caroline's total charge, in dollars, would be $17.00 + 10.00(4)$, or **57**, and Adam's total charge, in dollars, would be $20.00 + 9.50(4)$, or **58**. Since Caroline's total charge would be less than Adam's total charge, **4** can't be a possible value of x .

Question Difficulty: Medium

Question ID bda66e08

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: bda66e08



The functions f and g model the number of participants, in thousands, in two different programs x years since 2010. The graphs of $y = f(x)$ and $y = g(x)$ are shown. Which of the following could represent functions f and g ?

- A. $f(x) = -\frac{4}{11}x + 13$
 $g(x) = \frac{2}{11}x + 7$
- B. $f(x) = -\frac{4}{11}x + 7$
 $g(x) = \frac{2}{11}x + 13$
- C. $f(x) = -\frac{2}{11}x + 13$
 $g(x) = \frac{4}{11}x + 7$
- D. $f(x) = -\frac{2}{11}x + 7$
 $g(x) = \frac{4}{11}x + 13$

ID: bda66e08 Answer

Correct Answer: C

Rationale

Choice C is correct. A line in the xy -plane that passes through the points (x_1, y_1) and (x_2, y_2) has slope m , where $m = \frac{y_2 - y_1}{x_2 - x_1}$. The equation of the line can be written in the form $y = mx + b$, where b is the y -intercept represented by $(0, b)$. One of the lines shown in the graph passes through the points $(0, 7)$ and $(11, 11)$. Substituting 0 for x_1 , 7 for y_1 , 11 for x_2 , and 11 for y_2 in the equation $m = \frac{y_2 - y_1}{x_2 - x_1}$ yields $m = \frac{11 - 7}{11 - 0}$, or $m = \frac{4}{11}$. Substituting $\frac{4}{11}$ for m and 7 for b in the equation $y = mx + b$ yields $y = \frac{4}{11}x + 7$. Therefore, an equation of this line is $y = \frac{4}{11}x + 7$. Similarly, the other line shown in the graph passes through the points $(0, 13)$ and $(11, 11)$. Substituting 0 for x_1 , 13 for y_1 , 11 for x_2 , and 11 for y_2 in the equation $m = \frac{y_2 - y_1}{x_2 - x_1}$ yields $m = \frac{11 - 13}{11 - 0}$, or $m = -\frac{2}{11}$. Substituting $-\frac{2}{11}$ for m and 13 for b in the equation $y = mx + b$ yields $y = -\frac{2}{11}x + 13$. Therefore, an equation of this line is $y = -\frac{2}{11}x + 13$. So, the functions f and g could be represented by $f(x) = -\frac{2}{11}x + 13$ and

$$g(x) = \frac{4}{11}x + 7.$$

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 6a0fc796

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: 6a0fc796

$$\begin{aligned}y - 6 &= \frac{4}{5}(x + 5) \\ y - v &= \frac{4}{5}(x + 10)\end{aligned}$$

In the given system of equations, v is a constant. The system has infinitely many solutions. What is the value of v ?

- A. 0
- B. 2
- C. 4
- D. 12

ID: 6a0fc796 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that the system has infinitely many solutions. A system of two linear equations has infinitely many solutions when the two linear equations are equivalent. Adding **6** to both sides of $y - 6 = \frac{4}{5}(x + 5)$ yields $y = \frac{4}{5}(x + 5) + 6$. Applying the distributive property to this equation yields $y = \frac{4}{5}x + \frac{20}{5} + 6$, or $y = \frac{4}{5}x + 10$. Adding v to both sides of $y - v = \frac{4}{5}(x + 10)$ yields $y = \frac{4}{5}(x + 10) + v$. Applying the distributive property to this equation yields $y = \frac{4}{5}x + \frac{40}{5} + v$, or $y = \frac{4}{5}x + 8 + v$. For the equations $y = \frac{4}{5}x + 10$ and $y = \frac{4}{5}x + 8 + v$ to be equivalent, it follows that $10 = 8 + v$. Subtracting **8** from both sides of this equation yields $2 = v$. Therefore, the value of v is **2**.

Choice A is incorrect. If $v = 0$, the system of equations would have no solution, not infinitely many solutions.

Choice C is incorrect. If $v = 4$, the system of equations would have no solution, not infinitely many solutions.

Choice D is incorrect. If $v = 12$, the system of equations would have no solution, not infinitely many solutions.

Question Difficulty: Medium

Question ID fc49f3c9

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: fc49f3c9

$38(x - n) = 38y + 38n$

One of the equations in a system of two linear equations is given, where n is a positive constant. The system has no solution. Which equation could be the second equation in this system?

- A. $4x - 4y = 8n$
- B. $4x + 4y = 4n$
- C. $4x + 4y = 8n$
- D. $4x - 4y = 4n$

ID: fc49f3c9 Answer

Correct Answer: D

Rationale

Choice D is correct. A system of two linear equations in two variables, x and y , has no solution if the lines represented by the equations in the xy -plane are parallel and distinct. Lines represented by equations in standard form, $Ax + By = C$ and $Dx + Ey = F$, are parallel and distinct if the coefficients of x and y in one equation are proportional to the corresponding coefficients in the other equation, meaning $\frac{D}{A} = \frac{E}{B}$, and the constants are not proportional, meaning $\frac{F}{C}$ is not equal to $\frac{D}{A}$ nor $\frac{E}{B}$. It's given that one of the equations in a system of two linear equations is $38(x - n) = 38y + 38n$, where n is a positive constant. Distributing 38 on the left-hand side of this equation yields $38x - 38n = 38y + 38n$. Adding $38n$ to both sides of this equation yields $38x = 38y + 76n$. Subtracting $38y$ from both sides of this equation yields $38x - 38y = 76n$. Therefore, the given equation can be written in the form $Ax + By = C$, where $A = 38$, $B = -38$, and $C = 76n$. The equation in choice D, $4x - 4y = 4n$, is written in the form $Dx + Ey = F$, where $D = 4$, $E = -4$, and $F = 4n$. Using these values, it follows that the value of $\frac{D}{A}$ is $\frac{4}{38}$, or $\frac{2}{19}$; the value of $\frac{E}{B}$ is $\frac{-4}{-38}$, or $\frac{2}{19}$; and the value of $\frac{F}{C}$ is $\frac{4n}{76n}$, or $\frac{1}{19}$, since n is a positive constant. Since $\frac{D}{A} = \frac{E}{B}$ and since $\frac{F}{C}$ is not equal to $\frac{D}{A}$ nor $\frac{E}{B}$, it follows that the lines represented by these two equations are parallel and distinct. Therefore, the second equation in this system of equations with no solution could be $4x - 4y = 4n$.

Choice A is incorrect. The system consisting of this equation and the given equation has infinitely many solutions rather than no solution.

Choice B is incorrect. The system consisting of this equation and the given equation has one solution rather than no solution.

Choice C is incorrect. The system consisting of this equation and the given equation has one solution rather than no solution.

Question Difficulty: Hard

Question ID d3d42a8f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: d3d42a8f

The function f is defined by $f(x) = 6.2x$. What is the value of $f(x)$ when $x = 10$?

- A. 72
- B. 62
- C. 17.2
- D. 16.2

ID: d3d42a8f Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that the function f is defined by $f(x) = 6.2x$. The value of $f(x)$ when $x = 10$ can be found by substituting 10 for x in $f(x) = 6.2x$, which yields $f(10) = 6.2(10)$, or $f(10) = 62$. Therefore, when $x = 10$, the value of $f(x)$ is 62.

Choice A is incorrect. This is the value of $f(10) + 10$, not the value of $f(x)$ when $x = 10$.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the value of $6.2 + 10$, not the value of $f(x)$ when $x = 10$.

Question Difficulty: Easy

Question ID 08fd9e3a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	

ID: 08fd9e3a

If $2(5x) = 6$, what is the value of $5x$?

- A. $\frac{4}{5}$
- B. $\frac{5}{3}$
- C. **3**
- D. 4

ID: 08fd9e3a Answer

Correct Answer: C

Rationale

Choice C is correct. Dividing both sides of the given equation by **2** yields $5x = 3$. Therefore, the value of $5x$ is **3**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 23468c38

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	

ID: 23468c38

For a restaurant owner to purchase a pizza oven at a total price of **13,000** dollars, a onetime down payment is required, and then fixed monthly payments are made for the remaining amount owed for the pizza oven. The equation $13,000 = 2,800 + 200t$ represents this situation, where t is the number of fixed monthly payments that are made. Which of the following is the best interpretation of **200** in this context?

- A. The amount, in dollars, of the down payment
- B. The amount, in dollars, of each fixed monthly payment
- C. The total amount, in dollars, paid for the pizza oven after t fixed monthly payments
- D. The total number of fixed monthly payments

ID: 23468c38 Answer

Correct Answer: B

Rationale

Choice B is correct. For a restaurant owner to purchase a pizza oven at a total price of **13,000** dollars, a onetime down payment is required, and then fixed monthly payments are made for the remaining amount owed for the pizza oven. It's given that the equation $13,000 = 2,800 + 200t$ represents this situation, where t is the number of fixed monthly payments that are made. In this equation, **2,800** is a constant term, which represents the onetime down payment, in dollars, and **200** is the coefficient of t . Multiplying **200** by t gives the total amount, in dollars, paid by the fixed monthly payments. Therefore, the best interpretation of **200** in this context is the amount, in dollars, of each fixed monthly payment.

Choice A is incorrect. This is the best interpretation of **2,800**, not **200**, in this context.

Choice C is incorrect. This is the best interpretation of **13,000**, not **200**, in this context.

Choice D is incorrect. This is the best interpretation of the variable t , not **200**, in this context.

Question Difficulty: Easy

Question ID 2d703010

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 2d703010

A train traveled **173** miles in the first **3** hours of a trip and traveled at an average speed of **53** miles per hour for the remainder of the trip. Which equation gives the total number of miles, ***y***, the train traveled if the trip was ***x*** hours long and ***x* > 3**?

- A. **$y = 53x + 173$**
- B. **$y = 53x + 14$**
- C. **$y = 14x + 226$**
- D. **$y = 14x + 53$**

ID: 2d703010 Answer

Correct Answer: B

Rationale

Choice B is correct. It’s given that a train traveled **173** miles in the first **3** hours of a trip. It's also given that the train traveled at an average speed of **53** miles per hour for the remainder of the trip. Since the trip was ***x*** hours long, it follows that the remainder of the trip was ***x* − 3** hours. Therefore, **53(*x* − 3)** represents the distance, in miles, the train traveled for the remainder of the trip. Adding **173** to **53(*x* − 3)** gives the total number of miles, ***y***, the train traveled if the trip was ***x*** hours long and ***x* > 3**. It follows that **$y = 173 + 53(x - 3)$** , which is equivalent to **$y = 173 + 53x - 159$** , or **$y = 53x + 14$** .

Choice A is incorrect. This equation gives the total number of miles, ***y***, the train traveled if the trip was ***x* + 3**, not ***x***, hours long.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID acca8366

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	

ID: acca8366

Line p is defined by $2y + 8x = 11$. Line r is perpendicular to line p in the xy -plane. What is the slope of line r ?

- A. -4
- B. $-\frac{1}{4}$
- C. $\frac{1}{4}$
- D. 4

ID: acca8366 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that line p is defined by the equation $2y + 8x = 11$. The equation can be written in slope-intercept form $y = mx + b$, where m represents the slope of the line. Subtracting $8x$ from both sides of the equation $2y + 8x = 11$ yields $2y = -8x + 11$. Dividing both sides of this equation by 2 yields $y = -4x + \frac{11}{2}$. Therefore, the slope of line p in the xy -plane is -4 . It's also given that line p is perpendicular to line r . Perpendicular lines have slopes that are negative reciprocals of each other, so the slope of line r is $-\frac{1}{(-4)} = \frac{1}{4}$.

Choice A is incorrect. This is the slope of line p , not line r .

Choice B is incorrect. This is the reciprocal, not the negative reciprocal, of the slope of line p .

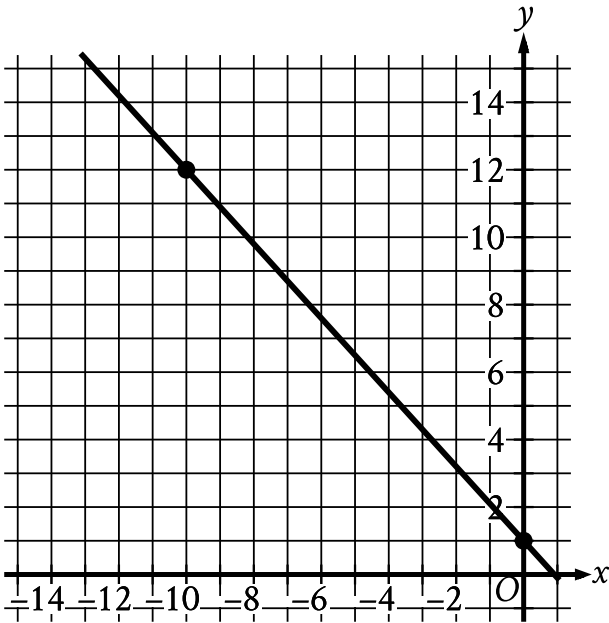
Choice D is incorrect. This is the negative, not the negative reciprocal, of the slope of line p .

Question Difficulty: Medium

Question ID 34e549b3

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: 34e549b3



The graph of line g is shown in the xy -plane. Line k is defined by $165x + py = w$, where p and w are constants. If line k is graphed in this xy -plane, resulting in the graph of a system of two linear equations, the system of two linear equations will have infinitely many solutions. What is the value of $p + w$?

ID: 34e549b3 Answer

Correct Answer: 300

Rationale

The correct answer is **300**. It's given that the graph of line g is shown in the xy -plane and that line k is defined by $165x + py = w$, where p and w are constants. It's also given that if line k is graphed in this xy -plane, the resulting system of two linear equations will have infinitely many solutions. A system of two linear equations has infinitely many solutions when the two linear equations are equivalent. Therefore, line g and line k have the same slope and the same y -intercept. The slope of a line that passes through the points (x_1, y_1) and (x_2, y_2) can be calculated using the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. The graph shows that line g passes through the points $(-10, 12)$ and $(0, 1)$. Substituting $(-10, 12)$ and $(0, 1)$ for (x_1, y_1) and (x_2, y_2) , respectively, in the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$ yields $m = \frac{1 - 12}{0 - (-10)}$, or $m = -\frac{11}{10}$. An equation representing a linear relationship can be written in the form $y = mx + b$, where m is the slope and $(0, b)$ is the y -intercept of the line in the xy -plane. Since line g passes through the point $(0, 1)$, it follows that $b = 1$. Thus, the equation of line g is $y = -\frac{11}{10}x + 1$. For line k , the equation $165x + py = w$ can be rewritten in $y = mx + b$ form by isolating y . Subtracting $165x$ from both sides of this equation yields $py = -165x + w$. Dividing both sides of this equation by p yields $y = \frac{-165}{p}x + \frac{w}{p}$. It follows that in the xy -plane, line k has a slope of $\frac{-165}{p}$ and $(0, \frac{w}{p})$ is the y -intercept. Since line g and line k have the same slope and the same y -intercept, it follows that $-\frac{165}{p} = -\frac{11}{10}$ and $\frac{w}{p} = 1$. Multiplying both sides of $-\frac{165}{p} = -\frac{11}{10}$ by $-10p$ yields $1,650 = 11p$. Dividing both sides of this equation by 11 yields

150 = p . Substituting **150** for **p** in the equation $\frac{w}{p} = 1$ yields $\frac{w}{150} = 1$. Multiplying both sides of this equation by **150** yields **$w = 150$** . Substituting **150** for **p** and **150** for **w** in the expression **$p + w$** yields **$150 + 150$** , which is equivalent to **300**. Therefore, the value of **$p + w$** is **300**.

Question Difficulty: Hard

Question ID 2c2e8b61

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: 2c2e8b61

$$\begin{aligned}y &= 4x \\ 5x - 3y &= -14\end{aligned}$$

The solution to the given system of equations is (x, y) . What is the value of $x + y$?

- A. 4
- B. 6
- C. 8
- D. 10

ID: 2c2e8b61 Answer

Correct Answer: D

Rationale

Choice D is correct. The first equation in the given system is $y = 4x$. Substituting $4x$ for y in the second equation yields $5x - 3(4x) = -14$, which is equivalent to $5x - 12x = -14$, or $-7x = -14$. Dividing both sides of this equation by -7 yields $x = 2$. Substituting 2 for x in the equation $y = 4x$ yields $y = 4(2)$, or $y = 8$. Substituting 2 for x and 8 for y in the expression $x + y$ yields $2 + 8$, or 10 .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the value of $y - x$, not the value of $x + y$.

Choice C is incorrect. This is the value of y , not the value of $x + y$.

Question Difficulty: Medium

Question ID 0fdece4f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	

ID: 0fdece4f

A tow truck travels at a speed of at least **25** miles per hour but no more than **45** miles per hour for a section of a road. Which inequality represents this situation, where x is the speed of the tow truck, in miles per hour, on this section of the road?

- A. $x \geq 25$
- B. $x \geq 45$
- C. $25 \leq x \leq 45$
- D. $x \leq 70$

ID: 0fdece4f Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that a tow truck travels at a speed of at least **25** miles per hour but no more than **45** miles per hour. It's also given that x represents the speed of the tow truck, in miles per hour. At least **25** miles per hour means $x \geq 25$, and no more than **45** miles per hour means $x \leq 45$. Therefore, the inequality that represents this situation is $25 \leq x \leq 45$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 78deebff

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	

ID: 78deebff

In the xy -plane, which of the following does NOT contain any points that are part of the solution set to $5x - 7y > 35$?

- A. The x -axis
- B. The region where $x > 0$ and $y > 0$
- C. The region where $x < 0$ and $y < 0$
- D. The region where $x < 0$ and $y > 0$

ID: 78deebff Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that the solution set consists of points in the xy -plane where $5x - 7y > 35$, which can also be written as $5x + (-7y) > 35$. If (x, y) is a point in the region where $x < 0$ and $y > 0$, then x is a negative number and y is a positive number. If x is a negative number, then the value of $5x$ is a negative number. If y is a positive number, then the value of $-7y$ is a negative number. It follows that the expression $5x + (-7y)$ is the sum of two negative numbers and is therefore negative. If the value of $5x + (-7y)$ is negative, then the value of $5x + (-7y)$ cannot be greater than 35 . Thus, the region where $x < 0$ and $y > 0$ does not contain any points that are part of the solution set to $5x - 7y > 35$.

Choice A is incorrect. The x -axis contains the point $(10, 0)$, which is part of the solution set to $5x - 7y > 35$, since $5(10) - 7(0) > 35$, or $50 > 35$, is true.

Choice B is incorrect. The region where $x > 0$ and $y > 0$ contains the point $(10, 1)$, which is part of the solution set to $5x - 7y > 35$, since $5(10) - 7(1) > 35$, or $43 > 35$, is true.

Choice C is incorrect. The region where $x < 0$ and $y < 0$ contains the point $(-1, -10)$, which is part of the solution set to $5x - 7y > 35$, since $5(-1) - 7(-10) > 35$, or $65 > 35$, is true.

Question Difficulty: Medium

Question ID b909ec41

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	

ID: b909ec41

$9r = 7(r + 7)$

What value of r is the solution to the given equation?

ID: b909ec41 Answer

Correct Answer: 24.5, 49/2

Rationale

The correct answer is $\frac{49}{2}$. It's given that $9r = 7(r + 7)$. Applying the distributive property on the right-hand side of this equation yields $9r = 7r + 49$. Subtracting $7r$ from both sides of this equation yields $2r = 49$. Dividing both sides of this equation by 2 yields $r = \frac{49}{2}$. Therefore, the solution to the given equation is $\frac{49}{2}$. Note that 49/2 and 24.5 are examples of ways to enter a correct answer.

Question Difficulty: Medium

Question ID 14319946

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 14319946

In the xy -plane, line k passes through the points $(8, 6)$ and $(9, 12)$ and is defined by the equation $y = mx + b$. What is the value of b ?

ID: 14319946 Answer

Correct Answer: -42

Rationale

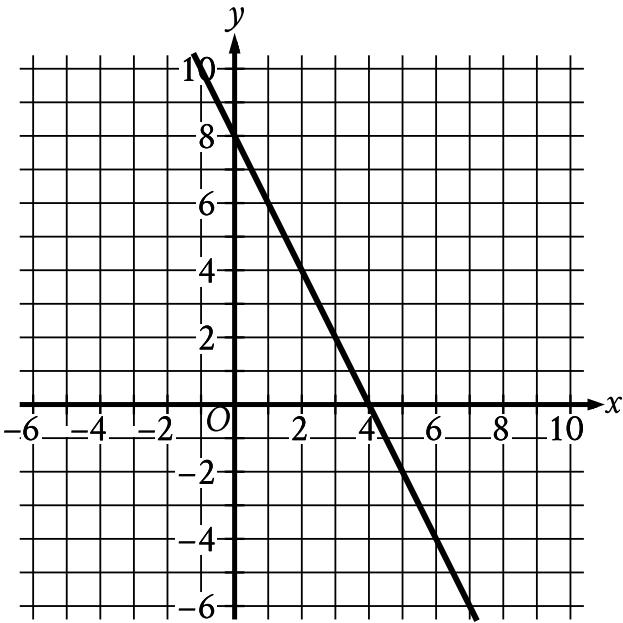
The correct answer is -42 . For a linear equation in the form $y = mx + b$, m represents the slope and b represents the y -coordinate of the y -intercept of the graph in the xy -plane. The slope of the graph of a line containing any two points, (x_1, y_1) and (x_2, y_2) , can be found using the slope formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. It's given that in the xy -plane, line k passes through the points $(8, 6)$ and $(9, 12)$. Substituting $(8, 6)$ and $(9, 12)$ for (x_1, y_1) and (x_2, y_2) , respectively, in the slope formula yields $m = \frac{12 - 6}{9 - 8}$, or $m = 6$. Since line k passes through the point $(9, 12)$ and the slope of the line is 6 , substituting 9 for x , 12 for y , and 6 for m in the equation $y = mx + b$ yields $12 = (6)(9) + b$, or $12 = 54 + b$. Subtracting 54 from both sides of this equation yields $-42 = b$. Therefore, the value of b is -42 .

Question Difficulty: Medium

Question ID 56efb86c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 56efb86c



The graph of the linear function f is shown, where $y = f(x)$. Which equation defines function f ?

- A. $f(x) = -2x - 8$
- B. $f(x) = -2x + 8$
- C. $f(x) = 2x - 8$
- D. $f(x) = 2x + 8$

ID: 56efb86c Answer

Correct Answer: B

Rationale

Choice B is correct. A linear function can be defined by the equation $f(x) = mx + b$, where m is the slope and $(0, b)$ is the y -intercept of the graph of $y = f(x)$ in the xy -plane. The slope of a line that passes through the points (x_1, y_1) and (x_2, y_2) can be calculated using the equation $m = \frac{y_2 - y_1}{x_2 - x_1}$. The line shown passes through the points $(4, 0)$ and $(0, 8)$. Substituting $(4, 0)$ for (x_1, y_1) and $(0, 8)$ for (x_2, y_2) in the equation $m = \frac{y_2 - y_1}{x_2 - x_1}$ yields $m = \frac{8 - 0}{0 - 4}$, or $m = -2$. Since the line shown passes through the point $(0, 8)$, it follows that $b = 8$. Substituting -2 for m and 8 for b in the equation $f(x) = mx + b$ yields $f(x) = -2x + 8$. Therefore, the equation $f(x) = -2x + 8$ defines function f .

Choice A is incorrect. This equation defines a linear function whose graph has a slope of -2 and a y -intercept of $(0, -8)$, not $(0, 8)$.

Choice C is incorrect. This equation defines a linear function whose graph has a slope of 2 , not -2 , and a y -intercept of $(0, -8)$,

not $(0, 8)$.

Choice D is incorrect. This equation defines a linear function whose graph has a slope of 2 , not -2 , and a y -intercept of $(0, 8)$.

Question Difficulty: Easy

Question ID 0b8cb842

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	

ID: 0b8cb842

If $9x + 4 = 67$, what is the value of $90x + 40$?

- A. 7
- B. 70
- C. 130
- D. 670

ID: 0b8cb842 Answer

Correct Answer: D

Rationale

Choice D is correct. Multiplying both sides of the given equation by 10 yields $10(9x + 4) = 10(67)$, or $90x + 40 = 670$. Therefore, the value of $90x + 40$ is 670.

Choice A is incorrect. This is the value of x , not $90x + 40$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID a286a6df

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: a286a6df

For the linear function h , $h(0) = 0$ and $h(11) = 10$. Which equation defines h ?

- A. $h(x) = \frac{10}{11}x$
- B. $h(x) = \frac{10}{11}x + 10$
- C. $h(x) = \frac{11}{10}x$
- D. $h(x) = \frac{11}{10}x + 11$

ID: a286a6df Answer

Correct Answer: A

Rationale

Choice A is correct. Since h is a linear function, it can be defined by an equation of the form $h(x) = mx + b$, where m and b are constants. It's given that $h(0) = 0$. Substituting 0 for x and 0 for $h(x)$ in the equation $h(x) = mx + b$ yields $0 = 0m + b$, or $0 = b$. Substituting 0 for b in the equation $h(x) = mx + b$ yields $h(x) = mx$. It's also given that $h(11) = 10$. Substituting 11 for x and 10 for $h(x)$ in the equation $h(x) = mx$ yields $10 = 11m$. Dividing both sides of this equation by 11 yields $m = \frac{10}{11}$. Substituting $\frac{10}{11}$ for m in the equation $h(x) = mx$ yields $h(x) = \frac{10}{11}x$. Therefore, an equation that defines h is $h(x) = \frac{10}{11}x$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID e4fd913d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: e4fd913d

A computer company pays customers for trading in old computers. The payment p , in dollars, is the original retail price of a computer minus \$150 for each year since the computer was released. Which equation represents this situation for a computer with an original retail price of \$690 when it was released t years ago, where t is a whole number and $1 \leq t \leq 4$?

- A. $p = 150 - 690t$
- B. $p = 690 - 150t$
- C. $t = 150 - 690p$
- D. $t = 690 - 150p$

ID: e4fd913d Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that the payment p , in dollars, for trading in an old computer is the original retail price of the computer minus \$150 for each year since the computer was released t years ago. It's also given that the original retail price of a computer was \$690. Thus, the equation that represents this situation is $p = 690 - 150t$.

Choice A is incorrect. This equation represents a situation for a computer with an original retail price of \$150 minus \$690 for each year since the computer was released.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

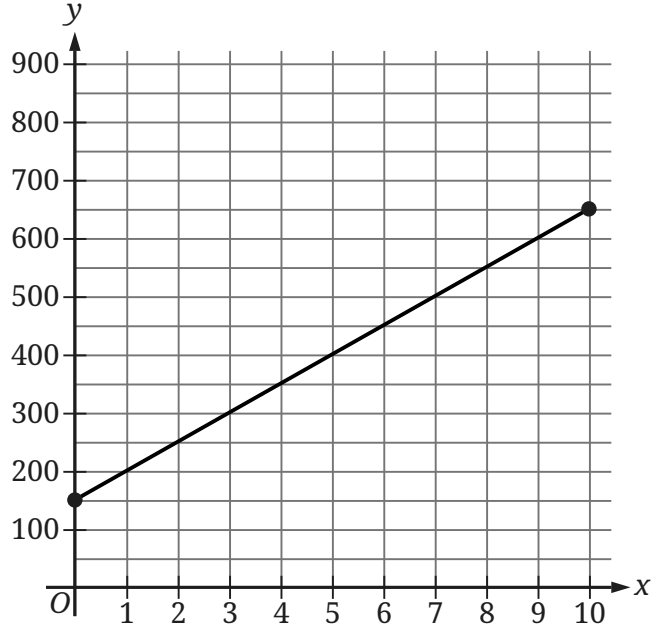
Question Difficulty: Easy

Question ID 8d886d46

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 8d886d46

While walking on a trail, Mary stopped to read a trail sign. The graph shows the total distance y , in meters, Mary had walked on the trail x minutes after leaving the trail sign.



What distance, in meters, did Mary walk on the trail in the **10** minutes after leaving the trail sign?

ID: 8d886d46 Answer

Correct Answer: 500

Rationale

The correct answer is **500**. It’s given that the graph shows the total distance y , in meters, Mary had walked on the trail x minutes after leaving the trail sign. The graph represents a linear relationship between time and distance, so the distance walked in a given time interval corresponds to the change in the y -values over that interval. At $x = 0$, the graph shows a y -value of **150**, and at $x = 10$, the graph shows a y -value of **650**. The distance Mary walked in the **10** minutes after leaving the trail sign is the difference between these y -values, which can be written as **650** $–$ **150**, or **500**. Therefore, the distance, in meters, Mary walked in the **10** minutes after leaving the trail sign is **500**.

Question Difficulty: Hard

Question ID daa7a4c7

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	

ID: daa7a4c7

Which expression is equivalent to $2\left(x - \frac{9}{2}\right)(x + 8)$?

- A. $2x^2 - 72$
- B. $2x^2 + 7x - 72$
- C. $2x^2 - x - 36$
- D. $x^2 + 7x - 36$

ID: daa7a4c7 Answer

Correct Answer: B

Rationale

Choice B is correct. In the given expression, $2\left(x - \frac{9}{2}\right)$ can be rewritten as $2x - 2\left(\frac{9}{2}\right)$, or $2x - 9$. Applying the distributive property to the expression $(2x - 9)(x + 8)$ yields $2x(x + 8) - 9(x + 8)$, or $2x^2 + 16x - 9x - 72$. Combining like terms in this expression yields $2x^2 + 7x - 72$. Thus, $2\left(x - \frac{9}{2}\right)(x + 8)$ is equivalent to $2x^2 + 7x - 72$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

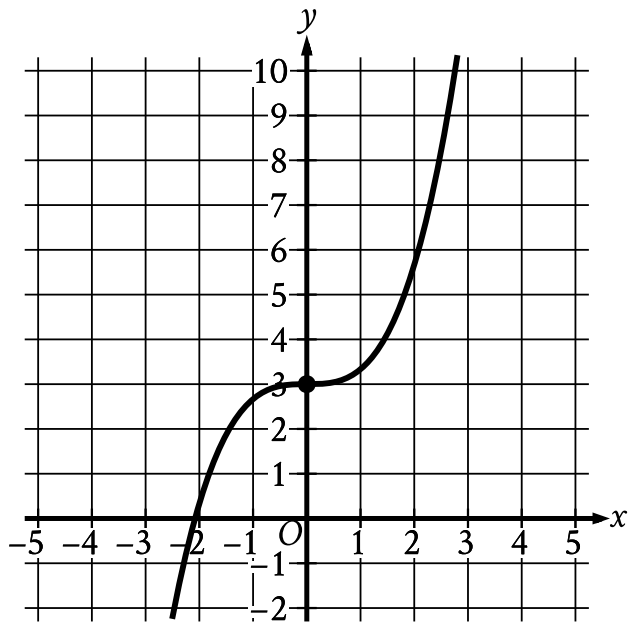
Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 0953fad0

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 0953fad0



The graph of $y = f(x)$ is shown. What is the y -intercept of $y = f(x) + 5$?

- A. $(0, 8)$
- B. $(0, 5)$
- C. $(0, 3)$
- D. $(0, 0)$

ID: 0953fad0 Answer

Correct Answer: A

Rationale

Choice A is correct. The y -intercept of a graph is the point at which the graph intersects the y -axis. The graph of $y = f(x)$ shown intersects the y -axis at $(0, 3)$, so the y -intercept of the graph of $y = f(x)$ is $(0, 3)$. The graph of $y = f(x) + 5$ is the result of shifting the graph of $y = f(x)$ up 5 units. Therefore, the y -intercept of the graph of $y = f(x) + 5$ is 5 units above the y -intercept of the graph of $y = f(x)$. It follows that the y -intercept of the graph of $y = f(x) + 5$ is $(0, 3 + 5)$, which is equivalent to $(0, 8)$.

Choice B is incorrect. The point $(0, 5)$ is 5 units above the point $(0, 0)$, not the point $(0, 3)$.

Choice C is incorrect. This is the y -intercept of the graph of $y = f(x)$, not the y -intercept of the graph of $y = f(x) + 5$.

Choice D is incorrect. This is the origin of the coordinate plane, not the y -intercept of the graph of $y = f(x) + 5$.

Question Difficulty: Medium

Question ID d444ba72

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: d444ba72

A certain bowl was sold online for prices varying between \$7 and \$13. The number of bowls sold can be modeled by a quadratic function based on the price, where 5 bowls were sold for \$12 each and the maximum of 9 bowls were sold for \$10 each. According to the model, what is the number of bowls that were sold for \$8 each?

ID: d444ba72 Answer

Correct Answer: 5

Rationale

The correct answer is 5. It's given that the number of bowls sold can be modeled by a quadratic function based on the price, where a maximum of 9 bowls were sold for \$10 each and 5 bowls were sold for \$12 each. This quadratic function can be defined by an equation of the form $N(p) = a(p - h)^2 + k$, where $N(p)$ is the number of bowls sold when the price is \$ p each, and a , h , and k are constants such that the function reaches its maximum value, k , when $p = h$. Since the model indicates the maximum number of bowls sold is 9 when the price is \$10 each, $N(p)$ reaches its maximum value, 9, when $p = 10$. Therefore, $k = 9$ and $h = 10$. Substituting 10 for h and 9 for k in the function $N(p) = a(p - h)^2 + k$ yields $N(p) = a(p - 10)^2 + 9$. Since the model indicates the number of bowls sold is 5 when the price is \$12 each, the value of $N(p)$ is 5 when $p = 12$. Substituting 12 for p and 5 for $N(p)$ in the equation $N(p) = a(p - 10)^2 + 9$ yields $5 = a(12 - 10)^2 + 9$, or $5 = 4a + 9$. Subtracting 9 from both sides of this equation yields $-4 = 4a$. Dividing both sides of this equation by 4 yields $-1 = a$. Therefore, the model can be represented by the equation $N(p) = -(p - 10)^2 + 9$. Substituting 8 for p in this equation yields $N(8) = -(8 - 10)^2 + 9$, or $N(8) = -4 + 9$, or $N(8) = 5$. Therefore, according to the model, the number of bowls sold for \$8 each was 5.

Question Difficulty: Hard

Question ID 0829527e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 0829527e

The functions p and r are defined by the equations shown, where a and b are integer constants such that $a < b$ and $b < 0$. If $y = p(x)$ and $y = r(x)$ are graphed in the xy -plane, which of the following equations displays, as a constant or coefficient, the y -coordinate of the y -intercept of the graph of the corresponding function?

- I. $p(x) = a(4.1)^x + b$
- II. $r(x) = a(4.1)^{x+b}$

- A. I only
- B. II only
- C. I and II
- D. Neither I nor II

ID: 0829527e Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that a and b are integer constants such that $a < b$ and $b < 0$. The y -coordinate of the y -intercept of the graph of a function is the value of the function when $x = 0$. Equation I is $p(x) = a(4.1)^x + b$. Substituting 0 for x in this equation yields $p(0) = a(4.1)^0 + b$, which can be rewritten as $p(0) = a(1) + b$, or $p(0) = a + b$. Therefore, the y -coordinate of the y -intercept of the graph of $y = p(x)$ is $a + b$. Equation I has a coefficient of a and a constant b . Thus, the value of the y -coordinate of the y -intercept, $a + b$, is not displayed as a constant or coefficient in equation I. Equation II is $r(x) = a(4.1)^{x+b}$. Similarly, substituting 0 for x in this equation yields $r(0) = a(4.1)^{0+b}$, which can be rewritten as $r(0) = a(4.1)^b$. Since $b < 0$, the value of $(4.1)^b$ is not equal to 1 . Thus, the value of $a(4.1)^b$ is not equal to a . The value of $a(4.1)^b$ is also not equal to b , since there is no pair of integers a and b such that $a(4.1)^b = b$. Thus, the value of the y -coordinate of the y -intercept for this equation is $a(4.1)^b$, which is not displayed in equation II as a constant or coefficient. The coefficient in equation II is a , and the only constant, b , appears in the exponent. Therefore, neither equation I nor equation II displays, as a constant or coefficient, the y -coordinate of the y -intercept of the graph of the corresponding function.

- Choice A is incorrect and may result from conceptual or calculation errors.
- Choice B is incorrect and may result from conceptual or calculation errors.
- Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 109042e4

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 109042e4

$r^2 + qr = 2r - 55$

In the given equation, q is an integer constant. The given equation has no real solutions. What is the largest possible value of q ?

ID: 109042e4 Answer

Correct Answer: 16

Rationale

The correct answer is **16**. A quadratic equation of the form $ax^2 + bx + c = 0$, where a , b , and c are constants, has no real solution if and only if its discriminant, $b^2 - 4ac$, is negative. It's given that in the equation $r^2 + qr = 2r - 55$, q is an integer constant. Subtracting $2r$ and adding 55 to both sides of this equation yields $r^2 + qr - 2r + 55 = 0$, or $r^2 + (q - 2)r + 55 = 0$. In this equation, $a = 1$, $b = q - 2$, and $c = 55$. Substituting 1 for a , $q - 2$ for b , and 55 for c in the discriminant expression yields $(q - 2)^2 - 4(1)(55)$, or $(q - 2)^2 - 220$. Since the value of the discriminant must be negative, it follows that $(q - 2)^2 - 220 < 0$, or $(q - 2)^2 < 220$. This is equivalent to $|q - 2| < \sqrt{220}$, or $-\sqrt{220} < q - 2 < \sqrt{220}$. Adding 2 to each part of this inequality yields $2 - \sqrt{220} < q < 2 + \sqrt{220}$. Since $\sqrt{220}$ is approximately **14.83**, it follows that $-12.83 < q < 16.83$. Since q is an integer, the largest possible value of q is **16**.

Question Difficulty: Hard

Question ID f9d58e07

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	

ID: f9d58e07

The expression $\frac{29x+102}{x(x+51)}$ is equivalent to $\frac{p}{x} + \frac{w}{x+51}$, where p and w are constants. What is the value of w ?

ID: f9d58e07 Answer

Correct Answer: 27

Rationale

The correct answer is **27**. The given expression $\frac{p}{x} + \frac{w}{x+51}$ can be rewritten with the common denominator $x(x + 51)$. To add the two terms of this expression, $\frac{p}{x}$ can be rewritten as $\frac{p(x+51)}{x(x+51)}$ and $\frac{w}{(x+51)}$ can be rewritten as $\frac{wx}{x(x+51)}$. Therefore, the expression $\frac{p}{x} + \frac{w}{(x+51)}$ can be rewritten as $\frac{p(x+51)}{x(x+51)} + \frac{wx}{x(x+51)}$, which is equivalent to $\frac{px+51p+wx}{x(x+51)}$, or $\frac{(p+w)x+51p}{x(x+51)}$. Since $\frac{29x+102}{x(x+51)}$ is equivalent to $\frac{(p+w)x+51p}{x(x+51)}$ and these expressions have equivalent denominators, these expressions must also have equivalent numerators. It follows that $29x + 102 = (p + w)x + 51p$. In this equation, the x-coefficients must be equal and the constant terms must be equal. Therefore, $p + w = 29$ and $51p = 102$. Dividing both sides of $51p = 102$ by 51 yields $p = 2$. Substituting 2 for p in $p + w = 29$ yields $2 + w = 29$. Subtracting 2 from both sides of this equation yields $w = 27$. Therefore, the value of w is **27**.

Question Difficulty: Hard

Question ID 29646434

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	

ID: 29646434

Which expression is equivalent to $(9x^2 + 6) - (8x^2)$?

- A. 7
- B. $17x^2 + 6$
- C. $x^2 - 6$
- D. $x^2 + 6$

ID: 29646434 Answer

Correct Answer: D

Rationale

Choice D is correct. The given expression is equivalent to $9x^2 + 6 - 8x^2$. The terms $9x^2$ and $-8x^2$ are like terms. Combining these like terms yields x^2 . It follows that the expression $(9x^2 + 6) - (8x^2)$ is equivalent to $x^2 + 6$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is equivalent to $(9x^2 + 6) + (8x^2)$, not the given expression.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID ba27a4c7

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: ba27a4c7

$$\frac{1}{7}(x - k)(16 - x^2) = 0$$

In the given equation, k is a positive constant. The sum of the solutions to the equation is **59**. What is the value of k ?

ID: ba27a4c7 Answer

Correct Answer: 59

Rationale

The correct answer is **59**. It's given that in the equation $\frac{1}{7}(x - k)(16 - x^2) = 0$, k is a positive constant. Multiplying both sides of this equation by **7** yields $(x - k)(16 - x^2) = 0$. By the zero product property, it follows that $x - k = 0$ or $16 - x^2 = 0$. Adding k to both sides of the equation $x - k = 0$ yields $x = k$. Adding x^2 to both sides of the equation $16 - x^2 = 0$ yields $16 = x^2$. Taking the square root of both sides of this equation yields $4 = x$ and $-4 = x$. Therefore, the solutions to the given equation are k , **4**, and **-4**. It's given that the sum of the solutions to the equation is **59**. Thus, $k + 4 + (-4) = 59$, or $k = 59$.

Question Difficulty: Hard

Question ID a3ffa25f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	

ID: a3ffa25f

$f(x) = 2x + 3$
 $g(x) = 7x - 2$
 $h(x) = 5x + 6$

The functions f , g , and h are defined as shown. If $f(x) \cdot g(x) - h(x) = ax^2 + bx + c$, where a , b , and c are constants, what is the value of b ?

- A. -5
- B. 12
- C. 20
- D. 22

ID: a3ffa25f Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that $f(x) = 2x + 3$, $g(x) = 7x - 2$, and $h(x) = 5x + 6$. It's also given that $f(x) \cdot g(x) - h(x) = ax^2 + bx + c$, where a , b , and c are constants. Substituting $2x + 3$ for $f(x)$, $7x - 2$ for $g(x)$, and $5x + 6$ for $h(x)$ in this equation yields $(2x + 3)(7x - 2) - (5x + 6) = ax^2 + bx + c$. Applying the distributive property to the left-hand side of this equation yields $(2x)(7x) + (2x)(-2) + (3)(7x) + (3)(-2) + (-1)(5x) + (-1)(6) = ax^2 + bx + c$, or $14x^2 - 4x + 21x - 6 - 5x - 6 = ax^2 + bx + c$. Combining like terms on the left-hand side of this equation yields $14x^2 + 12x - 12 = ax^2 + bx + c$. It follows that the value of b is 12 .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 2f230ca7

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 2f230ca7

$$y = 7,400(0.87)^x$$

The given equation estimates the value, in dollars, of a certain piece of equipment x years after it was purchased. According to the given equation, what was the estimated value, in dollars, of the piece of equipment at the time it was purchased?

- A. 740
- B. 6,438
- C. 7,400
- D. 8,700

ID: 2f230ca7 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that $y = 7,400(0.87)^x$ estimates the value, in dollars, of a certain piece of equipment x years after it was purchased. The estimated value of the piece of equipment at the time it was purchased corresponds to the value of y when $x = 0$. Substituting 0 for x in the equation $y = 7,400(0.87)^x$ yields $y = 7,400(0.87)^0$, which is equivalent to $y = 7,400(1)$, or $y = 7,400$. Therefore, the estimated value, in dollars, of the piece of equipment at the time it was purchased was 7,400.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This was the estimated value, in dollars, of the piece of equipment 1 year after it was purchased.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 48dee717

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 48dee717

The function C defined by $C(t) = 6(1.5)^t$ gives the estimated number of clients at a certain business, where t is the number of weeks since the business opened. Which of the following is the best interpretation of the number 6 in this context?

- A. The estimated number of clients when the business opened
- B. The increase in the estimated number of clients each week
- C. The number of weeks the business has been open
- D. The percent increase in the estimated number of clients each week

ID: 48dee717 Answer

Correct Answer: A

Rationale

Choice A is correct. For an exponential function of the form $C(t) = a(b)^t$, where a and b are constants, the initial value of the function, or the value of the function when $t = 0$, is a , and the value of the function changes by a factor of b each time t increases by 1 . It follows that in the given function, the initial value is 6 and the value of the function changes by a factor of 1.5 each time t increases by 1 . It's given that the function C gives the estimated number of clients at a certain business and t is the number of weeks since the business opened. Therefore, the best interpretation of the number 6 in this context is the estimated number of clients when $t = 0$, or when the business opened.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

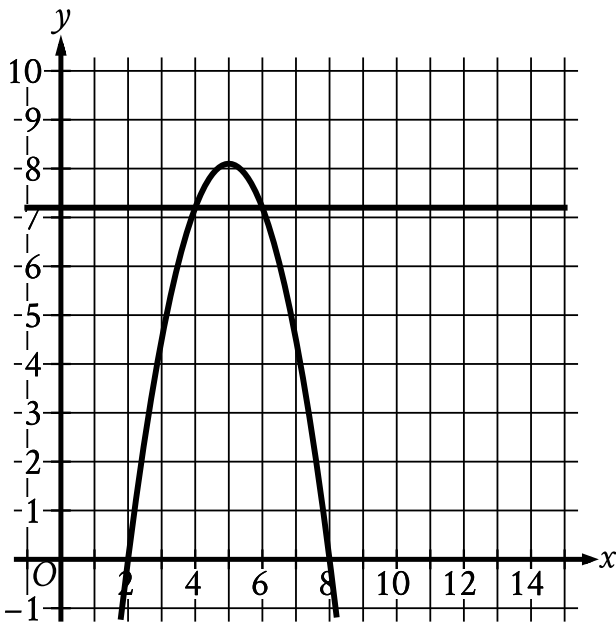
Choice D is incorrect and may result from conceptual errors.

Question Difficulty: Medium

Question ID 51857e96

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 51857e96



The graph of a system of a linear equation and a quadratic equation is shown. A solution to the system is (x, y) . What is a possible value of x ?

- A. 4
- B. 5
- C. 7.2
- D. 8

ID: 51857e96 Answer

Correct Answer: A

Rationale

Choice A is correct. The solutions to the system correspond to the points where the graphs of the equations intersect. The graphs of the linear equation and the quadratic equation shown intersect at approximately $(4, 7.2)$ and $(6, 7.2)$. It follows that the possible values of x are 4 and 6. Of these two possible values, only 4 is given as a choice.

Choice B is incorrect. This is the x-value of the vertex of the graph of the quadratic equation, not the x-value of a solution to the system.

Choice C is incorrect. This is the approximate y -value, not the x -value, of a solution to the system.

Choice D is incorrect. This is the x -value of an x -intercept of the graph of the quadratic equation.

Question Difficulty: Medium

Question ID 9bcb0516

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	

ID: 9bcb0516

The given expression can be rewritten as $\frac{a}{6}x^2 + \frac{b}{6}x + \frac{c}{6}$, where a , b , and c are constants. What is the value of $a + b + c$?

A. -612

B. -306

C. -102

D. -17

ID: 9bcb0516 Answer

Correct Answer: A

Rationale

Choice A is correct. Expanding each squared binomial in the given expression yields $-14(25x^2 - 20x + 4) + 6(25x^2 - 30x + 9)$. Applying the distributive property to both terms of this expression yields $(-350x^2 + 280x - 56) + (150x^2 - 180x + 54)$. Combining like terms in this expression yields $-200x^2 + 100x - 2$. Since the given expression can be rewritten as $\frac{a}{6}x^2 + \frac{b}{6}x + \frac{c}{6}$, it follows that $\frac{a}{6} = -200$, $\frac{b}{6} = 100$, and $\frac{c}{6} = -2$. Multiplying both sides of each of these equations by 6 yields $a = -1,200$, $b = 600$, and $c = -12$, respectively. Substituting $-1,200$ for a , 600 for b , and -12 for c in the expression $a + b + c$ yields $-1,200 + 600 - 12$, or -612 .

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the value of $\frac{a}{6} + \frac{b}{6} + \frac{c}{6}$, not $a + b + c$.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID ae5dc01e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: ae5dc01e

$f(x) = 27(1.20)^{\frac{x}{3}}$

For the given function f , the value of $f(x)$ increases by $p\%$ for every increase of x by 6. What is the value of p ?

ID: ae5dc01e Answer

Correct Answer: 44

Rationale

The correct answer is **44**. For a function defined by an equation of the form $f(x) = a\left(1 + \frac{r}{100}\right)^{\frac{x}{k}}$, where a , r , and k are positive constants, the value of $f(x)$ increases by $r\%$ for every increase of x by k . The given equation can be rewritten as $f(x) = 27(1.20)^{\frac{2x}{6}}$, or $f(x) = 27\left((1.2)^2\right)^{\frac{x}{6}}$, which is equivalent to $f(x) = 27(1.44)^{\frac{x}{6}}$, or $f(x) = 27\left(1 + \frac{44}{100}\right)^{\frac{x}{6}}$. This equation is of the form $f(x) = a\left(1 + \frac{r}{100}\right)^{\frac{x}{k}}$, where $a = 27$, $r = 44$, and $k = 6$. It follows that the value of $f(x)$ increases by **44%** for every increase of x by 6. Therefore, the value of p is **44**.

Alternate approach: It's given that $f(x) = 27(1.20)^{\frac{x}{3}}$. An increase in the value of x by 6 can be represented by the expression $x + 6$. The value of $f(x + 6)$ can be found by substituting $x + 6$ for x in the equation $f(x) = 27(1.20)^{\frac{x}{3}}$, which yields $f(x + 6) = 27(1.20)^{\frac{x+6}{3}}$. By the product property of exponents, this equation is equivalent to $f(x + 6) = 27(1.20)^{\frac{x}{3}} \cdot (1.20)^{\frac{6}{3}}$, or $f(x + 6) = 27(1.20)^{\frac{x}{3}} \cdot (1.20)^2$, which can be rewritten as $f(x + 6) = 27(1.20)^{\frac{x}{3}} \cdot (1.44)$. Since $f(x) = 27(1.20)^{\frac{x}{3}}$, it follows that $f(x + 6) = f(x) \cdot (1.44)$. Therefore, for any value of x , the value of $f(x + 6)$ is 1.44 times, or 44% greater than, the value of $f(x)$. It's given that the value of $f(x)$ increases by $p\%$ for every increase in the value of x by 6. It follows that the value of p is **44**.

Question Difficulty: Hard

Question ID 20ceb544

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 20ceb544

$4x^2 - 5x - 5 = 0$

What is the greatest solution to the given equation?

- A. $\frac{5}{8} - \frac{\sqrt{105}}{8}$
- B. $\frac{5}{8} + \frac{\sqrt{105}}{8}$
- C. $\frac{5}{4} - \frac{\sqrt{105}}{4}$
- D. $\frac{5}{4} + \frac{\sqrt{105}}{4}$

ID: 20ceb544 Answer

Correct Answer: B

Rationale

Choice B is correct. The solutions to a quadratic equation of the form $ax^2 + bx + c = 0$, where a , b , and c are constants, can be calculated using the quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$. The given equation is of this form, where $a = 4$, $b = -5$, and $c = -5$. Substituting these values into the quadratic formula yields $x = \frac{-(-5) \pm \sqrt{(-5)^2 - 4(4)(-5)}}{2(4)}$, or $x = \frac{5 \pm \sqrt{25 + 80}}{8}$. These solutions can be rewritten as $x = \frac{5 \pm \sqrt{105}}{8}$, or $x = \frac{5}{8} \pm \frac{\sqrt{105}}{8}$. Since $\frac{5}{8} + \frac{\sqrt{105}}{8}$ is greater than $\frac{5}{8} - \frac{\sqrt{105}}{8}$, it follows that the greatest solution to the given equation is $\frac{5}{8} + \frac{\sqrt{105}}{8}$.

Choice A is incorrect. This is the least, not the greatest, solution to the given equation.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID f27c4fa7

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	

ID: f27c4fa7

$$\frac{\sqrt[3]{n^{17}p^5}}{5n^2(\sqrt[8]{p^8})}$$

For all positive values of n and p , the given expression is equivalent to which of the following?

I. $\frac{(n^{\frac{14}{3}})(p^{\frac{5}{3}})}{5np}$

II. $\frac{\sqrt[3]{n^{11}p^2}}{5}$

- A. I only
- B. II only
- C. I and II
- D. Neither I nor II

ID: f27c4fa7 Answer

Correct Answer: C

Rationale

Choice C is correct. For positive values of a , $\sqrt[N]{a} = a^{\frac{1}{N}}$. Therefore, the given expression, $\frac{\sqrt[3]{n^{17}p^5}}{5n^2(\sqrt[8]{p^8})}$, can be rewritten as $\frac{(n^{17}p^5)^{\frac{1}{3}}}{5n^2(p^8)^{\frac{1}{8}}}$. Applying properties of exponents, this can be rewritten as $\frac{n^{\frac{17}{3}}p^{\frac{5}{3}}}{5n^2p^{\frac{8}{8}}}$, or $\frac{n^{\frac{17}{3}}p^{\frac{5}{3}}}{5n^2p}$. Dividing the numerator and the denominator of this expression by n yields $\frac{n^{\frac{14}{3}}p^{\frac{5}{3}}}{5np}$, which is statement I. Dividing the numerator and the denominator of this expression by np yields $\frac{n^{\frac{11}{3}}p^{\frac{2}{3}}}{5}$. This expression is equivalent to $\frac{(n^{11}p^2)^{\frac{1}{3}}}{5}$. It follows that this expression is equivalent to $\frac{\sqrt[3]{n^{11}p^2}}{5}$, which is statement II. Therefore, for all positive values of n and p , the given expression is equivalent to I and II.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 263b160d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 263b160d

A lab analyst observes a sample of a substance. An exponential model estimates that the mass, in grams, of the sample decreases by **24%** every **22.96** minutes. Which of the following equations could represent this model, where M is the estimated mass, in grams, of the sample t minutes after the lab analyst began observing the sample?

- A. $M = 100(0.24)^{t+22.96}$
- B. $M = 100(0.24)^{\frac{t}{22.96}}$
- C. $M = 100(0.76)^{t+22.96}$
- D. $M = 100(0.76)^{\frac{t}{22.96}}$

ID: 263b160d Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that the mass of a sample of a substance decreases by **24%** every **22.96** minutes. Therefore, after each **22.96**-minute interval the mass is multiplied by $1 - 0.24$, or **0.76**. It's also given that t represents the number of minutes after the lab analyst began observing the sample; therefore, $\frac{t}{22.96}$ represents the number of **22.96**-minute intervals. The equation that could represent this model is $M = 100(0.76)^{\frac{t}{22.96}}$, where M is the estimated mass, in grams, of the sample t minutes after the lab analyst began observing it.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 28d7dd59

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 28d7dd59

The function h is defined by $h(x) = a^x + b$, where a and b are positive constants. The graph of $y = h(x)$ in the xy -plane passes through the points $(0, 10)$ and $(2, 13)$. What is the value of ab ?

- A. 13
- B. 18
- C. 20
- D. 26

ID: 28d7dd59 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that function h is defined by $h(x) = a^x + b$, where a and b are positive constants. It's also given that the graph of $y = h(x)$ in the xy -plane passes through the point $(0, 10)$. Substituting 0 for x and 10 for $h(x)$ in the equation $h(x) = a^x + b$ yields $10 = a^0 + b$. Since $a^0 = 1$, this equation can be rewritten as $10 = 1 + b$. Subtracting 1 from both sides of this equation yields $9 = b$. Substituting 9 for b in the equation $h(x) = a^x + b$ yields $h(x) = a^x + 9$. It's also given that the graph of $y = h(x)$ in the xy -plane passes through the point $(2, 13)$. Substituting 2 for x and 13 for $h(x)$ in the equation $h(x) = a^x + 9$ yields $13 = a^2 + 9$. Subtracting 9 from both sides of this equation yields $4 = a^2$. Taking the square root of both sides of this equation yields $2 = a$ and $-2 = a$. Since a is positive, it follows that $2 = a$. Since $9 = b$ and $2 = a$, it follows that the value of ab is $(2)(9)$, or 18 .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 42a062ab

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 42a062ab

$$x^2 + y^2 = 36$$
$$y = mx + \frac{b}{4}$$

In the given system of equations, m and b are negative constants. In the xy -plane, the graphs of the equations in the given system intersect at the point $(-5, y)$, where $y < 0$. Which expression represents the value of b ?

- A. $-\frac{5m}{4} + \frac{\sqrt{11}}{4}$
- B. $\frac{5m}{4} - \frac{\sqrt{11}}{4}$
- C. $-20m + 4\sqrt{11}$
- D. $20m - 4\sqrt{11}$

ID: 42a062ab Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that the graphs of $x^2 + y^2 = 36$ and $y = mx + \frac{b}{4}$, where m and b are negative constants, intersect at the point $(-5, y)$ in the xy -plane. It follows that this intersection point represents a solution to the given system of equations. Substituting -5 for x in the equation $x^2 + y^2 = 36$ yields $(-5)^2 + y^2 = 36$, or $25 + y^2 = 36$. Subtracting 25 from each side of this equation yields $y^2 = 11$. It's also given that $y < 0$, so taking the negative square root of each side of this equation yields $y = -\sqrt{11}$. Substituting -5 for x and $-\sqrt{11}$ for y in the equation $y = mx + \frac{b}{4}$ yields $-\sqrt{11} = m(-5) + \frac{b}{4}$, or $-\sqrt{11} = -5m + \frac{b}{4}$. Adding $5m$ to each side of this equation yields $5m - \sqrt{11} = \frac{b}{4}$. Multiplying each side of this equation by 4 yields $4(5m - \sqrt{11}) = b$. Applying the distributive property to the left-hand side of this equation yields $20m - 4\sqrt{11} = b$. Therefore, the expression $20m - 4\sqrt{11}$ represents the value of b .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 71393e45

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 71393e45

The function f is defined by $f(x) = 3x^5$. In the xy -plane, the graph of $y = g(x)$ is the result of shifting the graph of $y = f(x)$ up 4 units. Which equation defines the function g ?

- A. $g(x) = 3x^5 + 4$
- B. $g(x) = 3x^5 - 4$
- C. $g(x) = 3x^{20}$
- D. $g(x) = 12x^5$

ID: 71393e45 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that in the xy -plane, the graph of $y = g(x)$ is the result of shifting the graph of $y = f(x)$ up 4 units. It follows that $g(x) = f(x) + 4$. It's also given that function f is defined by $f(x) = 3x^5$. Substituting $3x^5$ for $f(x)$ into $g(x) = f(x) + 4$ yields $g(x) = 3x^5 + 4$. Therefore, the equation that defines function g is $g(x) = 3x^5 + 4$.

Choice B is incorrect. This equation defines a function g for which the graph of $y = g(x)$ is the result of shifting the graph of $y = f(x)$ down, not up, 4 units.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Question Difficulty: Easy

Question ID f2753c52

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: f2753c52

The number of cells of a certain bacteria doubles every **25** minutes during its growth period when placed in a medium of milk in a controlled laboratory environment. If a sample of **830** cells of this bacteria is placed in milk in a controlled laboratory environment, which function f best represents the number of cells of the bacteria x minutes after the growth period begins, where x is less than the number of minutes in the growth period?

- A. $f(x) = 830\left(\frac{1}{2}\right)^{\frac{x}{25}}$
- B. $f(x) = 830(2)^{\frac{x}{25}}$
- C. $f(x) = 830\left(\frac{1}{3}\right)^{\frac{x}{25}}$
- D. $f(x) = 830(3)^{\frac{x}{25}}$

ID: f2753c52 Answer

Correct Answer: B

Rationale

Choice B is correct. For a bacteria with an initial population of P cells and a doubling time of m minutes, the number of cells, $f(x)$, of the bacteria x minutes after the growth period begins can be given by the equation $f(x) = P(2)^{\frac{x}{m}}$. It's given that there are **830** cells of the bacteria when the growth period begins. Therefore, $P = 830$. It's also given that the number of cells of the bacteria doubles every **25** minutes during the growth period. Therefore, $m = 25$. Substituting **830** for P and **25** for m in the equation $f(x) = P(2)^{\frac{x}{m}}$ yields $f(x) = 830(2)^{\frac{x}{25}}$.

Choice A is incorrect. This function represents a situation where the number of cells decreases by half, rather than doubles, every **25** minutes.

Choice C is incorrect. This function represents a situation where the number of cells decreases by two-thirds, rather than doubles, every **25** minutes.

Choice D is incorrect. This function represents a situation where the number of cells triples, rather than doubles, every **25** minutes.

Question Difficulty: Easy

Question ID 80cece7f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	

ID: 80cece7f

The expression $6x^4 + 31x^2 + 35$ can be rewritten as $(3x^2 + a)(2x^2 + b)$, where a and b are positive integers, or as $(3x^2 + c)(2x^2 + d)$, where c and d are positive nonintegers. What is the value of $a + c$?

ID: 80cece7f Answer

Correct Answer: 15.5, 31/2

Rationale

The correct answer is **15.5**. It's given that $6x^4 + 31x^2 + 35$ can be rewritten as $(3x^2 + a)(2x^2 + b)$, where a and b are positive integers. The expression $(3x^2 + a)(2x^2 + b)$ can be rewritten as $6x^4 + 3bx^2 + 2ax^2 + ab$, or $6x^4 + (3b + 2a)x^2 + ab$. Therefore, the expression $6x^4 + (3b + 2a)x^2 + ab$ is equivalent to $6x^4 + 31x^2 + 35$. It follows that $3b + 2a = 31$ and $ab = 35$. Since a and b are positive integers and $ab = 35$, the possible values of a and b are **1** and **35**, **5** and **7**, **7** and **5**, and **35** and **1**, respectively. Of these options, the only pair that satisfies the equation $3b + 2a = 31$ is $a = 5$ and $b = 7$, since $3(7) + 2(5) = 31$. It's also given that $6x^4 + 31x^2 + 35$ can be rewritten as $(3x^2 + c)(2x^2 + d)$, where c and d are positive nonintegers. The expression $(3x^2 + c)(2x^2 + d)$ can be rewritten as $6x^4 + 3dx^2 + 2cx^2 + cd$, or $6x^4 + (3d + 2c)x^2 + cd$. Therefore, the expression $6x^4 + (3d + 2c)x^2 + cd$ is equivalent to $6x^4 + 31x^2 + 35$. It follows that $3d + 2c = 31$ and $cd = 35$. Since it's given that c and d are positive nonintegers, it follows that c and d are not zero. Thus, dividing both sides of the equation $cd = 35$ by c yields $d = \frac{35}{c}$. Substituting $\frac{35}{c}$ for d in the equation $3d + 2c = 31$ yields $3(\frac{35}{c}) + 2c = 31$, or $\frac{105}{c} + 2c = 31$. Multiplying both sides of this equation by c yields $c(\frac{105}{c} + 2c) = c(31)$, or $105 + 2c^2 = 31c$. Subtracting $31c$ from both sides of this equation yields $105 + 2c^2 - 31c = 0$, or $2c^2 - 31c + 105 = 0$. This equation can be rewritten as $2c^2 - 10c - 21c + 105 = 0$. Since the first two terms of the expression $2c^2 - 10c - 21c + 105$ have a common factor of $2c$ and the last two terms of the expression $2c^2 - 10c - 21c + 105$ have a common factor of 21 , the expression $2c^2 - 10c - 21c + 105$ can be rewritten as $2c(c - 5) - 21(c - 5)$. Since each term of this expression has a common factor of $(c - 5)$, the expression $2c(c - 5) - 21(c - 5)$ can be rewritten as $(c - 5)(2c - 21)$. Thus, the equation $2c^2 - 31c + 105 = 0$ can be rewritten as $(c - 5)(2c - 21) = 0$. It follows that $c = 5$ or $c = \frac{21}{2}$. Since c is a positive noninteger, it follows that $c = \frac{21}{2}$, or $c = 10.5$. Substituting **5** for a and **10.5** for c into the expression $a + c$ yields **5 + 10.5**, or **15.5**. Therefore, the value of $a + c$ is **15.5**. Note that 15.5 and 31/2 are examples of ways to enter a correct answer.

Alternate approach: The correct answer is **15.5**. It's given that $6x^4 + 31x^2 + 35$ can be rewritten as $(3x^2 + a)(2x^2 + b)$, where a and b are positive integers. The expression $(3x^2 + a)(2x^2 + b)$ can be rewritten as $6x^4 + 3bx^2 + 2ax^2 + ab$, or $6x^4 + (3b + 2a)x^2 + ab$. Therefore, the expression $6x^4 + (3b + 2a)x^2 + ab$ is equivalent to $6x^4 + 31x^2 + 35$. It follows that $3b + 2a = 31$ and $ab = 35$. Since it's given that a and b are positive integers, it follows that a and b are not zero. Thus, dividing both sides of the equation $ab = 35$ by a yields $b = \frac{35}{a}$. Substituting $\frac{35}{a}$ for b in the equation $3b + 2a = 31$ yields $3(\frac{35}{a}) + 2a = 31$, or $\frac{105}{a} + 2a = 31$. Multiplying both sides of this equation by a yields $105 + 2a^2 = 31a$. Subtracting $31a$ from both sides of this equation yields $2a^2 - 31a + 105 = 0$. This equation can be rewritten as $2a^2 - 10a - 21a + 105 = 0$. Since the first two terms of the expression $2a^2 - 10a - 21a + 105$ have a common factor of $2a$ and the last two terms of the expression $2a^2 - 10a - 21a + 105$ have a common factor of 21 , the expression $2a^2 - 10a - 21a + 105$ can be rewritten as $2a(a - 5) - 21(a - 5)$. Since each term of this expression has a common factor of $(a - 5)$, the expression $2a(a - 5) - 21(a - 5)$ can be rewritten as $(a - 5)(2a - 21)$. Thus, the equation $2a^2 - 31a + 105 = 0$ can be rewritten as $(a - 5)(2a - 21) = 0$. It follows that $a = 5$ or $a = \frac{21}{2}$. Since a is a positive integer, it follows that $a = 5$. It's also given that $6x^4 + 31x^2 + 35$ can be rewritten as $(3x^2 + c)(2x^2 + d)$, where c and d are positive nonintegers. The expression

$(3x^2 + c)(2x^2 + d)$ can be rewritten as $6x^4 + 3dx^2 + 2cx^2 + cd$, or $6x^4 + (3d + 2c)x^2 + cd$. Therefore, the expression $6x^4 + (3d + 2c)x^2 + cd$ is equivalent to $6x^4 + 31x^2 + 35$. It follows that $3d + 2c = 31$ and $cd = 35$. Since it's given that c and d are positive nonintegers, it follows that c and d are not zero. Thus, dividing both sides of the equation $cd = 35$ by c yields $d = \frac{35}{c}$. Substituting $\frac{35}{c}$ for d in the equation $3d + 2c = 31$ yields $3\left(\frac{35}{c}\right) + 2c = 31$, or $\frac{105}{c} + 2c = 31$. Multiplying both sides of this equation by c yields $105 + 2c^2 = 31c$. Subtracting $31c$ from both sides of this equation yields $2c^2 - 31c + 105 = 0$. Since the equations $2a^2 - 31a + 105 = 0$ and $2c^2 - 31c + 105 = 0$ represent the same equation, it follows that a is equal to c . Therefore, $c = 5$ or $c = \frac{21}{2}$. Since c is a positive noninteger, it follows that $c = \frac{21}{2}$, or $c = 10.5$. Substituting 5 for a and 10.5 for c in the expression $a + c$ yields $5 + 10.5$, or 15.5 . Therefore, the value of $a + c$ is 15.5 . Note that 15.5 and $31/2$ are examples of ways to enter a correct answer.

Question Difficulty: Hard

Question ID 32024e13

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 32024e13

According to a model, the estimated volume of a certain tree in a forest increases by **13.9%** for every increase of **100** centimeters in the tree's height. The equation $y = ab^{\frac{x}{100}}$ represents this model, where y represents the estimated volume, in cubic centimeters, of this tree in the forest, x represents the tree's height, in centimeters, and a and b are constants. In this equation, what is the value of b ?

- A. 0.139
- B. 1
- C. 1.139
- D. 13.9

ID: 32024e13 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that equation $y = ab^{\frac{x}{100}}$ represents the model, where y is the estimated volume, in cubic centimeters, of a certain tree in a forest and x is the tree's height, in centimeters. For an exponential equation of this form to represent a model where the value of y increases as the value of x increases, it must follow that the values of a and b are positive. It's also given that the tree's estimated volume increases by **13.9%** for every increase of **100** centimeters in the tree's height. Therefore, for every **100**-centimeter increase in the tree's height, the estimated volume of the tree is **1.139** times larger. In other words, $ab^{\frac{x+100}{100}} = 1.139ab^{\frac{x}{100}}$, which is equivalent to $ab^{\frac{x}{100}}b^{\frac{100}{100}} = 1.139ab^{\frac{x}{100}}$. Since a and b are positive, dividing both sides of this equation by $ab^{\frac{x}{100}}$ yields $b^{\frac{100}{100}} = 1.139$, or $b = 1.139$. Therefore, the value of b is **1.139**.

Choice A is incorrect. This is the given percent increase in decimal form, not the value of b .

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the given percent increase, not the value of b .

Question Difficulty: Hard

Question ID 09745d77

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 09745d77

$$h(t) = -16t^2 + b$$

The function h estimates a ball's height, in feet, above a floor t seconds after the ball is released, where b is a constant. The function estimates that the ball is **19.36** feet above the floor when it is released at $t = 0$. How many seconds after being released does the function estimate the ball will reach the floor?

ID: 09745d77 Answer

Correct Answer: 1.1, 11/10

Rationale

The correct answer is **1.1**. It's given that $h(t) = -16t^2 + b$ estimates a ball's height, in feet, above a floor t seconds after the ball is released, where b is a constant. It's also given that the function estimates that the ball is **19.36** feet above the floor when it is released at $t = 0$. Substituting **19.36** for $h(t)$ and **0** for t in the given function yields $19.36 = -16(0)^2 + b$, or $19.36 = b$. Substituting **19.36** for b in function h yields $h(t) = -16t^2 + 19.36$. When the ball reaches the floor, its height is **0** feet above the floor. Substituting **0** for $h(t)$ yields $0 = -16t^2 + 19.36$. Adding $16t^2$ to both sides of this equation yields $16t^2 = 19.36$. Dividing both sides of this equation by **16** yields $t^2 = 1.21$. Taking the positive square root of both sides of this equation yields $t = 1.1$. Therefore, the function estimates that the ball will reach the floor **1.1** seconds after being released. Note that 1.1 and 11/10 are examples of ways to enter a correct answer.

Question Difficulty: Hard

Question ID 15a9384a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	

ID: 15a9384a

Which expression is equivalent to $\frac{4x+1}{x-9}$, where $x > 9$?

- A. $4 + \frac{37}{x-9}$
- B. $4 + \frac{35}{x-9}$
- C. $4 + \frac{1}{x-9}$
- D. $4 - \frac{1}{x-9}$

ID: 15a9384a Answer

Correct Answer: A

Rationale

Choice A is correct. The given expression $\frac{4x+1}{x-9}$ can be rewritten as $\frac{4x-36+36+1}{x-9}$, which is equivalent to $\frac{4x-36}{x-9} + \frac{37}{x-9}$. Factoring 4 from the numerator of the first fraction yields $\frac{4(x-9)}{x-9} + \frac{37}{x-9}$. Since $x > 9$, this expression is equivalent to $4 + \frac{37}{x-9}$. Therefore, $\frac{4x+1}{x-9}$ is equivalent to $4 + \frac{37}{x-9}$.

Choice B is incorrect. This expression is equivalent to $\frac{4x-1}{x-9}$, not $\frac{4x+1}{x-9}$.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 02fdffcf

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 02fdffcf

$\frac{1}{4xy} + xyz = \frac{1}{3yz}$

In the given equation, x , y , and z are positive numbers. Which expression is equivalent to y ?

- A. $\frac{4x-3z}{12x^2z^2}$
- B. $\sqrt{\frac{4x-3z}{12x^2z^2}}$
- C. $\frac{1}{3xz^2-4x^2z}$
- D. $\sqrt{\frac{1}{3xz^2-4x^2z}}$

ID: 02fdffcf Answer

Correct Answer: B

Rationale

Choice B is correct. It’s given that in the equation $\frac{1}{4xy} + xyz = \frac{1}{3yz}$, x , y , and z are positive numbers. Multiplying both sides of this equation by the least common denominator of $12xyz$ yields $12xyz\left(\frac{1}{4xy} + xyz\right) = 12xyz\left(\frac{1}{3yz}\right)$, which gives $\frac{12xyz}{4xy} + 12x^2y^2z^2 = \frac{12xyz}{3yz}$, or $3z + 12x^2y^2z^2 = 4x$. Subtracting $3z$ from both sides of this equation yields $12x^2y^2z^2 = 4x - 3z$. Dividing both sides of this equation by $12x^2z^2$ yields $y^2 = \frac{4x-3z}{12x^2z^2}$. Taking the square root of both sides of this equation yields $y = \pm\sqrt{\frac{4x-3z}{12x^2z^2}}$. Since y is a positive number, the expression $\sqrt{\frac{4x-3z}{12x^2z^2}}$ is equivalent to y .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID df0005c5

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: df0005c5

The function f is defined by $f(x) = x^3 + 15$. What is the value of $f(2)$?

ID: df0005c5 Answer

Correct Answer: 23

Rationale

The correct answer is **23**. The value of $f(2)$ is the value of $f(x)$ when $x = 2$. The function f is defined by $f(x) = x^3 + 15$. Substituting **2** for x in this equation yields $f(2) = 2^3 + 15$. This equation can be rewritten as $f(2) = 8 + 15$, or $f(2) = 23$. Therefore, the value of $f(2)$ is **23**.

Question Difficulty: Easy

Question ID 6e43bfd3

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 6e43bfd3

The function f is a quadratic function. In the xy -plane, the graph of $y = f(x)$ has a vertex at $(1, 9)$ and passes through the points $(2, 33)$ and $(-1, 105)$. What is the value of $f(-2) - f(0)$?

- A. 81
- B. 129
- C. 192
- D. 225

ID: 6e43bfd3 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that function f is a quadratic function, and in the xy -plane, the graph of $y = f(x)$ has a vertex at $(1, 9)$. In the xy -plane, the graph of a quadratic function with a vertex at (h, k) has an equation of the form $y = a(x - h)^2 + k$, where a is a constant. Therefore, function f can be written as $f(x) = a(x - 1)^2 + 9$. It's also given that the graph of $y = f(x)$ passes through the point $(2, 33)$. Substituting 2 for x and 33 for $f(x)$ in the equation $f(x) = a(x - 1)^2 + 9$ yields $33 = a(2 - 1)^2 + 9$, or $33 = a + 9$. Subtracting 9 from both sides of this equation yields $24 = a$. Substituting 24 for a in the equation $f(x) = a(x - 1)^2 + 9$ yields $f(x) = 24(x - 1)^2 + 9$. The value of $f(-2)$ can be found by substituting -2 for x in function f . Therefore, $f(-2) = 24(-2 - 1)^2 + 9$, which yields $f(-2) = 24(9) + 9$, or $f(-2) = 225$. The value of $f(0)$ can be found by substituting 0 for x in function f . Therefore, $f(0) = 24(0 - 1)^2 + 9$, which yields $f(0) = 24 + 9$, or $f(0) = 33$. It follows that $f(-2) - f(0) = 225 - 33$, or $f(-2) - f(0) = 192$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the value of $f(-2)$, not the value of $f(-2) - f(0)$.

Question Difficulty: Hard

Question ID 9b6614b4

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	

ID: 9b6614b4

Which expression is equivalent to $100x^2 + 2,600$?

- A. $100(26x^2)$
- B. $100(x^2 + 26)$
- C. $2,600(100x^2)$
- D. $2,600(100x^2 + 1)$

ID: 9b6614b4 Answer

Correct Answer: B

Rationale

Choice B is correct. Each term in the given expression, $100x^2 + 2,600$, has a common factor of **100**. Therefore, the expression can be rewritten as $100(x^2) + 100(26)$, or $100(x^2 + 26)$.

Choice A is incorrect. This expression is equivalent to $2,600x^2$, not $100x^2 + 2,600$.

Choice C is incorrect. This expression is equivalent to $260,000x^2$, not $100x^2 + 2,600$.

Choice D is incorrect. This expression is equivalent to $260,000x^2 + 2,600$, not $100x^2 + 2,600$.

Question Difficulty: Easy

Question ID 19f5c0f7

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 19f5c0f7

$|4x - 3| = -9$

How many solutions does the given equation have?

- A. Zero
- B. One
- C. Two
- D. Infinitely many

ID: 19f5c0f7 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that $|4x - 3| = -9$. By the definition of absolute value, the left-hand side of this equation, $|4x - 3|$, must be nonnegative. Since the right-hand side of this equation, -9 , is negative, it follows that no value of x can make this equation true; therefore, the given equation has zero solutions.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 32ef6f1a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 32ef6f1a

The function g is defined by $g(x) = 10^{5x-8}$. What is the value of x when $g(x)$ is equal to 10,000?

- A. $\frac{11}{5}$
- B. $\frac{12}{5}$
- C. 3
- D. 4

ID: 32ef6f1a Answer

Correct Answer: B

Rationale

Choice B is correct. It’s given that $g(x) = 10^{5x-8}$. Since 10,000 can be written as 10^4 , substituting 10^4 for $g(x)$ in this equation yields $10^4 = 10^{5x-8}$. Because the bases on the left-hand side and right-hand side of this equation are equal, it follows that the exponents must be equal. Therefore, $4 = 5x - 8$. Adding 8 to both sides of this equation yields $12 = 5x$. Dividing both sides of this equation by 5 yields $\frac{12}{5} = x$. Therefore, the value of x when $g(x)$ is equal to 10,000 is $\frac{12}{5}$.

Choice A is incorrect. This is the value of x when $g(x)$ is equal to 1,000, not 10,000.

Choice C is incorrect and may result from conceptual or calculation errors.

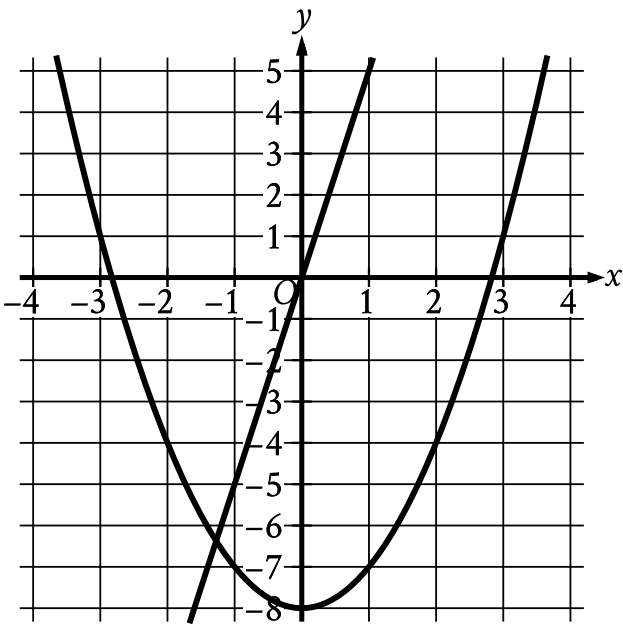
Choice D is incorrect. This is the value of $5x - 8$, not x , when $g(x)$ is equal to 10,000.

Question Difficulty: Medium

Question ID 68026521

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 68026521



The graph of a system of a linear and a quadratic equation is shown. Which system of equations is represented by the graph?

- A. $y = -5x$
 $y = x^2 + 8$
- B. $y = -5x$
 $y = x^2 - 8$
- C. $y = 5x$
 $y = x^2 + 8$
- D. $y = 5x$
 $y = x^2 - 8$

ID: 68026521 Answer

Correct Answer: D

Rationale

Choice D is correct. It’s given that the graph shows a system consisting of a linear and a quadratic equation in the xy -plane. In each of the given choices, the linear equation is in the form $y = mx$, where m is a constant, and the quadratic equation is in the form $y = x^2 + c$, where c is a constant. The graph of a linear equation in the form $y = mx$ has a slope of m and a y -intercept at

$(0, 0)$. Given two points on the line, (x_1, y_1) and (x_2, y_2) , the slope can be calculated using the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. The graph of the linear equation in this system has a y-intercept at $(0, 0)$ and passes through the point $(1, 5)$. Substituting the points $(0, 0)$ and $(1, 5)$ for (x_1, y_1) and (x_2, y_2) in the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$ yields $m = \frac{5 - 0}{1 - 0}$, which is equivalent to $m = \frac{5}{1}$, or $m = 5$. It follows that the linear equation is $y = 5x$. The graph of a quadratic equation in the form $y = x^2 + c$ has a vertex at $(0, c)$. The graph of the quadratic equation in this system has a vertex at $(0, -8)$. It follows that the quadratic equation is $y = x^2 + (-8)$, or $y = x^2 - 8$. Thus, the system of equations represented by the graph is $y = 5x$ and $y = x^2 - 8$.

Choice A is incorrect. The graph of this system of equations has a line with a slope of -5 , not 5 , and a parabola with a vertex at $(0, 8)$, not $(0, -8)$.

Choice B is incorrect. The graph of this system of equations has a line with a slope of -5 , not 5 .

Choice C is incorrect. The graph of this system of equations has a parabola with a vertex at $(0, 8)$, not $(0, -8)$.

Question Difficulty: Easy

Question ID 89646fb7

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 89646fb7

The value of a certain fishing boat was **\$24,000** when it was purchased. Each year, the value of the fishing boat decreased by **20%** of the previous year’s value. Which equation represents this situation, where ***v*** is the value of the fishing boat, in dollars, and ***y*** is the number of years since the fishing boat was purchased?

- A. $v = 0.8(24,000)^y$
- B. $v = 1.2(24,000)^y$
- C. $v = 24,000(0.8)^y$
- D. $v = 24,000(1.2)^y$

ID: 89646fb7 Answer

Correct Answer: C

Rationale

Choice C is correct. It’s given that the value of a certain fishing boat was **\$24,000** when it was purchased. It's also given that each year, the value of the fishing boat decreased by **20%** of the previous year's value. It follows that this situation can be represented by the equation $v = a(1 - \frac{r}{100})^y$, where ***v*** is the value of the boat, in dollars, ***y*** years since it was purchased, ***a*** is the value of the boat, in dollars, when it was purchased, and ***r*** is the percent by which the value of the boat decreases each year from the previous year's value. Substituting **24,000** for ***a*** and **20** for ***r*** in this equation yields $v = 24,000(1 - \frac{20}{100})^y$, or $v = 24,000(0.8)^y$. Therefore, $v = 24,000(0.8)^y$ represents this situation.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This equation represents a situation where each year the value of the boat increased, not decreased, by **20%** of the previous year's value.

Question Difficulty: Medium

Question ID 4885895b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 4885895b

$r(t) = 63t - 2t^2$

The function r is defined by the given equation. The function s is defined by $s(t) = r(t) + 1$. Which expression represents the maximum value of $s(t)$?

- A. $1 + \left(\frac{63}{2}\right)^2$
- B. $1 + 2\left(\frac{63}{4}\right)^2$
- C. $1 - 2\left(\frac{63}{4}\right)^2$
- D. $1 - \left(\frac{63}{2}\right)^2$

ID: 4885895b Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that $s(t) = r(t) + 1$. Since $r(t) = 63t - 2t^2$, it follows that $s(t) = 63t - 2t^2 + 1$, or $s(t) = -2t^2 + 63t + 1$. For a quadratic function s defined by an equation of the form $s(t) = a(t - h)^2 + k$, where a , h , and k are constants and a is negative, the value of $s(t)$ reaches its maximum, k , when the value of t is h . The equation $s(t) = -2t^2 + 63t + 1$ can be rewritten in the form $s(t) = a(t - h)^2 + k$ by completing the square. The equation $s(t) = -2t^2 + 63t + 1$ is equivalent to $s(t) = -2\left(t^2 - \frac{63}{2}t\right) + 1$, or $s(t) = -2\left(t^2 - \frac{63}{2}t + \left(\frac{63}{4}\right)^2 - \left(\frac{63}{4}\right)^2\right) + 1$. This equation can be rewritten as $s(t) = -2\left(\left(t - \frac{63}{4}\right)^2 - \left(\frac{63}{4}\right)^2\right) + 1$, or $s(t) = -2\left(t - \frac{63}{4}\right)^2 + 1 + 2\left(\frac{63}{4}\right)^2$. This equation is in the form $s(t) = a(t - h)^2 + k$, where $a = -2$, $h = \frac{63}{4}$, and $k = 1 + 2\left(\frac{63}{4}\right)^2$. Therefore, the maximum value of $s(t)$ is $1 + 2\left(\frac{63}{4}\right)^2$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID f71f7ccd

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: f71f7ccd

$2r = 9s + t$

The given equation relates the variables r , s , and t . Which equation correctly expresses s in terms of r and t ?

- A. $s = \frac{2}{9}r - t$
- B. $s = \frac{2r-t}{9}$
- C. $s = 18r - t$
- D. $s = 9(2r - t)$

ID: f71f7ccd Answer

Correct Answer: B

Rationale

Choice B is correct. Subtracting t from each side of the given equation yields $2r - t = 9s$. Dividing each side of this equation by 9 yields $\frac{2r-t}{9} = s$, which can be rewritten as $s = \frac{2r-t}{9}$. Therefore, the equation $s = \frac{2r-t}{9}$ correctly expresses s in terms of r and t .

Choice A is incorrect. This equation is equivalent to $2r = 9(s + t)$, not $2r = 9s + t$.

Choice C is incorrect. This equation is equivalent to $2r = \frac{s+t}{9}$, not $2r = 9s + t$.

Choice D is incorrect. This equation is equivalent to $2r = \frac{s}{9} + t$, not $2r = 9s + t$.

Question Difficulty: Easy

Question ID c6a3fc28

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: c6a3fc28

$$y = 9\left(\frac{a}{7}\right)^{x+c} - b$$

How many times does the graph of the given equation in the xy -plane cross the x -axis, where a , b , and c are positive constants such that $a > 7$ and $b > c$?

- A. Zero
- B. One
- C. Two
- D. Three

ID: c6a3fc28 Answer

Correct Answer: B

Rationale

Choice B is correct. It’s given that $y = 9\left(\frac{a}{7}\right)^{x+c} - b$, where a , b , and c are positive constants such that $a > 7$ and $b > c$. The graph of any equation in the xy -plane crosses the x -axis when there is a point on the graph for which the value of y is equal to 0 , and the value of y changes sign when the graph passes through that point. Since $a > 7$, the base $\frac{a}{7}$ is greater than 1 , so the expression $9\left(\frac{a}{7}\right)^{x+c} - b$ increases as x increases. Since b is positive, there is a point on the graph of $y = 9\left(\frac{a}{7}\right)^{x+c} - b$ for which the value of y is equal to 0 . When the graph passes through this point, the value of y changes sign from negative to positive. Therefore, the graph of the given equation crosses the x -axis exactly one time.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID c4fccc3c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: c4fccc3c

$$\frac{1}{x} + 8 = \frac{1}{6} + 8$$

What is the solution to the given equation?

- A. 1
- B. 6
- C. 8
- D. 14

ID: c4fccc3c Answer

Correct Answer: B

Rationale

Choice B is correct. Subtracting 8 from each side of the given equation yields $\frac{1}{x} = \frac{1}{6}$. Multiplying each side of this equation by $6x$ yields $6 = x$. Thus, the solution to the given equation is 6.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 92344c73

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 92344c73

The function f is defined by $f(x) = 56(0.19)^x$. For any positive integer n , the value of $f(n)$ is $p\%$ less than the value of $f(n - 1)$. What is the value of p ?

- A. 19
- B. 44
- C. 56
- D. 81

ID: 92344c73 Answer

Correct Answer: D

Rationale

Choice D is correct. For a function of the form $f(x) = a(b)^x$, where a and b are constants and $b < 1$, the value of $f(x)$ decreases by $100(1 - b)\%$ for every increase of x by 1. In the given function, $a = 56$ and $b = 0.19$. Therefore, for the given function, the value of $f(x)$ decreases by $100(1 - 0.19)\%$, or 81% , for every increase of x by 1. It's given that the value of $f(n)$ is $p\%$ less than the value of $f(n - 1)$. The increase from $x = n - 1$ to $x = n$ can be expressed as $n - (n - 1)$, which is equivalent to $n - n + 1$, or 1. Since there is an increase by 1 from $n - 1$ to n , it follows that the value of $f(n)$ is 81% less than the value of $f(n - 1)$. Thus, the value of p is 81.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID faaab034

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: faaab034

$(x - 16)(x - 10)(x + 7)(x + 17) = 0$
What is a positive solution to the given equation?

ID: faaab034 Answer

Correct Answer: 10, 16

Rationale

The correct answer is either **10** or **16**. Applying the zero product property to the given equation yields four equations: $x - 16 = 0$, $x - 10 = 0$, $x + 7 = 0$, and $x + 17 = 0$. Adding **16** to both sides of the equation $x - 16 = 0$ yields $x = 16$. Adding **10** to both sides of the equation $x - 10 = 0$ yields $x = 10$. Subtracting **7** from both sides of the equation $x + 7 = 0$ yields $x = -7$. Subtracting **17** from both sides of the equation $x + 17 = 0$ yields $x = -17$. Therefore, the solutions to the given equation are **16**, **10**, **-7**, and **-17**. It follows that there are two positive solutions to the given equation: **10** and **16**. Note that 10 and 16 are examples of ways to enter a correct answer.

Question Difficulty: Medium

Question ID 1d79a028

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 1d79a028

The ratio a to b is equivalent to the ratio 63 to 17 . What is the value of a when $b = 153$?

ID: 1d79a028 Answer

Correct Answer: 567

Rationale

The correct answer is 567 . It's given that the ratio a to b is equivalent to the ratio 63 to 17 , or $\frac{a}{b} = \frac{63}{17}$. When $b = 153$, the value of a can be calculated by substituting 153 for b in the equation $\frac{a}{b} = \frac{63}{17}$, which yields $\frac{a}{153} = \frac{63}{17}$. Multiplying both sides of this equation by 153 yields $a = 567$. Thus, when $b = 153$, the value of a is 567 .

Question Difficulty: Medium

Question ID d14eba41

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	

ID: d14eba41

At a recreation center, visitors used the exercise room for a total of t hours in July. At the same recreation center, visitors used the exercise room for $3.49t$ hours in August. What is the percent increase in the number of hours visitors used the exercise room from July to August?

- A. 2.49%
- B. 3.49%
- C. 249%
- D. 349%

ID: d14eba41 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that the total number of hours visitors used the exercise room increased from t hours in July to $3.49t$ hours in August. It follows that the number of hours increased by $3.49t - t$, or $2.49t$. Let $p\%$ represent the percent increase in the exercise room use from July to August, or the percentage that $2.49t$ is of t . The value of p can be found using the proportion $\frac{2.49t}{t} = \frac{p}{100}$, or $2.49 = \frac{p}{100}$. Multiplying both sides of this equation by 100 yields $p = 249$. Therefore, the percent increase in the number of hours visitors used the exercise room from July to August is 249% .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. The exercise room use in August is 349% of the exercise room use in July, but this doesn't represent the percent increase from July to August.

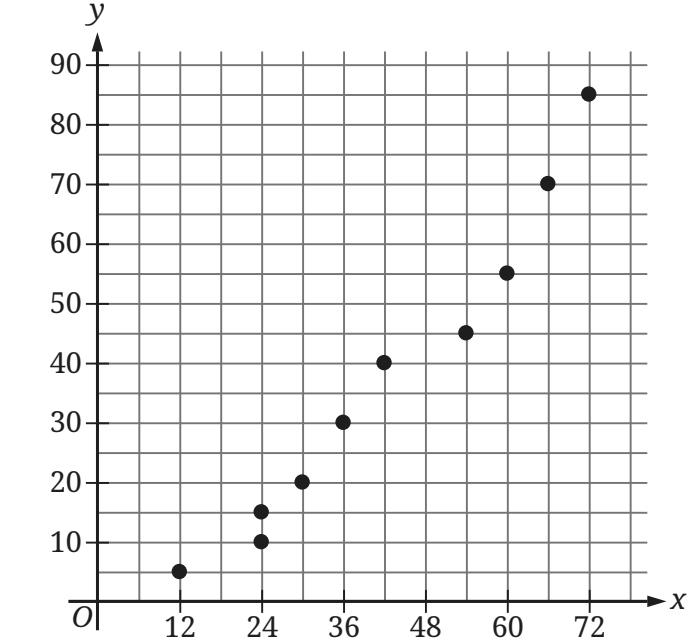
Question Difficulty: Hard

Question ID 8e6b2789

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	

ID: 8e6b2789

The owner of a restaurant supply store selects **10** cutting boards to put on sale. The graph shows the width x , in centimeters, of each cutting board and its corresponding sale price y , in dollars, where each value of y is an integer.



According to the graph, which statement is true?

- A. Cutting boards that have a width greater than **48** centimeters have an average sale price of **\$59.00**.
- B. Cutting boards that have a width greater than **48** centimeters have an average sale price of **\$40.00**.
- C. Cutting boards that have a width between **18** and **48** centimeters have an average sale price of **\$31.20**.
- D. Cutting boards that have a width between **18** and **48** centimeters have an average sale price of **\$23.00**.

ID: 8e6b2789 Answer

Correct Answer: D

Rationale

Choice D is correct. Each point (x, y) on the scatterplot gives the width x , in centimeters, and the sale price y , in dollars, of one of the **10** cutting boards put on sale at this store. According to the graph, **5** cutting boards have widths that are between **18** and **48** centimeters. These widths, in centimeters, are **24, 24, 30, 36, and 42**. The sale prices, in dollars, corresponding to these widths are **10, 15, 20, 30, and 40**, respectively. The average of these sale prices, in dollars, is $\frac{10+15+20+30+40}{5}$, or $\frac{115}{5}$, which is equivalent to **23**. Therefore, cutting boards that have a width between **18** and **48** centimeters have an average sale price of **\$23.00**.

Choice A is incorrect. Cutting boards that have a width greater than **48** centimeters have an average sale price of **\$63.75**, not **\$59.00**.

Choice B is incorrect. Cutting boards that have a width greater than **48** centimeters have an average sale price of **\$63.75**, not **\$40.00**.

Choice C is incorrect. Cutting boards that have a width between **18** and **48** centimeters have an average sale price of **\$23.00**, not **\$31.20**.

Question Difficulty: Medium

Question ID a871939f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Evaluating statistical claims: Observational studies and experiments	

ID: a871939f

A statistician conducted a study to investigate the relationship between the average number of minutes piano students practiced each week and the percentage of correct notes the students played during a recital. For the study, the statistician selected a sample of **85** piano students at random from all the piano students in a certain region who were in their second year with their conservatory. The table shows the information for **75** piano students in this sample who practiced at least **50** minutes on average each week.

	Percentage of correct notes the students played during a recital			
Average number of minutes practiced each week	Less than 50%	50% to 80%	Greater than 80%	Total
50 to 150	7	28	5	40
151 or more	3	11	21	35
Total	10	39	26	75

Which of the following is the largest population to which the results of the study can be generalized?

- A. All piano students who were in their second year with their conservatory
- B. All piano students in the region
- C. All piano students in the region who were in their second year with their conservatory
- D. The **75** piano students in the sample who practiced at least **50** minutes on average each week

ID: a871939f Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that the statistician selected a sample of **85** piano students at random from all piano students in a certain region who were in their second year with their conservatory. It's also given that the table shows the information for **75** piano students in the sample who practiced at least **50** minutes on average each week. Selecting a sample at random from a population allows the results of the study to be generalized to that population, but not beyond it. Therefore, the largest population to which the results of the study can be generalized is the population of all piano students in the region who were in their second year with their conservatory.

Choice A is incorrect. The results can't be generalized to all piano students who were in their second year with their conservatory since only students in a certain region were sampled.

Choice B is incorrect. The results can't be generalized to all piano students in the region since only students in their second year with their conservatory were sampled.

Choice D is incorrect. The results of the study can be generalized beyond the sample itself to the population from which it was drawn.

Question Difficulty: Hard

Question ID c86bc0c9

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: c86bc0c9

- A rectangle has an area of **3,168** square inches. What is the area, in **square feet**, of this rectangle? (**1 foot = 12 inches**)
- A. **22**
 - B. **56.3**
 - C. **264**
 - D. **675.4**

ID: c86bc0c9 Answer

Correct Answer: A

Rationale

- Choice A is correct. It’s given that **1 foot = 12 inches**. It follows that **1² square foot = 12² square inches**, or **1 square foot = 144 square inches**. Thus, **3,168** square inches is equivalent to **$\left(3,168 \text{ square inches}\right)\left(\frac{1 \text{ square foot}}{144 \text{ square inches}}\right)$** , or **22** square feet.
- Choice B is incorrect and may result from conceptual or calculation errors.
- Choice C is incorrect. This is the value of $\frac{3,168}{12}$, not $\frac{3,168}{144}$.
- Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 4c14d003

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 4c14d003

There is a proportional relationship between p and t . If $p = 5,130$ when $t = 6,840$, then what is the value of p when $t = 5,472$?

- A. 7,296
- B. 7,182
- C. 4,104
- D. 3,762

ID: 4c14d003 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that there is a proportional relationship between p and t . This relationship can be written as $\frac{p}{t}$. If $p = 5,130$ when $t = 6,840$, the value of p when $t = 5,472$ can be calculated by solving the equation $\frac{5,130}{6,840} = \frac{p}{5,472}$. Multiplying each side of this equation by $5,472$ yields $(5,472) \frac{5,130}{6,840} = \frac{p}{5,472} (5,472)$, or $p = 4,104$. Therefore, when $t = 5,472$, the value of p is $4,104$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 5ba166cd

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: 5ba166cd

$a, 26, 29, b, 31, 47, c$

For the given data set, the data values are listed in ascending order, where a , b , and c are constants. For this data set, the mean is **36**, the median is **29**, and the range is **72**. What is the value of c ?

- A. **54**
- B. **72**
- C. **81**
- D. **98**

ID: 5ba166cd Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that the data set containing $a, 26, 29, b, 31, 47, c$ is listed in ascending order and has a mean of **36**, a median of **29**, and a range of **72**. Since there are **7** values in the given data set, it follows that the median is the value in the fourth position. It's given that b is in the fourth position in the given data set, so it follows that $b = 29$. Since the mean of the data set is **36**, it follows that $\frac{(a+26+29+29+31+47+c)}{7} = 36$. Multiplying both sides of this equation by **7** yields $a + 26 + 29 + 29 + 31 + 47 + c = 252$, which is equivalent to $a + c + 162 = 252$. Subtracting **162** from both sides of this equation yields $a + c = 90$. The range is the difference between the greatest and least values in a data set. Since c is the greatest value and a is the least value in the given data set, it follows that $c - a = 72$, or $-a + c = 72$. Adding the equations $a + c = 90$ and $-a + c = 72$ yields $2c = 162$. Dividing both sides of this equation by **2** yields $c = 81$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID f70dd899

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: f70dd899

For data set A, the table summarizes the distribution of the number of deliveries received by an office each day during a period of **11** days.

Deliveries	Days
0	2
3	2
4	2
5	2
6	1
7	1
13	1

The data value **13** was recorded in error and is removed from data set A to create data set B, which consists of the remaining **10** data values. Which statement best compares the median of data set A and the median of data set B?

- A. The median of data set B is less than the median of data set A.
- B. The median of data set B is greater than the median of data set A.
- C. The median of data set B is equal to the median of data set A.
- D. There is not enough information to compare the medians of the two data sets.

ID: f70dd899 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that for data set A, the table summarizes the number of deliveries received by an office each day over a period of **11** days. If a data set contains an odd number of data values, the median is represented by the middle data value in the list when the data values are listed in ascending or descending order. Since data set A contains **11** data values in ascending order, the median of data set A is the **6th** value. Based on the number of days for each value shown in the table, the **6th** value is **4**, so the median of data set A is **4**. It's also given that the data value **13** was recorded in error and is removed from data set A to create data set B. It follows that data set B contains **10** data values in ascending order. If a data set contains an even number of data values, the median is between the two middle data values when the values are listed in ascending or descending order. It follows that the median of data set B is between the **5th** and **6th** values. Based on the number of days for each value shown in the table, the **5th** and **6th** values are both **4**, so the median of data set B is **4**. Therefore, the median of data set B is equal to the median of data set A.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID aaff9ba9

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: aaff9ba9

On average, a certain plant grows **87** millimeters every ***m*** months. At this rate, which expression represents the number of millimeters, on average, the plant grows every ***k*** years?

- A. $\frac{29m}{4k}$
- B. $\frac{29k}{4m}$
- C. $\frac{1,044m}{k}$
- D. $\frac{1,044k}{m}$

ID: aaff9ba9 Answer

Correct Answer: D

Rationale

Choice D is correct. It’s given that on average, a certain plant grows **87** millimeters every ***m*** months. This average rate can be written as $\frac{87}{m}$ millimeters per month. Since there are **12** months in **1** year, it follows that there are **12*k*** months in ***k*** years. Therefore, the number of millimeters, on average, the plant grows every ***k*** years can be calculated by multiplying the rate $\frac{87}{m}$ millimeters per month by **12*k*** months, which yields $(\frac{87}{m})(12k)$, or $\frac{1,044k}{m}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 6744ece7

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Inference from sample statistics and margin of error	

ID: 6744ece7

Researchers studied a sample of a species of fish from a certain location. The researchers found that the mean length of fish in the sample was **16.59** millimeters. Based on the sample mean and margin of error, the researchers estimated that the mean length of all fish of this species from this location was between **15.86** and **17.32** millimeters. What is the margin of error, in millimeters?

- A. **0.73**
- B. **2.19**
- C. **15.86**
- D. **16.59**

ID: 6744ece7 Answer

Correct Answer: A

Rationale

Choice A is correct. It’s given that the mean length of fish in the sample was **16.59** millimeters. It’s also given that based on the sample mean and margin of error, the researchers estimated that the mean length of all fish of this species from this location was between **15.86** and **17.32** millimeters. These values are calculated by subtracting the margin of error from **16.59** millimeters to get the lower bound, **15.86** millimeters, and by adding the margin of error to **16.59** millimeters to get the upper bound, **17.32** millimeters. It follows that the margin of error, in millimeters, can be calculated as the difference between **16.59** and **15.86** or between **17.32** and **16.59**. The difference between **16.59** and **15.86** is **16.59 — 15.86**, or **0.73**. Therefore, the margin of error, in millimeters, is **0.73**.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the lower bound of the researchers’ estimate of the mean length of all fish of this species from this location, not the margin of error.

Choice D is incorrect. This is the mean length of fish in the sample, not the margin of error.

Question Difficulty: Medium

Question ID 9d8572fb

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Inference from sample statistics and margin of error	

ID: 9d8572fb

A researcher surveyed a random sample of **1,020** people from a certain city to estimate the percentage of people who support a recently announced proposal. From the survey, the researcher estimated that **86%** of people from the city support this proposal, with an associated margin of error of **2.13%**. The researcher repeated the survey with a random sample of people from the city that was double the original sample size. Assuming the margins of error were calculated in the same way, which of the following is true about the margin of error associated with the estimate from the larger sample?

- A. The margin of error is between **2.13%** and **3.13%**.
- B. The margin of error is between **4.13%** and **5.13%**.
- C. The margin of error is greater than **5.13%**.
- D. The margin of error is less than **2.13%**.

ID: 9d8572fb Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that from the initial survey, the margin of error associated with the estimate was **2.13%**. A larger sample size generally leads to a smaller margin of error. Therefore, assuming the margins of error were calculated in the same way, the margin of error associated with the estimate from the larger sample is less than **2.13%**.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Question Difficulty: Hard

Question ID 2f182fc9

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Inference from sample statistics and margin of error	

ID: 2f182fc9

A researcher is designing a study to investigate the average number of hours participants in a training course spend on a particular activity per day. The researcher will report an estimated average number of hours participants in the training course spend on this activity per day with an associated margin of error. The researcher is considering using a random sample of either **115** or **230** participants from the training course. Which of the following would be the most likely effect of using the larger random sample compared to the smaller random sample?

- A. The reported margin of error would be lower.
- B. The reported margin of error would be higher.
- C. The reported average number of hours would be lower.
- D. The reported average number of hours would be higher.

ID: 2f182fc9 Answer

Correct Answer: A

Rationale

Choice A is correct. When using a random sample to report an estimated average, a larger sample size, such as **230** participants instead of **115** participants, generally leads to a smaller margin of error. Therefore, the most likely effect of using the larger random sample compared to the smaller random sample is that the reported margin of error would be lower.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

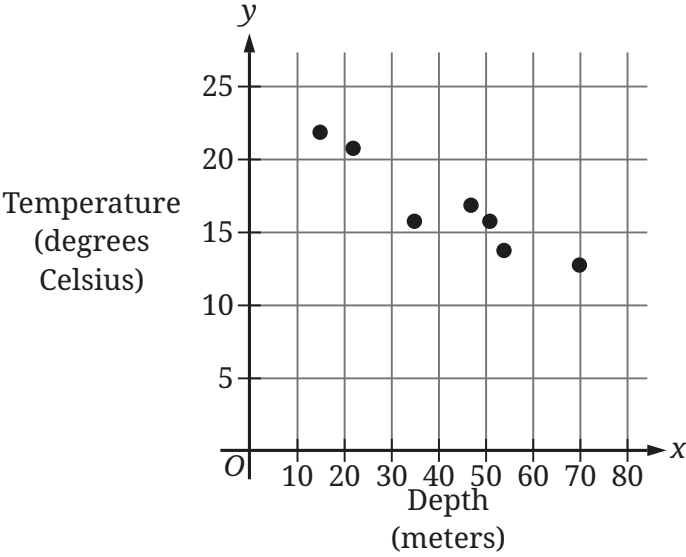
Question Difficulty: Medium

Question ID de601cf1

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	

ID: de601cf1

As part of a study of lake conditions, the water temperature was recorded at different depths below the surface of a certain lake. The data shown in the scatterplot give the recorded temperature, in degrees Celsius, for 7 depths, in meters, below the surface of the lake.



Which of the following is closest to the slope of a line of best fit for the data shown?

- A. -0.17
- B. -2.17
- C. -6.02
- D. -8.02

ID: de601cf1 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that the data shown in the scatterplot give the recorded temperature, in degrees Celsius, for 7 depths, in meters, below the surface of a lake. The graph of an appropriate line of best fit for the data shown passes approximately through the points $(15, 21.8)$ and $(70, 12.7)$. The slope of a line in the xy -plane that passes through the points (x_1, y_1) and (x_2, y_2) can be calculated as $\frac{y_2 - y_1}{x_2 - x_1}$. Substituting $(15, 21.8)$ and $(70, 12.7)$ for (x_1, y_1) and (x_2, y_2) , respectively, in the expression $\frac{y_2 - y_1}{x_2 - x_1}$ yields $\frac{12.7 - 21.8}{70 - 15}$, or $\frac{-9.1}{55}$, which is approximately -0.165 . Therefore, of the given choices, -0.17 is closest to the slope of a line of best fit for the data shown.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID b238e77f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: b238e77f

For the last three years, a pig at a farm gave birth to one litter per year. The table shows the number of piglets in each of these three litters.

Litter number	Number of piglets
1	8
2	7
3	9

What is the mean number of piglets per litter for these three litters?

- A. 2
- B. 7
- C. 8
- D. 9

ID: b238e77f Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that the numbers of piglets in litters **1**, **2**, and **3** are **8**, **7**, and **9**, respectively. The mean number of piglets per litter for these three litters is $\frac{8+7+9}{3}$, which is equivalent to $\frac{24}{3}$, or **8**.

Choice A is incorrect. This is the range of the number of piglets per litter for these three litters.

Choice B is incorrect. This is the minimum number of piglets per litter for these three litters.

Choice D is incorrect. This is the maximum number of piglets per litter for these three litters.

Question Difficulty: Easy

Question ID 52ee0e47

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	

ID: 52ee0e47

The number of audiobooks in a library increased by **131%** from last year to this year. The number of audiobooks in the library last year was **n** , and the number of audiobooks in the library this year is **bn** , where **b** and **n** are constants. What is the value of **b** ?

ID: 52ee0e47 Answer

Correct Answer: 2.31

Rationale

The correct answer is **2.31**. It's given that the number of audiobooks in the library increased by **131%** from last year to this year. Converting **131%** to a decimal yields **1.31**. It's also given that the number of audiobooks in the library last year was **n** , where **n** is a constant. It follows that the number of audiobooks in the library this year can be represented by **n** plus **131%** of **n** , or **$n + 1.31n$** . Combining like terms in this expression yields **2.31n**. Since the number of audiobooks in the library this year is represented as **bn** , where **b** is a constant, it follows that the value of **b** is **2.31**.

Question Difficulty: Hard

Question ID 42d2fb21

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Probability and conditional probability	

ID: 42d2fb21

The table shows the distribution of rooms in a certain facility by seating capacity.

Seating capacity	Proportion
Less than 18 seats	26%
18–40 seats	21%
41–65 seats	29%
Greater than 65 seats	24%

If a room in this facility is selected at random, which of the following is closest to the probability of selecting a room that has a seating capacity greater than **65** seats, given that the room has a seating capacity of at least **18** seats?

- A. **0.24**
- B. **0.32**
- C. **0.50**
- D. **0.92**

ID: 42d2fb21 Answer

Correct Answer: B

Rationale

Choice B is correct. It’s given that the proportion of rooms with less than **18** seats is **26%**, which is equivalent to **0.26**. It’s also given that the proportion of rooms with **18–40** seats is **21%**, or **0.21**; that the proportion of rooms with **41–65** seats is **29%**, or **0.29**; and that the proportion of rooms with greater than **65** seats is **24%**, or **0.24**. The probability of selecting a room that has a seating capacity greater than **65** seats given that the room has a seating capacity of at least **18** seats is equal to the proportion of rooms that have a seating capacity of greater than **65** seats divided by the proportion of rooms that have a seating capacity of at least **18** seats. Rooms with a seating capacity of at least **18** seats include the categories **18–40** seats, **41–65** seats, and greater than **65** seats, so the total proportion for this condition can be written as **0.21 + 0.29 + 0.24**, or **0.74**. Therefore, the probability of selecting a room that has a seating capacity greater than **65** seats given that it has a seating capacity of at least **18** seats is $\frac{0.24}{0.74}$, or approximately **0.324**. Of the given choices, **0.32** is closest to **0.324**.

Choice A is incorrect. This is the probability of selecting a room that has a seating capacity greater than **65** seats from all rooms in the facility, not just rooms that meet the condition of having a seating capacity of at least **18** seats.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 0c0f7f86

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: 0c0f7f86

Which of the following lists represents a data set with the smallest standard deviation?

- A. 51, 52, 54, 56, 57
- B. 52, 52, 54, 56, 56
- C. 52, 53, 54, 55, 56
- D. 53, 54, 54, 54, 55

ID: 0c0f7f86 Answer

Correct Answer: D

Rationale

Choice D is correct. Standard deviation is a measure of the spread of the values in a data set from the mean of the data set, and the smaller the spread, the smaller the standard deviation. Each of the four choices represents a data set that is symmetric with respect to the value **54**. Therefore, **54** is the mean of each of these data sets. All the values in the data set in choice D differ from **54** by either **0** or **1**. In each of the other three data sets, the values in the data set, other than the value of **54**, differ from **54** by at least **1**. Therefore, the data set in choice D has the smallest spread and thus the smallest standard deviation.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 2897a804

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	

ID: 2897a804

The speeds of objects A, B, and C are a meters per second, b meters per second, and c meters per second, respectively. If the speed of object A is **5,900%** of the speed of object C and the speed of object C is **0.008%** of the speed of object B, which expression represents the value of $a + b$ in terms of c ?

- A. $1,840c$
- B. $5,908c$
- C. $6,025c$
- D. $12,559c$

ID: 2897a804 Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that the speeds of objects A, B, and C are a meters per second, b meters per second, and c meters per second, respectively. If the speed of object A is **5,900%** of the speed of object C, this means that $a = \frac{5,900}{100}c$ or $a = 59c$. If the speed of object C is **0.008%** of the speed of object B, this means that $c = \frac{0.008}{100}b$, or $c = 0.00008b$. Dividing both sides of this equation by **0.00008** yields $\frac{1}{0.00008}c = b$, or $12,500c = b$. Since $a = 59c$ and $b = 12,500c$, it follows that $a + b = 59c + 12,500c$, or $a + b = 12,559c$. Therefore, the expression **12,559c** represents the value of $a + b$ in terms of c .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 74337020

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	

ID: 74337020

A square scale drawing has a side length of **55** inches, and **1** inch on the drawing represents an actual distance of **19** miles. A larger version of the same drawing is printed as a square with the side length **65%** longer than the side length of the previous drawing. On the larger drawing, which of the following is closest to the actual distance, in miles, represented by **1** inch?

- A. **6.65**
- B. **11.52**
- C. **31.35**
- D. **35.75**

ID: 74337020 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that **1** inch on the drawing represents an actual distance of **19** miles and that the larger drawing has a side length that is **65%** longer than the side length of the previous drawing. It follows that **(1.65)(1 inch)**, or **1.65** inches, on the larger drawing also represents an actual distance of **19** miles. Therefore, the scale of the larger drawing can be represented as $\frac{19 \text{ miles}}{1.65 \text{ inches}}$, which is approximately **11.52** miles per inch. Therefore, **11.52** is closest to the actual distance, in miles, represented by **1** inch on the larger drawing.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 3ddd3168

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Probability and conditional probability	

ID: 3ddd3168

The table gives the distribution of topping and type of crust for customer orders at a pizza place.

Topping	Type of crust	
	Thin	Thick
Extra cheese	60	30
Pepperoni	20	30
Tomato	80	20

If one of these customer orders is selected at random, what is the probability of selecting an order with thin crust, given the topping is pepperoni? (Express your answer as a decimal or fraction, not as a percent.)

ID: 3ddd3168 Answer

Correct Answer: 0.4, 2/5

Rationale

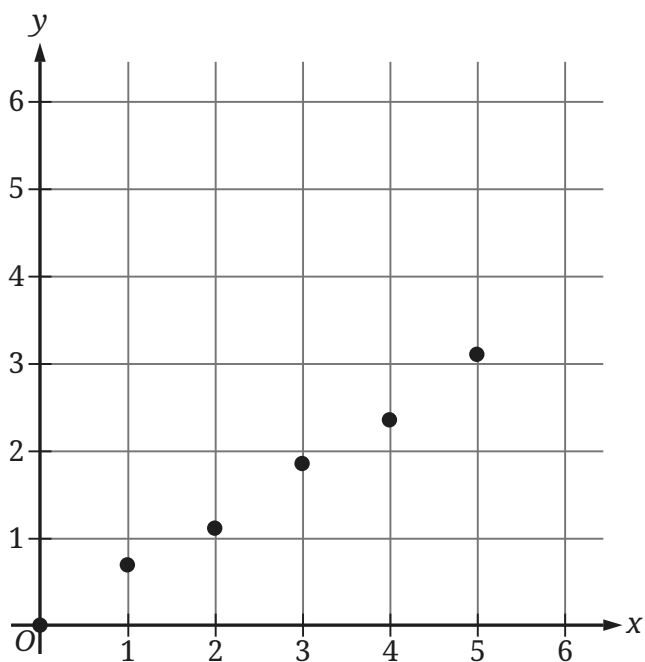
The correct answer is $\frac{2}{5}$. Since the order will be selected at random, the probability of selecting an order with thin crust, given the topping is pepperoni, is equal to the number of orders with thin crust where the topping is pepperoni divided by the total number of orders where the topping is pepperoni. According to the table, there are **20** orders with thin crust where the topping is pepperoni. According to the table, there are **30** orders with thick crust where the topping is pepperoni. Therefore, there are a total of **20 + 30**, or **50**, orders where the topping is pepperoni. Thus, the probability of selecting an order with thin crust, given the topping is pepperoni, is $\frac{20}{50}$, or $\frac{2}{5}$. Note that 2/5 and .4 are examples of ways to enter a correct answer.

Question Difficulty: Hard

Question ID b8e8440e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	

ID: b8e8440e



Which of the following equations is the most appropriate linear model for the data shown in the scatterplot?

- A. $y = -0.606x$
- B. $y = 0.606 - 1.001x$
- C. $y = 0.606x$
- D. $y = 0.606 + 1.001x$

ID: b8e8440e Answer

Correct Answer: C

Rationale

Choice C is correct. A linear model can be written in the form $y = mx + b$, where m is the slope of the graph of the model in the xy -plane and $(0, b)$ is the y -intercept. The graph of an appropriate linear model for the data shown in the scatterplot passes near the points $(0, 0)$ and $(4, 2.4)$ in the xy -plane. Two points on a line, (x_1, y_1) and (x_2, y_2) , can be used to find the slope of the line using the slope formula, $m = \frac{y_2 - y_1}{x_2 - x_1}$. Substituting the points $(0, 0)$ and $(4, 2.4)$ for (x_1, y_1) and (x_2, y_2) , respectively, in the slope formula yields $m = \frac{2.4 - 0}{4 - 0}$, or $m = 0.60$. Therefore, the value of m for an appropriate linear model is approximately 0.60 . Substituting 0.60 for m in $y = mx + b$ yields $y = 0.60x + b$. Since an appropriate linear model passes near the point $(0, 0)$, it follows that b is approximately 0 and an appropriate linear model for the data is $y = 0.60x + 0$. Thus, of the given choices, $y = 0.606x$ is the most appropriate linear model for these data.

Alternate approach: A linear model can be written in the form $y = mx + b$, where m is the slope of the graph of the model in the

xy-plane and $(0, b)$ is the y-intercept. The scatterplot shows that as the x-values of the data points increase, the y-values of the data points increase, which means the graph of an appropriate linear model has a positive slope. The scatterplot also shows that an appropriate linear model would pass close to the point $(0, 0)$. Of the given choices, $y = 0.606x$ is the only linear model whose graph has a positive slope and a y-intercept of $(0, 0)$.

Choice A is incorrect. The graph of this model has a negative slope, not a positive slope.

Choice B is incorrect. The graph of this model has a negative slope, not a positive slope, and a y-intercept that is not $(0, 0)$.

Choice D is incorrect. The graph of this model has a slope that is too steep to fit the data, and a y-intercept that is not $(0, 0)$.

Question Difficulty: Easy

Question ID caaf7d9b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Probability and conditional probability	

ID: caaf7d9b

A box holds **104** tokens, of which **13** are brown. If a token is selected at random from the box, what is the probability of selecting a token that is brown?

- A. $\frac{1}{13}$
- B. $\frac{1}{10}$
- C. $\frac{1}{8}$
- D. $\frac{1}{4}$

ID: caaf7d9b Answer

Correct Answer: C

Rationale

Choice C is correct. If a token is selected at random from the box, the probability of selecting a token that is brown is equal to the number of tokens that are brown divided by the total number of tokens in the box. Since there are **13** brown tokens and **104** total tokens in the box, the probability of selecting a token that is brown is $\frac{13}{104}$, or $\frac{1}{8}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 7e9cded1

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Inference from sample statistics and margin of error	

ID: 7e9cded1

A middle school faculty surveyed students to determine interest in an after-school chess club. Of the **100** students who were selected at random from all students in the school, **32%** expressed interest in the club. Based on the sample mean and the margin of error, the faculty expects that between **24%** and **40%** of all students in the school would express interest in the club. What is the margin of error?

- A. **0%**
- B. **8%**
- C. **16%**
- D. **32%**

ID: 7e9cded1 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that of the **100** students who were selected at random from all students in the school, **32%** expressed interest in the club. It's also given that based on the sample mean and the margin of error, the faculty expects that between **24%** and **40%** of all students in the school would express interest in the club. These values are calculated by subtracting the margin of error from **32%** to get the lower bound, **24%**, and by adding the margin of error to **32%** to get the upper bound, **40%**. It follows that the margin of error can be calculated as the difference between **32%** and **24%** or between **40%** and **32%**. The difference between **32%** and **24%** is $(32 - 24)\%$, or **8%**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the difference between **40%** and **24%**, not the difference between **32%** and **24%** or between **40%** and **32%**.

Choice D is incorrect. This is the percentage of the students selected at random who expressed interest in the club, not the margin of error.

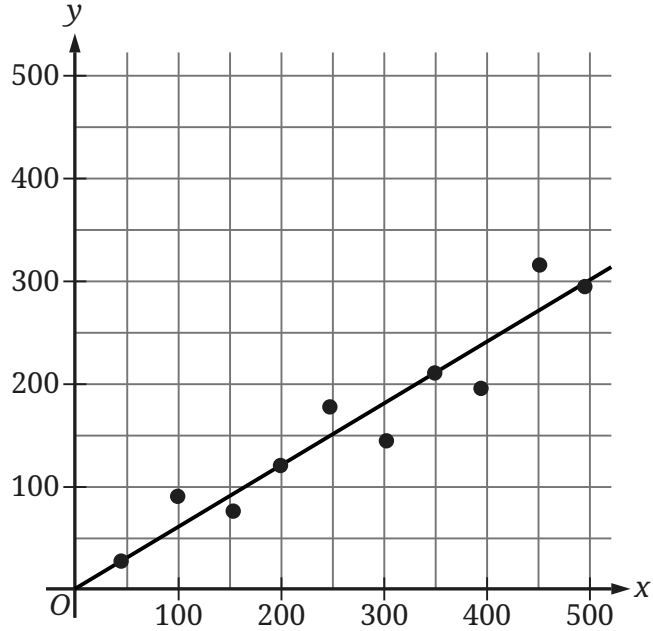
Question Difficulty: Medium

Question ID 63dad844

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	

ID: 63dad844

A group of **10** botanists recorded data on the germination rates of a certain type of seed for one growing season. The scatterplot shows the relationship between the number of seeds planted, x , and the number of seeds that germinated, y , for each of the botanists. A line of best fit is also shown.



Which of the following is the best interpretation of the slope of the line of best fit in this context?

- A. The number of seeds planted is predicted to increase by **60** seeds every **100** days.
- B. The number of seeds planted is predicted to increase by **300** seeds every **100** days.
- C. The number of seeds that germinate is predicted to increase by **60** seeds for every additional **100** seeds that are planted.
- D. The number of seeds that germinate is predicted to increase by **300** seeds for every additional **100** seeds that are planted.

ID: 63dad844 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that the scatterplot shows the relationship between the number of seeds planted, x , and the number of seeds that germinated, y . The slope of a line of best fit represents the change in the y -value for a given change in the x -value. According to the graph, the y -value of the line of best fit increases by about **60** when the x -value increases by about **100**, which represents an increase of about **60** germinated seeds for every additional **100** seeds planted. Therefore, the best interpretation of the slope of the line of best fit is that the number of seeds that germinate is predicted to increase by **60** seeds for every additional **100** seeds that are planted.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

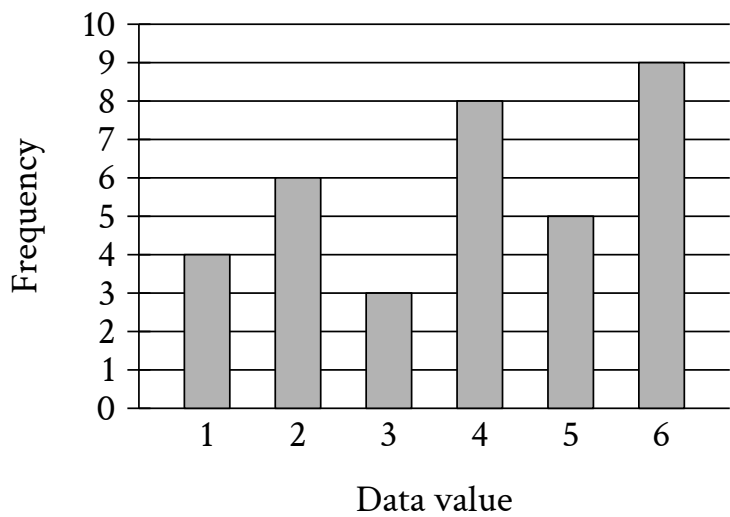
Question Difficulty: Medium

Question ID 53c6da75

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: 53c6da75

The bar graph shows the frequency of each data value in a data set.



What is the frequency of data value **2** in this data set?

ID: 53c6da75 Answer

Correct Answer: 6

Rationale

The correct answer is **6**. It's given that the bar graph shows the frequency of each data value in a data set. The bar corresponding to the data value **2** has a height of **6**. Therefore, the frequency of the data value **2** in this data set is **6**.

Question Difficulty: Easy

Question ID b43a2006

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Probability and conditional probability	

ID: b43a2006

For **100** buttons, the table summarizes the distribution by group and diameter.

Group	Diameter (millimeters)		
	Less than 20	20 to 30	Greater than 30
Group 1	12	8	3
Group 2	0	17	18
Group 3	10	32	0

One of these buttons will be selected at random. What is the probability of selecting a button with a diameter that is less than or equal to **30** millimeters, given that it is **not** in group 2? (Express your answer as a decimal or fraction, not as a percent.)

ID: b43a2006 Answer

Correct Answer: .9538, 62/65

Rationale

The correct answer is $\frac{62}{65}$. It's given that one button will be selected at random out of a total of **100** buttons and that the table shows the distribution of buttons by group and diameter. The probability of selecting a button with a diameter less than or equal to **30** millimeters, given that it is not in group 2, is the number of buttons with a diameter less than or equal to **30** millimeters that are not in group 2 divided by the total number of buttons not in group 2. Based on the table, the buttons not in group 2 are those in group 1 and group 3. In group 1, there are **12** buttons with a diameter less than **20** millimeters and **8** buttons with a diameter from **20** to **30** millimeters, which yields **12 + 8**, or **20**, buttons with a diameter less than or equal to **30** millimeters. In group 3, there are **10** buttons with a diameter less than **20** millimeters and **32** buttons with a diameter from **20** to **30** millimeters, which yields **10 + 32**, or **42**, buttons with a diameter less than or equal to **30** millimeters. Therefore, the total number of buttons with a diameter less than or equal to **30** millimeters that are not in group 2 is **20 + 42**, or **62**. The total number of buttons that are not in group 2 is the sum of all buttons in group 1 and group 3. There are **12 + 8 + 3**, or **23**, buttons in group 1 and **10 + 32 + 0**, or **42**, buttons in group 3. Therefore, the total number of buttons in groups 1 and 3 is **23 + 42**, or **65**. It follows that the probability of selecting a button with a diameter less than or equal to **30** millimeters, given that it is not in group 2, is $\frac{62}{65}$. Note that 62/65 or .9538 are examples of ways to enter a correct answer.

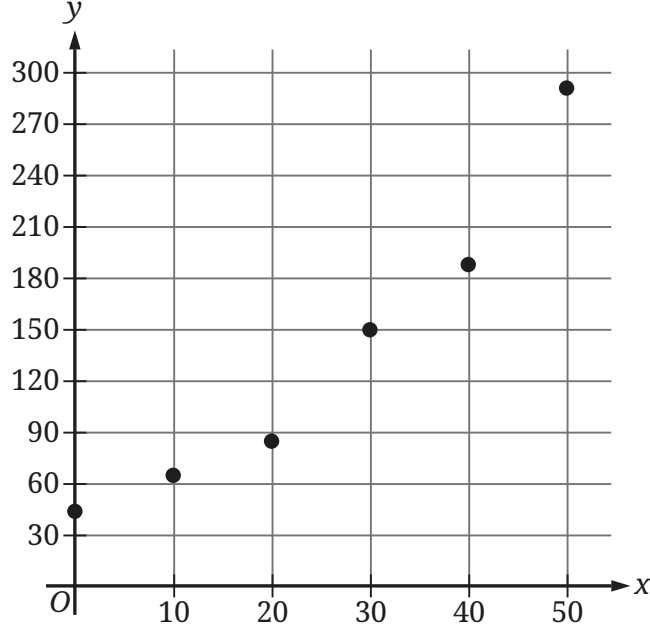
Question Difficulty: Hard

Question ID 0d7cbea0

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	

ID: 0d7cbea0

The scatterplot shows the relationship between the number of swans, y , in a certain population and the number of years since 1970, x , where $0 \leq x \leq 50$.



Which of the following equations is the most appropriate exponential model for the data shown in the scatterplot?

- A. $y = 43(1.04)^{-x}$
- B. $y = 43(1.04)^x$
- C. $y = 288(1.04)^{-x}$
- D. $y = 288(1.04)^x$

ID: 0d7cbea0 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that the scatterplot shows the relationship between the number of swans, y , and the number of years since 1970, x , where $0 \leq x \leq 50$. According to the scatterplot, the number of swans increases as the number of years since 1970 increases, and the data shows an exponential growth pattern rather than a linear pattern. When $a > 0$ and $b > 1$, an exponential model of the form $y = a(b)^x$ represents exponential growth and a model of the form $y = a(b)^{-x}$ represents exponential decay. The models in all four choices show $a > 0$ and $b > 1$. Choices B and D show models of the form $y = a(b)^x$ and thus represent exponential growth. The initial value a should approximate the y -value when $x = 0$, which is a little more than 40, so $a = 43$ is reasonable and consistent with the graph. Therefore, the equation $y = 43(1.04)^x$ is the most appropriate exponential model for the data.

Choice A is incorrect. The model $y = 43(1.04)^{-x}$ represents exponential decay, rather than exponential growth, which doesn't match the pattern in the scatterplot.

Choice C is incorrect. The model $y = 288(1.04)^{-x}$ represents exponential decay rather than exponential growth, which doesn't match the pattern in the scatterplot.

Choice D is incorrect. The model $y = 288(1.04)^x$ has an initial value of **288** when $x = 0$, so the initial value doesn't match the scatterplot.

Question Difficulty: Easy

Question ID 515a1bc4

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 515a1bc4

How many cups are equivalent to **83** quarts? (**4 cups = 1 quart**)

- A. **21**
- B. **79**
- C. **87**
- D. **332**

ID: 515a1bc4 Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that **4** cups is equal to **1** quart. Multiplying $\frac{4 \text{ cups}}{1 \text{ quart}}$ by **83** quarts yields **4** · **83** cups, or **332** cups.

Choice A is incorrect. This is the number of quarts in **83** cups, rounded to the nearest whole number.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID a2cadbcc

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: a2cadbcc

A cluster fly is traveling at a velocity of **76** centimeters per second. What is this velocity, in **millimeters** per second? (**1 centimeter = 10 millimeters**)

- A. **7,600**
- B. **836**
- C. **760**
- D. **86**

ID: a2cadbcc Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that **1 centimeter = 10 millimeters**. It follows that a velocity of **76** centimeters per second is equivalent to $\left(\frac{76 \text{ centimeters}}{1 \text{ second}}\right)\left(\frac{10 \text{ millimeters}}{1 \text{ centimeter}}\right)$, which is equivalent to **(76)(10)**, or **760**, millimeters per second.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 950ead0c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: 950ead0c

Data Set Q

Value	Frequency
a	110
b	30
c	0

Data Set R

Value	Frequency
a	0
b	3
c	11

Data sets Q and R are summarized in the tables shown, where a , b , and c are positive, consecutive integers and $a < b < c$. Which statement comparing the means of the data sets is true?

- A. The mean of data set R is $\frac{2(11)}{14}$ less than the mean of data set Q.
- B. The mean of data set R is $\frac{2(11)}{14}$ greater than the mean of data set Q.
- C. The mean of data set Q is 10 times the mean of data set R.
- D. The means of the two data sets are equal.

ID: 950ead0c Answer

Correct Answer: B

Rationale

Choice B is correct. The mean of a data set is the sum of the values of the data set divided by the total number of values in the data set. When a data set is represented in a frequency table, the sum of the values in the data set is the sum of the products of each value and its frequency. For data set Q, the sum of the products of each value and its frequency is $110a + 30b + 0c$, or $110a + 30b$, and the number of values is $110 + 30 + 0$, or 140. Therefore, the mean of data set Q is $\frac{110a+30b}{140}$. For data set R, the sum of the products of each value and its frequency is $0a + 3b + 11c$, or $3b + 11c$, and the number of values is $0 + 3 + 11$, or 14. Therefore, the mean of data set R is $\frac{3b+11c}{14}$. Since a , b , and c are positive, consecutive integers and $a < b < c$, it follows that $b = a + 1$ and $c = a + 2$. Substituting $a + 1$ for b in the mean of data set Q yields $\frac{110a+30(a+1)}{140}$, or $\frac{110a+30a+30}{140}$, which is equivalent to $\frac{140a+30}{140}$, or $a + \frac{3}{14}$. Substituting $a + 1$ for b and $a + 2$ for c in the mean of data set R yields $\frac{3(a+1)+11(a+2)}{14}$, or

$\frac{3a+3+11a+2(11)}{14}$, which is equivalent to $\frac{14a+3+2(11)}{14}$, or $a + \frac{3}{14} + \frac{2(11)}{14}$. Subtracting the mean of data set Q from the mean of data set R yields $\left(a + \frac{3}{14} + \frac{2(11)}{14}\right) - \left(a + \frac{3}{14}\right)$, or $\frac{2(11)}{14}$. Therefore, the mean of data set R is $\frac{2(11)}{14}$ greater than the mean of data set Q.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID b67847b7

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: b67847b7

Shape	Number of cards
Square	5
Circle	9
Triangle	14
Star	12

The table summarizes the distribution of shape printed on one side for cards in a deck of cards. How many cards in the deck have a triangle printed on one side?

ID: b67847b7 Answer

Correct Answer: 14

Rationale

The correct answer is **14**. The table summarizes the distribution of shape printed on one side for cards in a deck of cards. The table shows that the number of cards when the shape is a triangle is **14**. Therefore, there are **14** cards in the deck that have a triangle printed on one side.

Question Difficulty: Easy

Question ID 142fc517

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	

ID: 142fc517

0.1152 is 0.08% of the number b . The number b is 25% of the number c . What is the value of c ?

ID: 142fc517 Answer

Correct Answer: 576

Rationale

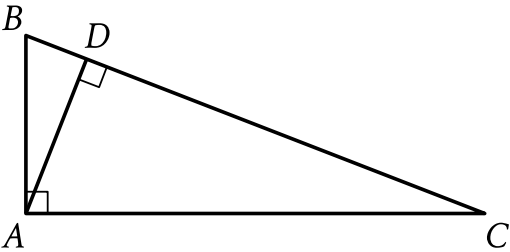
The correct answer is 576. It's given that 0.1152 is 0.08% of the number b . It follows that $0.1152 = \frac{0.08}{100}b$, or $0.1152 = 0.0008b$. Dividing both sides of this equation by 0.0008 yields $144 = b$. It's also given that the number b is 25% of the number c . It follows that $b = \frac{25}{100}c$, or $b = 0.25c$. Substituting 144 for b in this equation yields $144 = 0.25c$. Dividing both sides of this equation by 0.25 yields $576 = c$. Therefore, the value of c is 576.

Question Difficulty: Hard

Question ID bb237037

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	

ID: bb237037



Note: Figure not drawn to scale.

In the figure, $\overline{AD}$ and $\overline{BC}$ intersect at point D . If the length of $\overline{BD}$ is 5 and the tangent of angle CAD is 1.3 , what is the length of $\overline{AD}$?

ID: bb237037 Answer

Correct Answer: 6.5, 13/2

Rationale

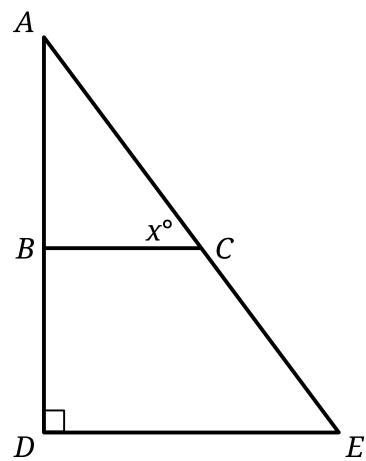
The correct answer is **6.5**. The figure shows that $\triangle BAC$ is a right triangle. In the figure, $\overline{AD}$ is perpendicular to $\overline{BC}$, which forms two additional right triangles: $\triangle BDA$ and $\triangle ADC$. If two pairs of corresponding angles in two triangles are congruent, then the triangles are similar. $\triangle BDA$ shares angle B with $\triangle BAC$ and $\triangle ADC$ shares angle C with $\triangle BAC$. Therefore, the figure contains three similar right triangles: $\triangle BAC$, $\triangle BDA$, and $\triangle ADC$. In similar triangles, the tangents of corresponding angles are equivalent. It's given that the tangent of angle CAD is 1.3 . Since angle CAD in $\triangle ADC$ corresponds to angle ABD in $\triangle BDA$, it follows that the tangent of angle ABD is also 1.3 . The tangent of an acute angle in a right triangle is equal to the length of the leg opposite the angle divided by the length of the leg adjacent to the angle. Therefore, $\tan ABD = \frac{AD}{BD}$. It's given that the length of $\overline{BD}$ is 5 . Substituting 5 for BD and 1.3 for $\tan ABD$ in the equation $\tan ABD = \frac{AD}{BD}$ yields $1.3 = \frac{AD}{5}$. Multiplying both sides of this equation by 5 yields $6.5 = AD$. Therefore, the length of $\overline{AD}$ is **6.5**. Note that 6.5 and 13/2 are examples of ways to enter a correct answer.

Question Difficulty: Hard

Question ID f6d848d7

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	

ID: f6d848d7



Note: Figure not drawn to scale.

In the figure, $\overline{BC}$ is parallel to $\overline{DE}$. If the length of $\overline{DE}$ is 147 and the length of $\overline{AE}$ is 245, what is the value of $\tan x^\circ$?

- A. $\frac{147}{196}$
- B. $\frac{147}{245}$
- C. $\frac{196}{147}$
- D. $\frac{245}{147}$

ID: f6d848d7 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that the length of $\overline{DE}$ is 147, and the length of $\overline{AE}$ is 245. Since angle D is a right angle, triangle ADE is a right triangle with hypotenuse $\overline{AE}$ and legs $\overline{AD}$ and $\overline{DE}$. By the Pythagorean theorem, if a right triangle has a hypotenuse with length c and legs with lengths a and b , then $a^2 + b^2 = c^2$. Substituting AD for a , 147 for b , and 245 for c in the equation $a^2 + b^2 = c^2$ yields $AD^2 + 147^2 = 245^2$. Subtracting 147^2 from both sides of this equation yields $AD^2 = 245^2 - 147^2$. Evaluating this difference yields $AD^2 = 60,025 - 21,609$, or $AD^2 = 38,416$. Taking the positive square root of both sides of this equation yields $AD = 196$. It's given that $\overline{BC}$ is parallel to $\overline{DE}$, and $\overline{AE}$ intersects $\overline{BC}$ and $\overline{DE}$. It follows that angle ACB is congruent to angle AED . Since angle A is common to triangles ABC and ADE , it follows that triangle ABC is similar to triangle ADE . In triangle ABC , $\tan x^\circ$ is the ratio of the leg opposite angle ACB to the leg adjacent to angle ACB , so $\tan x^\circ = \frac{AB}{BC}$. Since triangle ABC is similar to triangle ADE and similar triangles have proportional side lengths, it follows that $\frac{AB}{BC} = \frac{AD}{DE}$. Therefore, $\tan x^\circ = \frac{AD}{DE}$. Substituting 196 for AD and 147 for DE in this equation yields $\tan x^\circ = \frac{196}{147}$. Therefore, the value of $\tan x^\circ$ is $\frac{196}{147}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the value of $\cos x^\circ$, not $\tan x^\circ$.

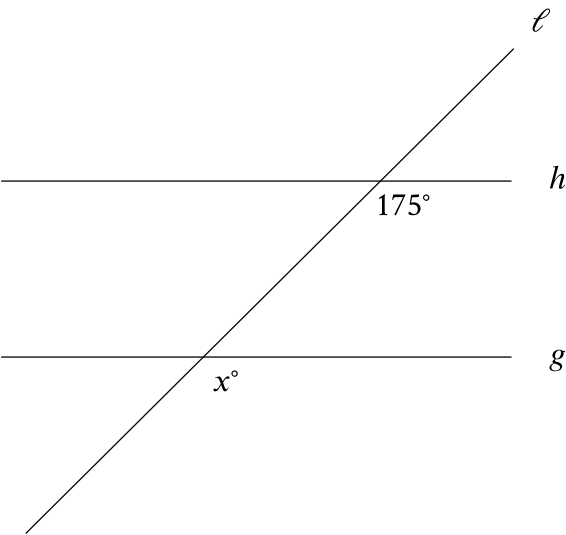
Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 2e7d9874

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 2e7d9874



Note: Figure not drawn to scale.

In the figure shown, line g is parallel to line h . Line l intersects both lines. What is the value of x ?

ID: 2e7d9874 Answer

Correct Answer: 175

Rationale

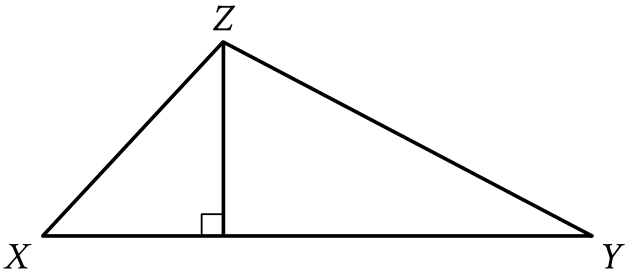
The correct answer is **175**. It's given that line g is parallel to line h . Since line l intersects both lines, it's a transversal. The angles in the figure marked as 175° and x° are on the same side of the transversal, where one is an interior angle with line h as a side, and the other is an exterior angle with line g as a side, respectively. Thus, the marked angles are corresponding angles. When two parallel lines are intersected by a transversal, corresponding angles have equal measure. Therefore, the value of x is **175**.

Question Difficulty: Easy

Question ID 7b73bb80

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	

ID: 7b73bb80



Note: Figure not drawn to scale.

In the figure shown, the measure of angle X is 54° . The length of $\overline{XY}$ is 26 units and the length of $\overline{XZ}$ is 19 units. What is the area, in square units, of triangle XYZ ?

- A. 247
- B. 494
- C. $247 \sin 54^\circ$
- D. $494 \sin 54^\circ$

ID: 7b73bb80 Answer

Correct Answer: C

Rationale

Choice C is correct. The area, A , of a triangle is given by the formula $A = \frac{1}{2}bh$, where b is the length of a base of the triangle and h is its corresponding height. It's given that the length of $\overline{XY}$ is 26 units. This length represents a base of the triangle, so $b = 26$. Let h represent its corresponding height. It's given that the measure of angle X is 54° and the length of $\overline{XZ}$ is 19 units. It follows that $\sin 54^\circ = \frac{h}{19}$. Multiplying both sides of this equation by 19 yields $19 \sin 54^\circ = h$. Substituting 26 for b and $19 \sin 54^\circ$ for h in the formula $A = \frac{1}{2}bh$ yields $A = \frac{1}{2}(26)(19 \sin 54^\circ)$, or $A = 247 \sin 54^\circ$. Therefore, the area, in square units, of triangle XYZ is $247 \sin 54^\circ$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

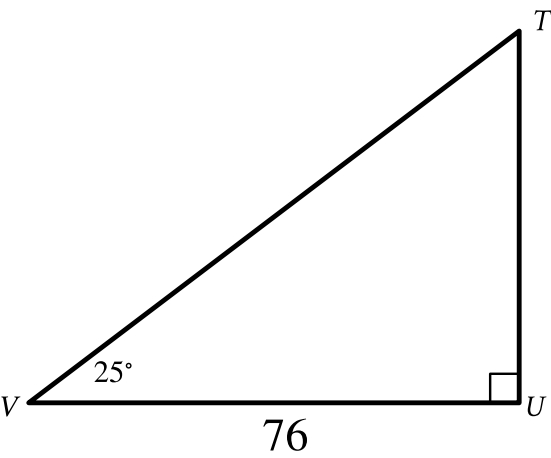
Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 8ed4acfd

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 8ed4acfd



Note: . igure not drawn to scale.

Triangle TUV is dilated by a scale factor of $\frac{1}{2}$ to obtain triangle $T'U'V'$ (not shown), where T corresponds to T' and U corresponds to U' . What is the measure, in degrees, of angle V' ?

- A. 25
- B. 38
- C. 50
- D. 76

ID: 8ed4acfd Answer

Correct Answer: A

Rationale

Choice A is correct. It’s given that triangle TUV is dilated by a scale factor of $\frac{1}{2}$ to obtain triangle $T'U'V'$, where T corresponds to T' and U corresponds to U' . A dilation is a transformation that multiplies all side lengths by the same scale factor while preserving angle measures. Therefore, corresponding angles in triangles TUV and $T'U'V'$ are equal. Since V corresponds to V' , the measure of angle V' is equal to the measure of angle V . The figure shows that the measure of angle V is 25° , so it follows that the measure of angle V' is also 25° .

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

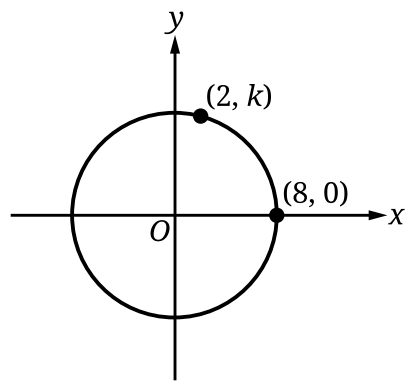
Choice D is incorrect. This is the length of $\overline{UV}$, not the measure, in degrees, of angle V' .

Question Difficulty: Medium

Question ID 8d1a790d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	

ID: 8d1a790d



Note: Figure not drawn to scale.

The circle shown has its center at (0, 0). What is the value of *k*?

- A. 6
- B. 7
- C. 8
- D. $\sqrt{60}$

ID: 8d1a790d Answer

Correct Answer: D

Rationale

Choice D is correct. It’s given that the circle shown has its center at (0, 0). The equation of a circle in the *xy*-plane with its center at (0, 0) can be expressed as $x^2 + y^2 = r^2$, where *r* is the radius of the circle. It’s shown that (8, 0) lies on the circle. Since the circle has its center at (0, 0) and (8, 0) lies on the circle, it follows that the radius of the circle is 8. Substituting 8 for *r* in the equation $x^2 + y^2 = r^2$ yields $x^2 + y^2 = 64$. It’s also shown that (2, *k*) lies on the circle. Substituting 2 for *x* and *k* for *y* in this equation yields $2^2 + k^2 = 64$, or $4 + k^2 = 64$. Subtracting 4 from both sides of this equation yields $k^2 = 60$. Taking the square root of both sides of this equation yields $k = \pm\sqrt{60}$. Since (2, *k*) lies above the *x*-axis, *k* must be positive. Therefore $k = \sqrt{60}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

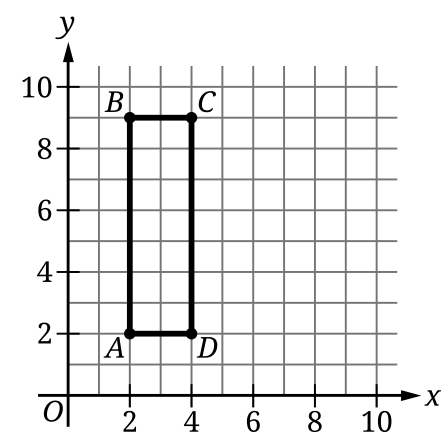
Choice C is incorrect. This is the *x*-coordinate of one of the circle’s *x*-intercepts, not the value of *k*.

Question Difficulty: Hard

Question ID bf4631fc

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: bf4631fc



Rectangle $ABCD$ shown has a length of **7** units and a width of **2** units. What is the area, in square units, of rectangle $ABCD$?

- A. **5**
- B. **9**
- C. **14**
- D. **28**

ID: bf4631fc Answer

Correct Answer: C

Rationale

Choice C is correct. The area, A , of a rectangle is given by the formula $A = l w$, where l is the length and w is the width of the rectangle. It's given that rectangle $ABCD$ shown has a length of **7** units and a width of **2** units. Substituting **7** for l and **2** for w in the formula $A = l w$ yields $A = (7)(2)$, or $A = 14$. Therefore, the area, in square units, of rectangle $ABCD$ is **14**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

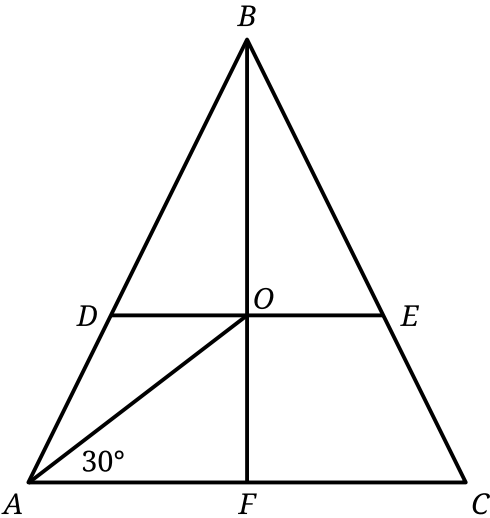
Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID b982dedb

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: b982dedb



Note: Figure not drawn to scale.

In triangle ABC , $AB = BC$. Points D and E lie on line segments AB and BC , respectively, such that line segments DE and AC are parallel. Point O is the midpoint of line segment DE , and the measure of $\angle OAC$ is 30° . Point F lies on line segment AC , and line segment BF passes through point O . If $AB = 81$ and $AO = BO$, what is the length of line segment BE ?

ID: b982dedb Answer

Correct Answer: 54

Rationale

The correct answer is **54**. It's given that $AB = BC$, so triangle ABC is isosceles. It's also given that line segments DE and AC are parallel. Since line segments DE and AC are parallel, triangle BDE is similar to triangle BAC , which means triangle BDE is also isosceles. It's given that point O is the midpoint of line segment DE . Since triangle BDE is isosceles, line segment BO splits triangle BDE into two congruent triangles, where $\angle BOD$ is a right angle. Therefore, $\angle BFA$ is also a right angle. The figure shows that the measure of $\angle OAF$ is 30° . It follows that triangle AOF is a 30-60-90 triangle, where the measure of $\angle AOF$ is 60° . In a 30-60-90 triangle, the side opposite the 30° angle is half the length of the side opposite the 90° angle. Let x represent the length of line segment OF and let $2x$ represent the length of line segment AO . It's given that $AO = BO$. Since $AO = BO$, the length of line segment BO is also $2x$. It follows that the length of line segment BF is $BO + OF$, which is $2x + x$, or $3x$. Since triangle BDE is similar to triangle BAC , the ratio of corresponding side lengths is equal to the ratio of the heights of the triangles. This means that $\frac{BE}{BC} = \frac{BO}{BF}$. Substituting $2x$ for BO and $3x$ for BF in this equation yields $\frac{BE}{BC} = \frac{2x}{3x}$, or $\frac{BE}{BC} = \frac{2}{3}$. It's given that $AB = 81$ and $AB = BC$, so $BC = 81$. Substituting 81 for BC in the equation $\frac{BE}{BC} = \frac{2}{3}$ yields $\frac{BE}{81} = \frac{2}{3}$. Multiplying both sides of this equation by 81 yields $BE = \frac{2}{3}(81)$, or $BE = 54$.

Question Difficulty: Hard

Question ID 50c577c7

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	

ID: 50c577c7

The measure of angle L is **60** degrees. If the measure of angle L is $\frac{\pi}{n}$ radians, where n is a constant, what is the value of n ?

ID: 50c577c7 Answer

Correct Answer: 3

Rationale

The correct answer is **3**. The measure of an angle in radians is equal to the measure of the angle in degrees multiplied by $\frac{\pi}{180}$. It's given that the measure of angle L is **60** degrees. Multiplying **60** by $\frac{\pi}{180}$ yields $\frac{60\pi}{180}$, or $\frac{\pi}{3}$. It's also given that the measure of angle L is $\frac{\pi}{n}$ radians, so $\frac{\pi}{n} = \frac{\pi}{3}$. Dividing both sides of this equation by π yields $\frac{1}{n} = \frac{1}{3}$. Taking the reciprocal of both sides of this equation yields $\frac{n}{1} = \frac{3}{1}$, or $n = 3$. Therefore, the value of n is **3**.

Question Difficulty: Hard

Question ID 8dbb23f8

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: 8dbb23f8

The length of the base of a triangle is **8** inches, and the height of the triangle is **12** inches. What is the area, in square inches, of the triangle?

- A. **20**
- B. **48**
- C. **60**
- D. **96**

ID: 8dbb23f8 Answer

Correct Answer: B

Rationale

Choice B is correct. The formula for the area, ***A***, of a triangle is **$A = \frac{1}{2}bh$** , where ***b*** is the length of the base of the triangle and ***h*** is the height of the triangle. It's given that the length of the base of a triangle is **8** inches and the height of the triangle is **12** inches. Substituting **8** for ***b*** and **12** for ***h*** in the formula **$A = \frac{1}{2}bh$** yields **$A = \frac{1}{2}(8)(12)$** , or **$A = 48$** . Therefore, the area, in square inches, of the triangle is **48**.

Choice A is incorrect. This is the sum of the length of the base of the triangle and the height of the triangle, not the area of the triangle.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the product of the length of the base of the triangle and the height of the triangle, not the area of the triangle.

Question Difficulty: Easy

Question ID 81416a34

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 81416a34

In $\triangle JKL$, the measures of both $\angle J$ and $\angle K$ are equal and the measure of $\angle L$ is 118° . What is the measure of $\angle J$?

- A. 31°
- B. 59°
- C. 62°
- D. 90°

ID: 81416a34 Answer

Correct Answer: A

Rationale

Choice A is correct. The sum of the measures of the three angles in a triangle is 180° . It's given that in $\triangle JKL$, the measure of $\angle L$ is 118° . Let x represent the measure of $\angle J$. It's also given that the measures of $\angle J$ and $\angle K$ are equal, so x also represents the measure of $\angle K$. It follows that $x + x + 118 = 180$, or $2x + 118 = 180$. Subtracting 118 from both sides of this equation yields $2x = 62$. Dividing both sides of this equation by 2 yields $x = 31$. Therefore, the measure of $\angle J$ is 31° .

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the sum of the measures of $\angle J$ and $\angle K$, not the measure of $\angle J$.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID b3c01259

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	

ID: b3c01259

$$x^2 + x\sqrt{t} + y^2 - y - 65 = 0$$

The given equation, where t is a constant, defines a circle in the xy -plane. The radius of this circle is $\sqrt{83}$. What is the value of t ?

- A. 73
- B. 71
- C. 37
- D. 35

ID: b3c01259 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that the equation $x^2 + x\sqrt{t} + y^2 - y - 65 = 0$, where t is a constant, defines a circle in the xy -plane. The equation of a circle in the xy -plane can be written in the form $(x - h)^2 + (y - k)^2 = r^2$, where (h, k) is the center and r is the radius of the circle. Adding 65 to both sides of the given equation yields $x^2 + x\sqrt{t} + y^2 - y = 65$. By completing the square, this equation can be rewritten as $\left(x^2 + x\sqrt{t} + \left(\frac{\sqrt{t}}{2}\right)^2\right) + \left(y^2 - y + \left(-\frac{1}{2}\right)^2\right) = 65 + \left(\frac{\sqrt{t}}{2}\right)^2 + \left(-\frac{1}{2}\right)^2$. This equation can be rewritten as $\left(x^2 + x\sqrt{t} + \frac{t}{4}\right) + \left(y^2 - y + \frac{1}{4}\right) = 65 + \frac{t}{4} + \frac{1}{4}$, or $\left(x + \frac{\sqrt{t}}{2}\right)^2 + \left(y - \frac{1}{2}\right)^2 = \frac{261}{4} + \frac{t}{4}$. Therefore, $r^2 = \frac{261}{4} + \frac{t}{4}$. It's given that the radius of this circle is $\sqrt{83}$; therefore, $r = \sqrt{83}$. Squaring both sides of this equation yields $r^2 = 83$. Since $r^2 = \frac{261}{4} + \frac{t}{4}$ and $r^2 = 83$, it follows that $\frac{261}{4} + \frac{t}{4} = 83$. Multiplying both sides of this equation by 4 yields $261 + t = 332$. Subtracting 261 from both sides of this equation yields $t = 71$. Therefore, the value of t is 71 .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 76558b72

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: 76558b72

The edge length, in inches, of cube Y is $\frac{3}{88}$ the edge length, in inches, of cube X. The surface area, in square inches, of cube Y is n times the surface area, in square inches, of cube X. What is the value of n ?

- A. $\frac{9}{7,744}$
- B. $\frac{27}{3,872}$
- C. $\frac{3}{88}$
- D. $\frac{9}{44}$

ID: 76558b72 Answer

Correct Answer: A

Rationale

Choice A is correct. Since all cubes are similar to each other, cube X and cube Y are similar figures. In similar figures, if lengths differ by a scale factor of k , then areas differ by a scale factor of k^2 . It's given that the edge length, in inches, of cube Y is $\frac{3}{88}$ the edge length, in inches, of cube X. It follows that the surface area, in square inches, of cube Y is $\left(\frac{3}{88}\right)^2$, or $\frac{9}{7,744}$, times the surface area, in square inches, of cube X. Therefore, the value of n is $\frac{9}{7,744}$.

Alternate approach: The surface area of a cube is $6s^2$, where s is the edge length. Let x represent the edge length, in inches, of cube X. Substituting x for s in $6s^2$ yields $6x^2$. Therefore, the surface area, in square inches, of cube X can be expressed as $6x^2$. It's given that the edge length, in inches, of cube Y is $\frac{3}{88}$ the edge length, in inches, of cube X. Therefore, the edge length, in inches, of cube Y can be expressed as $\left(\frac{3}{88}\right)x$, or $\frac{3x}{88}$. Substituting $\frac{3x}{88}$ for s in $6s^2$ yields $6\left(\frac{3x}{88}\right)^2$, which is equivalent to $6\left(\frac{9x^2}{7,744}\right)$, or $\frac{27x^2}{3,872}$. Therefore, the surface area, in square inches, of cube Y can be expressed as $\frac{27x^2}{3,872}$. It's given that the surface area, in square inches, of cube Y is n times the surface area, in square inches, of cube X. It follows that $\frac{27x^2}{3,872} = n\left(6x^2\right)$. Dividing both sides of this equation by $6x^2$ yields $\frac{27x^2}{3,872(6x^2)} = n$, which is equivalent to $\frac{27x^2}{23,232x^2} = n$, or $\frac{9}{7,744} = n$. Therefore, the value of n is $\frac{9}{7,744}$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID a413c1d9

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: a413c1d9

The area of a triangle is **36** square units. If the height of this triangle is **2** units, what is the length, in units, of the base of this triangle?

ID: a413c1d9 Answer

Correct Answer: 36

Rationale

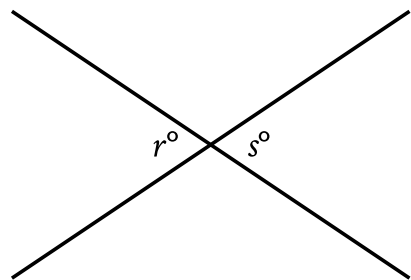
The correct answer is **36**. The area, ***A***, of a triangle is given by the formula **$A = \frac{1}{2}bh$** , where ***b*** represents the length of the base of the triangle and ***h*** represents its height. It's given that the triangle has an area of **36** square units and a height of **2** units. Substituting **36** for ***A*** and **2** for ***h*** in the formula **$A = \frac{1}{2}bh$** yields **$36 = \frac{1}{2}b(2)$** , or **$36 = b$** . Thus, the length, in units, of the base of this triangle is **36**.

Question Difficulty: Medium

Question ID 576363d9

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 576363d9



Note: Figure not drawn to scale.
In the figure, two lines intersect at a point. If $r = 53$, what is the value of s ?

- A. 37
- B. 53
- C. 143
- D. 233

ID: 576363d9 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that two lines intersect at a point. The angles with measures of r° and s° are vertical angles, so it follows that they are equal in measure. Therefore, $s = r$. If $r = 53$, it follows that $s = 53$.

Choice A is incorrect. This is the measure, in degrees, of the complement of an angle with a measure of 53° , not the value of s .

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 1cfe07d3

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: 1cfe07d3

A right prism has a square base with side lengths of **10** inches. The height of the prism is **58** inches. What is the volume, in cubic inches, of the prism?

ID: 1cfe07d3 Answer

Correct Answer: 5800

Rationale

The correct answer is **5,800**. The volume, V , of a right prism with a square base is given by the formula $V = s^2h$, where s is the side length of the square base and h is the height of the prism. It's given that a right prism has a square base with side lengths of **10** inches. It's also given that the height of the prism is **58** inches. Substituting **10** for s and **58** for h in the formula $V = s^2h$ yields $V = (10)^2(58)$, or $V = 5,800$. Therefore, the volume, in cubic inches, of the prism is **5,800**.

Question Difficulty: Hard

Question ID ad3dd08e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: ad3dd08e

The sum of the measures of two angles in a triangle is **64°**. What is the measure of the third angle?

- A. **26°**
- B. **52°**
- C. **116°**
- D. **128°**

ID: ad3dd08e Answer

Correct Answer: C

Rationale

Choice C is correct. It’s given that the sum of the measures of two angles in a triangle is **64°**. The sum of the measures of all three angles in a triangle is **180°**. It follows that the measure of the third angle is the difference between **180°** and **64°**, or **180 – 64**, which is **116°**. Therefore, the measure of the third angle is **116°**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 716ba193

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: 716ba193

The length of a side of square X is **9** centimeters. The area of rectangle Y is **32** square centimeters. What is the total area, in square centimeters, of square X and rectangle Y?

- A. **145**
- B. **113**
- C. **82**
- D. **81**

ID: 716ba193 Answer

Correct Answer: B

Rationale

Choice B is correct. The area of a square is s^2 , where s is the side length of the square. It's given that the length of a side of square X is **9** centimeters. It follows that the area of square X is **9²**, or **81** square centimeters. It's also given that the area of rectangle Y is **32** square centimeters. Therefore, the total area, in square centimeters, of square X and rectangle Y is **81 + 32**, or **113**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the area of square X, not the total area of square X and rectangle Y, in square centimeters.

Question Difficulty: Easy

Question ID 5ce95910

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: 5ce95910

A right circular cone has a height of **22 centimeters (cm)** and a base with a diameter of **18 cm**. The volume of this cone is $n\pi \text{ cm}^3$. What is the value of n ?

ID: 5ce95910 Answer

Correct Answer: 594

Rationale

The correct answer is **594**. It's given that the right circular cone has a height of **22 cm** and a base with a diameter of **18 cm**. Since the diameter of the base of the cone is **18 cm**, the radius of the base is **9 cm**. The volume V , in cm^3 , of a right circular cone can be found using the formula $V = \frac{1}{3}\pi r^2 h$, where h is the height, in **cm**, of the cone and r is the radius, in **cm**, of the base of the cone. Substituting **22** for h and **9** for r in this formula yields $V = \frac{1}{3}\pi(9)^2(22)$, or $V = 594\pi$. Therefore, the volume of the cone is **$594\pi \text{ cm}^3$** . It's given that the volume of the cone is $n\pi \text{ cm}^3$. Therefore, the value of n is **594**.

Question Difficulty: Hard

Question ID efb1f2e4

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: efb1f2e4

The base of a right rectangular pyramid has a length of **27** centimeters and a width of **2** centimeters. The height of this pyramid is **7** centimeters. What is the volume, in cubic centimeters, of this pyramid?

ID: efb1f2e4 Answer

Correct Answer: 126

Rationale

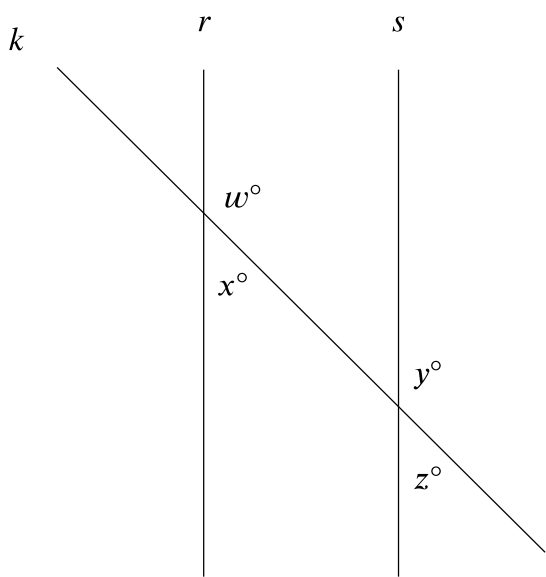
The correct answer is **126**. It's given that the base of the right rectangular pyramid has a length of **27** centimeters and a width of **2** centimeters and that the height of the pyramid is **7** centimeters. The volume, V , of a right rectangular pyramid is given by the formula $V = \frac{1}{3}lwh$, where l and w are the length and width, respectively, of the pyramid's base and h is the height of the pyramid. Substituting **27** for l , **2** for w , and **7** for h in this formula yields $V = \frac{1}{3}(27)(2)(7)$, or $V = 126$. Therefore, the volume, in cubic centimeters, of the pyramid is **126**.

Question Difficulty: Medium

Question ID 5db431fe

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 5db431fe



Note: . igure not drawn to scale.

In the figure shown, line *k* intersects lines *r* and *s*. If $w = 147$, which additional piece of information is sufficient to prove that lines *r* and *s* are parallel?

- A. $x = 33$
- B. $y = 147$
- C. $w + y = 180$
- D. $y + z = 180$

ID: 5db431fe Answer

Correct Answer: B

Rationale

Choice B is correct. The figure shows transversal *k* intersecting lines *r* and *s*, where the angles labeled w° and y° are two corresponding angles and the angles labeled x° and z° are two corresponding angles. When a transversal intersects two lines, the lines are parallel if and only if two corresponding angles are congruent. It’s given that $w = 147$. If $y = 147$, then the two corresponding angles labeled w° and y° are congruent, which is sufficient to prove that lines *r* and *s* are parallel.

Choice A is incorrect. The angle labeled x° forms a linear pair with the angle labeled w° , so $x = 180 - 147$, or $x = 33$, regardless of whether lines *r* and *s* are parallel.

Choice C is incorrect. Substituting 147 for w in this equation yields $147 + y = 180$, or $y = 33$. Since the angle labeled w° and the angle labeled y° are corresponding angles, if w and y aren’t equal, then lines *r* and *s* can’t be parallel.

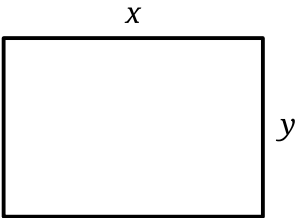
Choice D is incorrect. The relationship $y + z = 180$ describes a linear pair at line s , which is true regardless of whether lines r and s are parallel.

Question Difficulty: Medium

Question ID 39a005a5

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: 39a005a5



Note: Figure not drawn to scale.

In the rectangle shown, $x = 7$ centimeters and $y = 5$ centimeters. What is the area, in square centimeters, of the rectangle?

- A. 7
- B. 12
- C. 35
- D. 40

ID: 39a005a5 Answer

Correct Answer: C

Rationale

Choice C is correct. It’s given that $x = 7$ centimeters and $y = 5$ centimeters for the rectangle. The area, A , of a rectangle can be found using the formula $A = l w$, where l represents the length and w represents the width. In this rectangle, x represents the length and y represents the width. Substituting 7 for l and 5 for w in the equation yields $A = (7)(5)$, or $A = 35$. Therefore, the area of the rectangle is 35 square centimeters.

Choice A is incorrect. This is the length, not the area, of the rectangle.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 0aaaff02

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	

ID: 0aaaff02

The difference between the measure of angle A and the measure of angle B is $-\frac{5}{6}\pi$ radians. Which expression shows the difference between the measure of angle A and the measure of angle B , in degrees?

- A. $-\frac{5}{6}\pi \cdot \frac{360^\circ}{\pi}$
- B. $-\frac{5}{6}\pi \cdot \frac{180^\circ}{\pi}$
- C. $-\frac{5}{6}\pi \cdot \frac{\pi}{180^\circ}$
- D. $-\frac{5}{6} \cdot \frac{180^\circ}{\pi}$

ID: 0aaaff02 Answer

Correct Answer: B

Rationale

Choice B is correct. Since π radians $= 180^\circ$, the measure of an angle, in degrees, can be found by multiplying its measure, in radians, by $\frac{180^\circ}{\pi}$. It's given that the difference between the measure of angle A and the measure of angle B is $-\frac{5}{6}\pi$ radians. It follows that the expression $-\frac{5}{6}\pi \cdot \frac{180^\circ}{\pi}$ shows the difference between the measure of angle A and the measure of angle B , in degrees.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 8e46b01c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: 8e46b01c

Which expression represents the area, in square inches, of a square with a side length of **35** inches?

- A. $\sqrt{35}$
- B. $35 + 35$
- C. $(4)(35)$
- D. 35^2

ID: 8e46b01c Answer

Correct Answer: D

Rationale

Choice D is correct. The area of a square is given by s^2 , where s is the length of each side of the square. It's given that the length of each side of a square is **35** inches. Substituting **35** for s in the expression s^2 yields 35^2 . Therefore, the expression that represents the area, in square inches, of the square is 35^2 .

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

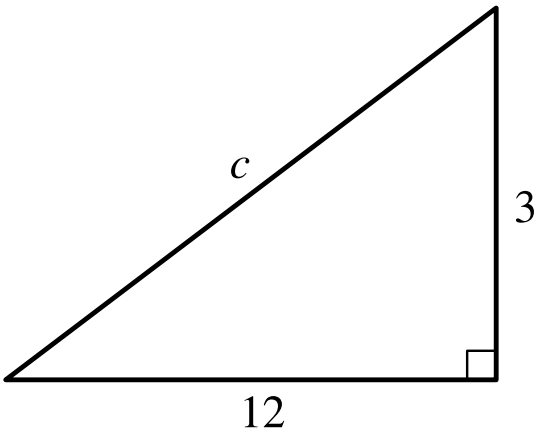
Choice C is incorrect. This expression represents the perimeter, in inches, of the square rather than the area, in square inches.

Question Difficulty: Easy

Question ID 3cd43205

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	

ID: 3cd43205



Note: . igure not drawn to scale.

Which equation shows the relationship between the side lengths of the triangle shown?

- A. $12 \cdot 3 = c$
- B. $12 + 3 = c$
- C. $12^2 + 3^2 = c^2$
- D. $12^2 - 3^2 = c^2$

ID: 3cd43205 Answer

Correct Answer: C

Rationale

Choice C is correct. The Pythagorean theorem states that for a right triangle, $a^2 + b^2 = c^2$, where a and b represent the lengths of the legs and c represents the length of the hypotenuse. It's given that the right triangle shown has legs with lengths **12** and **3** and a hypotenuse with length c . Substituting **12** for a and **3** for b in the formula $a^2 + b^2 = c^2$ yields $12^2 + 3^2 = c^2$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

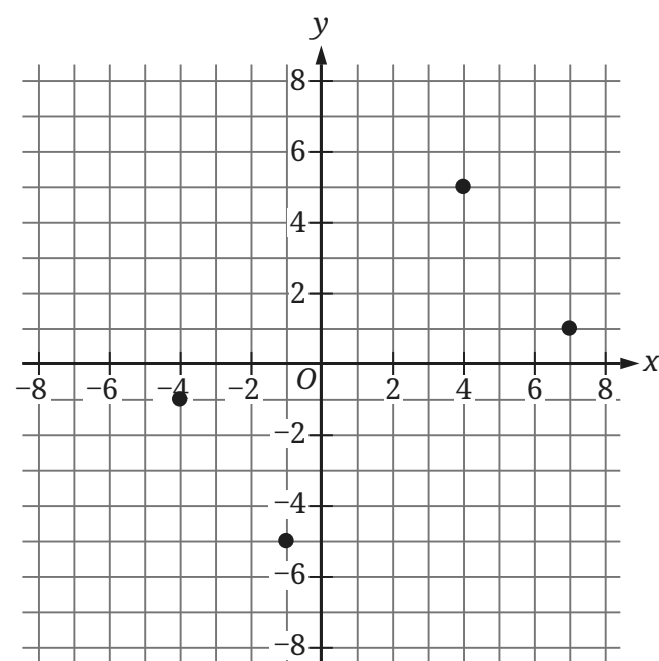
Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID adedb9dc

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: adedb9dc



A rectangle is formed by the four points shown. What is the area, in square units, of the rectangle?

ID: adedb9dc Answer

Correct Answer: 50

Rationale

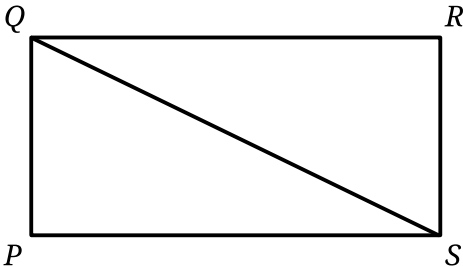
The correct answer is **50**. The area of a rectangle is $l \cdot w$, where l and w are the lengths of two adjacent sides of the rectangle. It's given that the points $(-4, -1)$, $(4, 5)$, $(7, 1)$, and $(-1, -5)$ form a rectangle. The points $(-4, -1)$ and $(4, 5)$ form a side of the rectangle that is adjacent to the side of the rectangle formed by the points $(4, 5)$ and $(7, 1)$. In the xy -plane, the distance between two points, (x_1, y_1) and (x_2, y_2) , can be found as $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$. Thus, the length, in units, of the side formed by the points $(-4, -1)$ and $(4, 5)$ can be found as $\sqrt{(4 - (-4))^2 + (5 - (-1))^2}$, which can be simplified to $\sqrt{100}$, or **10**. The length, in units, of the side formed by the points $(4, 5)$ and $(7, 1)$ can be found as $\sqrt{(7 - 4)^2 + (1 - 5)^2}$, which can be simplified to $\sqrt{25}$, or **5**. Therefore, the area, in square units, of the rectangle formed by the four points shown is **(10)(5)**, or **50**.

Question Difficulty: Hard

Question ID 203ea312

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	

ID: 203ea312



Note: Figure not drawn to scale.

In rectangle $PQRS$, the length of line segment QS is $\sqrt{2,344}$ and the length of side PS is 8 more than the length of side RS . What is the length of side QR ?

ID: 203ea312 Answer

Correct Answer: 38

Rationale

The correct answer is **38**. It's given that in rectangle $PQRS$, the length of side PS is 8 more than the length of side RS . In a rectangle, opposite sides are congruent. Since side QR is opposite side PS , it follows that side QR is congruent to side PS . Thus, the length of side QR is 8 more than the length of side RS . Let x represent the length of side RS . It follows that the length of side QR is represented by the expression $x + 8$. Since all angles in a rectangle are right angles, angle R is a right angle. It follows that side QR , side RS , and line segment QS form a right triangle, where line segment QS is the hypotenuse and sides QR and RS are the legs. It's given that the length of line segment QS is $\sqrt{2,344}$. By the Pythagorean theorem, if a right triangle has a hypotenuse with length c and legs with lengths a and b , then $a^2 + b^2 = c^2$. Substituting x for a , $x + 8$ for b , and $\sqrt{2,344}$ for c in this equation yields $x^2 + (x + 8)^2 = (\sqrt{2,344})^2$, or $x^2 + (x + 8)^2 = 2,344$. Expanding the left-hand side of this equation yields $x^2 + x^2 + 16x + 64 = 2,344$, or $2x^2 + 16x + 64 = 2,344$. Dividing both sides of this equation by 2 yields $x^2 + 8x + 32 = 1,172$. To complete the square, subtracting 16 from both sides of this equation yields $x^2 + 8x + 16 = 1,156$. Since the left-hand side of this equation is a perfect square trinomial, it follows that $x^2 + 8x + 16 = 1,156$ is equivalent to $(x + 4)^2 = 1,156$. Taking the positive square root of both sides of this equation yields $x + 4 = 34$. Subtracting 4 from both sides of this equation yields $x = 30$. Since the length of side QR is represented by the expression $x + 8$, it follows that the length of side QR is $30 + 8$, or **38**.

Question Difficulty: Hard

Question ID 3ea51996

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	

ID: 3ea51996

What is the perimeter of an equilateral triangle with a height of $39\sqrt{3}$?

- A. 117
- B. $117\sqrt{3}$
- C. 234
- D. $1,521\sqrt{3}$

ID: 3ea51996 Answer

Correct Answer: C

Rationale

Choice C is correct. It’s given that the equilateral triangle has a height of $39\sqrt{3}$. In equilateral triangles, all three angles have a measure of 60° and the height is equal to the length of an altitude from one vertex to the midpoint of the side across from the vertex. Thus, an altitude in this equilateral triangle has a length of $39\sqrt{3}$. The altitude divides the equilateral triangle into two congruent 30-60-90 right triangles, where the altitude is the side across from the 60° angle in each 30-60-90 right triangle. Since the altitude has a length of $39\sqrt{3}$, it follows from the properties of 30-60-90 right triangles that the side across from each 30° angle has a length of **39** and each hypotenuse has a length of **2(39)**, or **78**. Therefore, each side of the equilateral triangle has a length of **78**. In equilateral triangles, all three sides have equal length. Therefore, the perimeter of this equilateral triangle is **78 + 78 + 78**, or **234**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

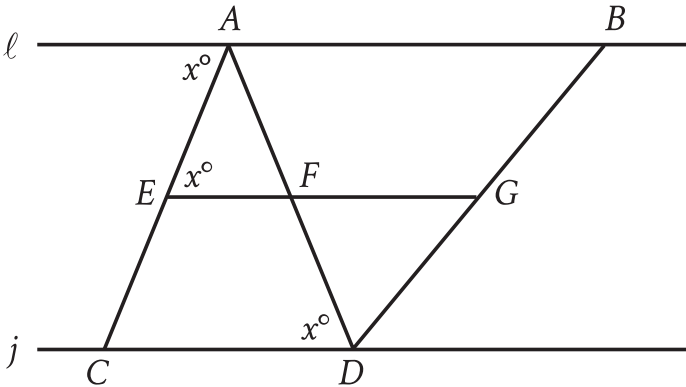
Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 09ff8833

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 09ff8833



Note: Figure not drawn to scale.

In the figure, $\overline{EG}$ and $\overline{AD}$ intersect at point F , and line l is parallel to line j . If $EF = 0.5CD$ and $FG = 2.25CD$, which of the following must be true?

- I. $AB = 2.25EF$
- II. $FG = 0.5AB$
- III. $\frac{EF}{FG} = \frac{AF}{FD}$

- A. I only
- B. II only
- C. I and II
- D. II and III

ID: 09ff8833 Answer

Correct Answer: B

Rationale

Choice B is correct. In the figure, $\overline{AE}$ intersects line l and $\overline{EG}$, and alternate interior angles are both marked x° . It follows that $\overline{EG}$ is parallel to line l . It's given that line l is parallel to line j , so $\overline{EG}$ is also parallel to line j . Thus, the following angles also have measure x° : $\angle EFA$, $\angle ACD$, $\angle BAD$, and $\angle GFD$. Triangles AEF and ACD each have two angles with measure x° , so triangles AEF and ACD are similar. It's also given that $EF = 0.5CD$, so $AF = 0.5AD$. It follows that $AF = FD$. Triangles DFG and DAB each have one angle with measure x° and share $\angle FDG$, so triangles DFG and DAB are similar. Since $AF = FD$, it follows that $AD = 2FD$, so $AB = 2FG$. Substituting $2FG$ for AB in statement I yields $2FG = 2.25EF$. It's given that $FG = 2.25CD$. Substituting $2.25CD$ for FG and $0.5CD$ for EF in the equation $2FG = 2.25EF$ yields $2(2.25CD) = 2.25(0.5CD)$, or $4.5CD = 1.125CD$. Thus, statement I is not true. Since $AB = 2FG$, it follows that $0.5AB = FG$, or $FG = 0.5AB$. Therefore, statement II must be true. Substituting $0.5CD$ for EF , $2.25CD$ for FG , and FD for AF in statement III yields $\frac{0.5CD}{2.25CD} = \frac{FD}{FD}$, or $\frac{2}{9} = 1$. Thus, statement III is not true. Therefore,

statement II only must be true.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

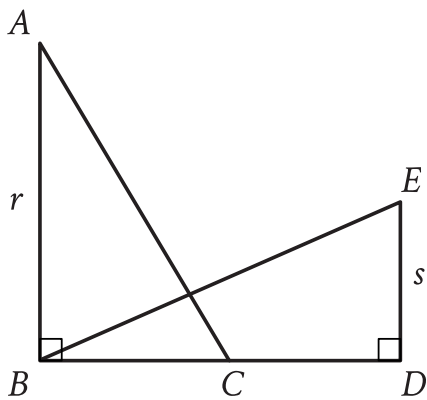
Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 74bd95c0

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 74bd95c0



Note: Figure not drawn to scale.

In the figure shown, $\triangle ABC$ is congruent to $\triangle BDE$. If $r = 54$ and $s = 26$, what is the length of $\overline{CD}$?

- A. 26
- B. 27
- C. 28
- D. 54

ID: 74bd95c0 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that $\triangle ABC$ is congruent to $\triangle BDE$. Corresponding sides of congruent triangles are equal in length. It's given that r , which represents the length of $\overline{AB}$, is equal to 54 and s , which represents the length of $\overline{DE}$, is equal to 26. Since $\overline{AB}$ and $\overline{DE}$ aren't equal in length, it follows that $\overline{AB}$ corresponds to $\overline{BD}$ and $\overline{BC}$ corresponds to $\overline{DE}$. Therefore, the length of $\overline{BD}$ is 54 and the length of $\overline{BC}$ is 26. The points B , C , and D lie on a straight line, so the length of $\overline{BD}$ can be expressed as the sum of the lengths of $\overline{BC}$ and $\overline{CD}$. This relationship can be written as $BD = BC + CD$. Substituting 54 for BD and 26 for BC in this equation yields $54 = 26 + CD$. Subtracting 26 from both sides of this equation yields $CD = 28$. Therefore, the length of $\overline{CD}$ is 28.

Choice A is incorrect. This is the length of $\overline{BC}$, not $\overline{CD}$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the length of $\overline{BD}$, not $\overline{CD}$.

Question Difficulty: Medium

Question ID b3787bbd

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: b3787bbd

Trapezoid A and trapezoid B are similar. The length of each side of trapezoid A is **41** times the length of the corresponding side of trapezoid B. The area of trapezoid A is how many times as large as the area of trapezoid B?

ID: b3787bbd Answer

Correct Answer: 1681

Rationale

The correct answer is **1,681**. It's given that trapezoid A and trapezoid B are similar. When two trapezoids are similar, the scale factor between their corresponding side lengths is ***k***, and the scale factor between their areas is ***k*²**. It's given that the length of each side of trapezoid A is **41** times the length of the corresponding side of trapezoid B, so the scale factor between their corresponding side lengths is **41**. It follows that the scale factor between the areas is **41²**, or **1,681**. Therefore, the area of trapezoid A is **1,681** times as large as the area of trapezoid B.

Question Difficulty: Hard

Question ID d17e4678

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: d17e4678

A right circular cylinder has a height of **16** inches and a volume of **$3,136\pi$** cubic inches. What is the radius, in inches, of this cylinder?

ID: d17e4678 Answer

Correct Answer: 14

Rationale

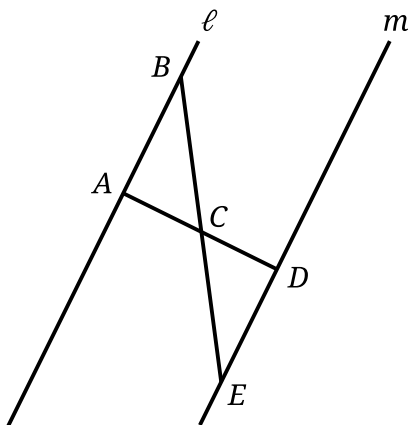
The correct answer is **14**. The volume, V , of a right circular cylinder is given by the formula $V = \pi r^2 h$, where r is the radius of the base of the cylinder and h is the height of the cylinder. It's given that the right circular cylinder has a height of **16** inches and a volume of **$3,136\pi$** cubic inches. Substituting **16** for h and **$3,136\pi$** for V in the formula $V = \pi r^2 h$ yields **$3,136\pi = \pi r^2 (16)$** , or **$3,136\pi = 16\pi r^2$** . Dividing both sides of this equation by **16π** yields **$196 = r^2$** . Since r represents the radius of the cylinder, the value of r is positive. Taking the positive square root of both sides of this equation yields **$r = 14$** . Therefore, the radius, in inches, of the cylinder is **14**.

Question Difficulty: Medium

Question ID 255b6f4e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 255b6f4e



Note: Figure not drawn to scale.
In the figure, segment AD and segment BE intersect at point C . Segment AD is perpendicular to line l at point A . Which of the following additional pieces of information is(are) sufficient to determine that triangle ABC is congruent to triangle DEC ?

- I. Segment AD is perpendicular to line m at point D .
- II. $AC = 13$ and $DC = 13$

- A. I is sufficient by itself, but II is not.
- B. II is sufficient by itself, but I is not.
- C. I is sufficient by itself, and II is sufficient by itself.
- D. Neither I nor II is sufficient by itself.

ID: 255b6f4e Answer

Correct Answer: D

Rationale

Choice D is correct. In the figure, it's given that segment AD is perpendicular to line l at point A . Statement I is that segment AD is perpendicular to line m at point D . It follows that angles BAC and EDC are right angles. Therefore, triangles ABC and DEC are right triangles. At least one pair of congruent corresponding sides is needed to determine triangle congruence by the side-angle-side (SAS) condition or the angle-side-angle (ASA) condition, so statement I is not sufficient to determine that triangle ABC is congruent to triangle DEC . Statement II is that $AC = 13$ and $DC = 13$. It follows that triangles ABC and DEC share one pair of congruent corresponding sides. Two additional pairs of congruent corresponding parts are needed to determine triangle congruence by SAS condition, ASA condition, or side-side-side (SSS) condition, so statement II is not sufficient to determine that triangle ABC is congruent to triangle DEC . Therefore, neither I nor II is sufficient by itself to determine that triangle ABC is congruent to triangle DEC .

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Question Difficulty: Hard

Question ID ed144e13

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: ed144e13

Right rectangular prism P is similar to right rectangular prism Q. The volume of prism P is **972 cubic centimeters (cm^3)**, and the volume of prism Q is **36 cm^3** . One edge of prism P has a length of **12 cm**. What is the length, in **cm**, of the corresponding edge of prism Q?

ID: ed144e13 Answer

Correct Answer: 4

Rationale

The correct answer is **4**. It's given that right rectangular prism P is similar to right rectangular prism Q. Therefore, there is a positive constant **k** such that the length of each edge of prism P is **k** times the length of the corresponding edge of prism Q and the volume of prism P is **k^3** times the volume of prism Q. It's given that the volume of prism P is **972 cm^3** and the volume of prism Q is **36 cm^3** . It follows that **$972 = 36k^3$** . Dividing both sides of this equation by **36** yields **$\frac{972}{36} = k^3$** , or **$k^3 = 27$** . Taking the cube root of both sides of this equation yields **$k = 3$** . It follows that the length of each edge of prism P is **3** times the length of the corresponding edge of prism Q. It's given that one edge of prism P has a length of **12 cm**. It follows that **12 cm** is **3** times the length of the corresponding edge of prism Q. Thus, the length, in **cm**, of the edge of prism Q corresponding to the edge of prism P with a length of **12 cm** is **$\frac{12}{3}$** , or **4**.

Question Difficulty: Hard

Question ID 50251b82

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	

ID: 50251b82

In right triangle XYZ , angles X and Z are acute angles. Point R lies on $\overline{XZ}$ such that $\overline{YR}$ is perpendicular to $\overline{XZ}$. The measure of $\angle XZY$ is a° , the measure of $\angle XYR$ is b° , and the length of $\overline{XY}$ is 19. If $\sin a^\circ = \frac{12}{13}$, what is the value of $\tan b^\circ$?

ID: 50251b82 Answer

Correct Answer: 2.4, 12/5

Rationale

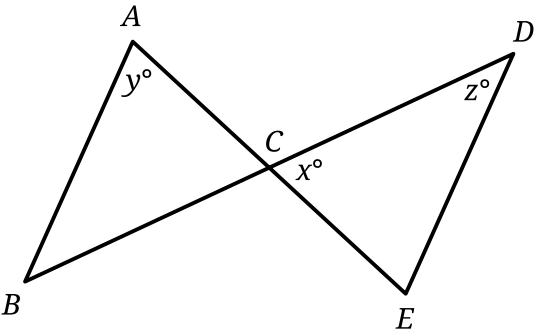
The correct answer is $\frac{12}{5}$. It's given that in right triangle XYZ , $\angle X$ and $\angle Z$ are acute angles. It follows that $\angle Y$ is a right angle. It's given that point R lies on $\overline{XZ}$ such that $\overline{YR}$ is perpendicular to $\overline{XZ}$. Since $\angle Y$ is a right angle and is comprised of $\angle XYR$ and $\angle ZYR$, it follows that the sum of the measures of $\angle XYR$ and $\angle ZYR$ is 90° . It's given that the measure of $\angle XYR$ is b° . It follows that the measure of $\angle ZYR$ is $(90 - b)^\circ$. Since $\overline{YR}$ is perpendicular to $\overline{XZ}$, it follows that $\angle XRY$ and $\angle ZRY$ are right angles. It follows that $\triangle XRY$ and $\triangle ZRY$ are right triangles. It's given that the measure of $\angle XZY$ is a° . Because the sum of the measures of the interior angles of a triangle is 180° , it follows that the measures of $\angle ZRY$, $\angle XZY$, and $\angle ZYR$ have a sum of 180° . Since the measure of $\angle ZRY$ is 90° , the measure of $\angle YZR$ is a° , and the measure of $\angle ZYR$ is $(90 - b)^\circ$, it follows that $90 + a + 90 - b = 180$, or $a - b + 180 = 180$. Subtracting 180 from both sides of this equation yields $a - b = 0$. Adding b to both sides of this equation yields $a = b$. It's given that $\sin a^\circ = \frac{12}{13}$. Since $a = b$, it follows that $\sin b^\circ = \frac{12}{13}$. In a right triangle, the sine of an acute angle is the ratio of the length of the leg opposite the angle to the length of the hypotenuse. It follows that, for a positive constant k , $12k$ represents the length of the leg opposite $\angle XYR$ and $13k$ represents the length of the hypotenuse of $\triangle XRY$. The Pythagorean theorem states that in a right triangle, the square of the length of the hypotenuse is equal to the sum of the squares of the lengths of the other two sides. Let nk represent the length of the leg adjacent to $\angle XYR$. It follows that $(13k)^2 = (12k)^2 + (nk)^2$, or $169k^2 = 144k^2 + n^2k^2$. Subtracting $144k^2$ from both sides of this equation yields $25k^2 = n^2k^2$. Taking the positive square root of both sides of this equation yields $5k = nk$. Therefore, the length of the leg adjacent to $\angle XYR$ is $5k$. In a right triangle, the tangent of an acute angle is the ratio of the length of the leg opposite the angle to the length of the leg adjacent to the angle. It follows that $\tan b^\circ = \frac{12k}{5k}$, or $\tan b^\circ = \frac{12}{5}$. Note that 12/5 and 2.4 are examples of ways to enter a correct answer.

Question Difficulty: Hard

Question ID aef1d52b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: aef1d52b



Note: Figure not drawn to scale.

In the figure, $\overline{AE}$ intersects $\overline{DB}$ at point C and $\overline{AB}$ is parallel to $\overline{DE}$, creating triangles ABC and CDE . If $x = 63$ and $y = 2x - 47$, what is the value of z ?

- A. 38
- B. 47
- C. 63
- D. 79

ID: aef1d52b Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that $\overline{AE}$ intersects $\overline{DB}$ at point C and that $\overline{AB}$ is parallel to $\overline{DE}$, creating triangles ABC and CDE . Since $\overline{AB}$ is parallel to $\overline{DE}$, angles BAC and DEC are opposite interior angles and are congruent. It follows that the measure of angle DEC is y° . This means the three interior angles of triangle CDE have measures of x° , z° , and y° , respectively. It's given that $x = 63$ and $y = 2x - 47$. Substituting 63 for x in the equation $y = 2x - 47$ yields $y = 2(63) - 47$, or $y = 79$. Therefore, the three interior angles of triangle CDE have measures 63° , z° , and 79° , respectively. The sum of the interior angles of a triangle is 180° . Therefore, $63 + z + 79 = 180$, or $z + 142 = 180$. Subtracting 142 from both sides of this equation yields $z = 38$. Therefore, the value of z is 38 .

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the value of x , not z .

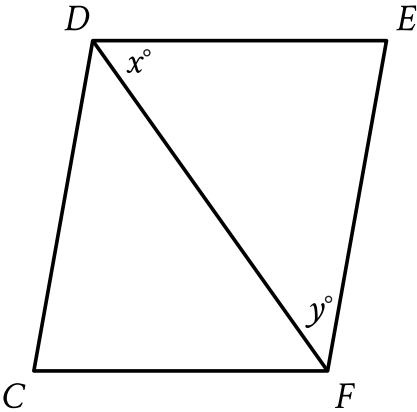
Choice D is incorrect. This is the value of y , not z .

Question Difficulty: Medium

Question ID 55bfc881

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 55bfc881



Note: Figure not drawn to scale.

In the figure shown, $CD = EF$ and $FC = DE$. If $x = 65$ and $y = 30$, what is the measure of $\angle CDE$?

- A. 30°
- B. 65°
- C. 95°
- D. 130°

ID: 55bfc881 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that $CD = EF$ and $FC = DE$, and the figure shows that $\overline{DF}$ is a side of both $\triangle CDF$ and $\triangle EFD$. Since $\triangle CDF$ and $\triangle EFD$ have three pairs of congruent sides, the triangles are congruent, where $\angle CFD$ corresponds to $\angle EDF$ and $\angle CDF$ corresponds to $\angle EFD$. It's given that the measure of $\angle EDF$ is 65° and the measure of $\angle EFD$ is 30° . Since corresponding angles of congruent triangles are congruent, $\angle CFD$ is congruent to $\angle EDF$ and $\angle CDF$ is congruent to $\angle EFD$. Therefore, the measure of $\angle CFD$ is 65° and the measure of $\angle CDF$ is 30° . Since $\angle CDE$ is made up of $\angle CDF$ and $\angle EDF$, it follows that the measure of $\angle CDE$ is equal to the sum of the measures of $\angle CDF$ and $\angle EDF$. Therefore, the measure of $\angle CDE$ is equal to $(30 + 65)^\circ$, or 95° .

Choice A is incorrect. This is the measure of $\angle CDF$, not the measure of $\angle CDE$.

Choice B is incorrect. This is the measure of $\angle CFD$, not the measure of $\angle CDE$.

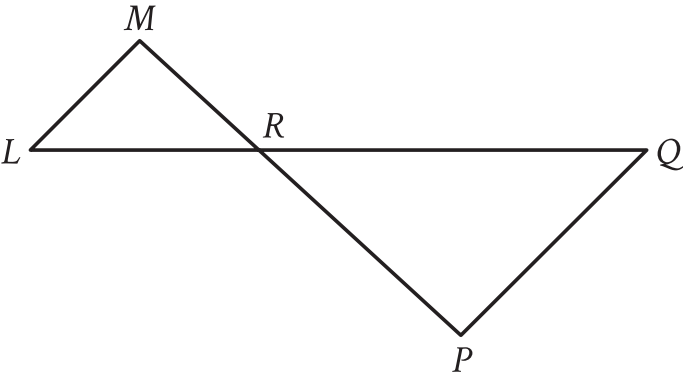
Choice D is incorrect. This is the sum of the measures of $\angle CFD$ and $\angle EDF$, not $\angle CDF$ and $\angle EDF$.

Question Difficulty: Medium

Question ID b63926ad

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: b63926ad



Note: Figure not drawn to scale.

In the figure, $\overline{LQ}$ intersects $\overline{MP}$ at point R , and $\overline{LM}$ is parallel to $\overline{PQ}$. The lengths of $\overline{MR}$ and $\overline{RP}$ are 8 and 14 units, respectively. The area of $\triangle LMR$ is 36 square units. What is the area of $\triangle PQR$, in square units?

- A. $\frac{576}{49}$
- B. $\frac{144}{7}$
- C. 63
- D. $\frac{441}{4}$

ID: b63926ad Answer

Correct Answer: D

Rationale

Choice D is correct. The figure shows that $\angle MRL$ and $\angle PRQ$ are vertical angles. Since vertical angles are congruent, $\angle MRL$ and $\angle PRQ$ are congruent. It's given that $\overline{LM}$ is parallel to $\overline{PQ}$. The figure also shows that $\overline{LQ}$ intersects $\overline{LM}$ and $\overline{PQ}$. If two parallel segments are intersected by a third segment, alternate interior angles are congruent. Thus, $\angle MLR$ and $\angle PQR$ are congruent. Since $\triangle LMR$ and $\triangle PQR$ have two pairs of congruent angles, the triangles are similar, where $\overline{MR}$ in $\triangle LMR$ corresponds to $\overline{PR}$ in $\triangle PQR$. Since the areas of similar triangles are proportional to the squares of the lengths of corresponding sides, it follows that $\frac{\text{area of } \triangle PQR}{\text{area of } \triangle LMR} = \left(\frac{RP}{MR}\right)^2$. It's given that the lengths of $\overline{MR}$ and $\overline{RP}$ are 8 and 14 units, respectively, and the area of $\triangle LMR$ is 36 square units. Substituting 14 for RP , 8 for MR , and 36 for area of $\triangle LMR$ in the equation $\frac{\text{area of } \triangle PQR}{\text{area of } \triangle LMR} = \left(\frac{RP}{MR}\right)^2$ yields $\frac{\text{area of } \triangle PQR}{36} = \left(\frac{14}{8}\right)^2$, which is equivalent to $\frac{\text{area of } \triangle PQR}{36} = \left(\frac{7}{4}\right)^2$, or $\frac{\text{area of } \triangle PQR}{36} = \frac{49}{16}$. Multiplying each side of this equation by 36 yields area of $\triangle PQR = 36\left(\frac{49}{16}\right)$, or area of $\triangle PQR = \frac{441}{4}$. Therefore, the area of $\triangle PQR$ is $\frac{441}{4}$ square units.

Choice A is incorrect. This is the result of multiplying 36 by $\left(\frac{8}{14}\right)^2$ instead of by $\left(\frac{14}{8}\right)^2$.

Choice B is incorrect. This is the result of multiplying 36 by $\frac{8}{14}$ instead of by $\left(\frac{14}{8}\right)^2$.

Choice C is incorrect. This is the result of multiplying **36** by $\frac{14}{8}$ instead of by $\left(\frac{14}{8}\right)^2$.

Question Difficulty: Hard

Question ID 33e7b14a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: 33e7b14a

Rectangle $ABCD$ has an area of **280** square centimeters. The length of the rectangle is **10** centimeters. What is the width, in centimeters, of the rectangle?

- A. **10**
- B. **28**
- C. **76**
- D. **140**

ID: 33e7b14a Answer

Correct Answer: B

Rationale

Choice B is correct. The area A of a rectangle is the product of its length l and its width w . It follows that $A = l \cdot w$. It's given that rectangle $ABCD$ has an area of **280** square centimeters and the length of the rectangle is **10** centimeters. Substituting **280** for A and **10** for l in the equation $A = l \cdot w$ yields $280 = 10w$. Dividing both sides of this equation by **10** yields $28 = w$. Therefore, the width, in centimeters, of the rectangle is **28**.

Choice A is incorrect. This is the length, in centimeters, not the width, in centimeters, of the rectangle.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is half the area, in square centimeters, not the width, in centimeters, of the rectangle.

Question Difficulty: Easy

Question ID d041c471

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	

ID: d041c471

For two acute angles, $\angle Q$ and $\angle R$, $\cos(Q) = \sin(R)$. The measures, in degrees, of $\angle Q$ and $\angle R$ are $x + 52$ and $4x + 13$, respectively. What is the value of x ?

- A. 5
- B. 13
- C. 23
- D. 38

ID: d041c471 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that for two acute angles, $\angle Q$ and $\angle R$, $\cos(Q) = \sin(R)$. For two acute angles, if the sine of one angle is equal to the cosine of the other, the angles are complementary. Therefore, the sum of the measures, in degrees, of $\angle Q$ and $\angle R$ is 90. It's given that the measures, in degrees, of $\angle Q$ and $\angle R$ are $x + 52$ and $4x + 13$, respectively. Therefore, $(x + 52) + (4x + 13) = 90$. Combining like terms in this equation yields $5x + 65 = 90$. Subtracting 65 from each side of this equation results in $5x = 25$. Dividing each side of this equation by 5 yields $x = 5$.

Choice B is incorrect. This would be the value of x if $\angle Q$ and $\angle R$ were congruent rather than complementary.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 3b574e3a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	

ID: 3b574e3a

In the xy -plane, circle F is defined by the equation $(x + 8)^2 + (y + 8)^2 = 25$. Circle G has the same center as circle F , and the radius of circle G is 2 units greater than the radius of circle F . Circle G is defined by the equation $(x + 8)^2 + (y + 8)^2 = p$, where p is a constant. What is the value of p ?

ID: 3b574e3a Answer

Correct Answer: 49

Rationale

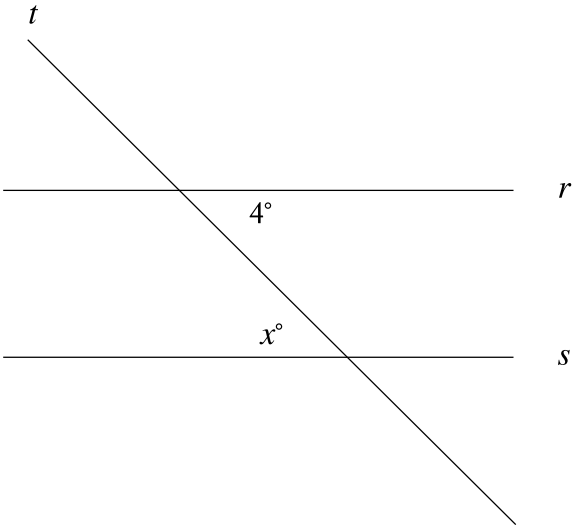
The correct answer is 49. An equation of a circle in the xy -plane can be written as $(x - h)^2 + (y - k)^2 = r^2$, where the center of the circle is (h, k) and the radius of the circle is r . It's given that in the xy -plane, circle F is defined by the equation $(x + 8)^2 + (y + 8)^2 = 25$, which can be rewritten as $(x - (-8))^2 + (y - (-8))^2 = 5^2$. It follows that $h = -8$, $k = -8$, and $r = 5$. Therefore, the center of circle F is $(-8, -8)$ and the radius of circle F is 5. It's given that circle G has the same center as circle F , and the radius of circle G is 2 units greater than the radius of circle F . Therefore, the center of circle G is $(-8, -8)$ and the radius of circle G is $5 + 2$, or 7. Substituting -8 for h , -8 for k , and 7 for r into the equation $(x - h)^2 + (y - k)^2 = r^2$ yields $(x + 8)^2 + (y + 8)^2 = 7^2$, which is equivalent to $(x + 8)^2 + (y + 8)^2 = 49$. It follows that circle G is defined by the equation $(x + 8)^2 + (y + 8)^2 = 49$; therefore, the value of p is 49.

Question Difficulty: Hard

Question ID 565b85ff

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 565b85ff



Note: . igure not drawn to scale.

In the figure, line t intersects lines r and s . Which additional piece of information is sufficient to prove that lines r and s are parallel?

- A. $x < 24$
- B. $x = 24$
- C. $x > 24$
- D. $x - 24 = 360$

ID: 565b85ff Answer

Correct Answer: B

Rationale

Choice B is correct. In the figure shown, transversal t intersects lines r and s . Two lines intersected by a transversal are parallel if and only if a pair of alternate interior angles have equal measure. The angle marked 24° and the angle marked x° form a pair of alternate interior angles. Therefore, $x = 24$ is sufficient to prove that lines r and s are parallel.

Choice A is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Question Difficulty: Easy

Question ID 3c96377b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: 3c96377b

A right square pyramid has a total surface area of **70,560** square inches, and the combined surface area of the four lateral faces of this pyramid is **38,160** square inches. What is the height, in inches, of this pyramid?

- A. 56
- B. 90
- C. 106
- D. 180

ID: 3c96377b Answer

Correct Answer: A

Rationale

Choice A is correct. The total surface area of a right square pyramid is the sum of the combined surface area of its four lateral faces and the area of its square base. It's given that the right square pyramid has a total surface area of **70,560** square inches, and the combined surface area of the four lateral faces of this pyramid is **38,160** square inches. It follows that the area of the square base is the difference between the total surface area and the combined surface area of the four lateral faces, or **(70,560 – 38,160)** square inches, which is equal to **32,400** square inches. The area, ***B***, of the base of a square pyramid is given by the equation **$B = s^2$** , where ***s*** is the side length of the base. Substituting **32,400** for ***B*** in this equation yields **$32,400 = s^2$** . Taking the positive square root of both sides of this equation yields **$180 = s$** . The four lateral faces of a square pyramid are congruent triangles. It follows that the area of each lateral face is $\frac{38,160}{4}$, or **9,540**, square inches. The area, ***A***, of each lateral face of a square pyramid is given by the equation **$A = \frac{1}{2} s \ell$** , where ***s*** is the side length of the pyramid's base and ***ℓ*** is the pyramid's slant height. Substituting **9,540** for ***A*** and **180** for ***s*** in this equation yields **$9,540 = \frac{1}{2} (180) \ell$** , or **$9,540 = 90 \ell$** . Dividing both sides of this equation by **90** yields **$106 = \ell$** . In the right square pyramid, a right triangle can be drawn, where the lengths of the legs correspond to the pyramid's height, ***h***, and half the side length of its base, $\frac{1}{2} s$, and the length of the hypotenuse corresponds to the pyramid's slant height, ***ℓ***. It follows that the relationship between ***h***, ***s***, and ***ℓ*** is given by the equation **$h^2 + (\frac{1}{2} s)^2 = \ell^2$** . Substituting **180** for ***s*** and **106** for ***ℓ*** in this equation yields **$h^2 + ((\frac{1}{2})(180))^2 = (106)^2$** , or **$h^2 + 8,100 = 11,236$** . Subtracting **8,100** from both sides of this equation yields **$h^2 = 3,136$** . Taking the positive square root of both sides of the equation yields **$h = 56$** . Therefore, the height, in inches, of this pyramid is **56**.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the pyramid's slant height, not its height, in inches.

Choice D is incorrect. This is the side length of the pyramid's base, not its height, in inches.

Question Difficulty: Hard

Question ID d27973f9

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: d27973f9

For triangles ABC and FGH , angles B and G each measure 51° , the length of $\overline{AB}$ is 6 , and the length of $\overline{FG}$ is 18 . Which additional piece(s) of information is(are) sufficient to prove that triangle ABC is similar to triangle FGH ?

- I. The measure of angle A is equal to the measure of angle F .
- II. The measure of angle C is equal to the measure of angle H .
- III. The length of $\overline{FH}$ is 3 times the length of $\overline{AC}$.

- A. I is sufficient, II is sufficient, and III is sufficient.
- B. I is sufficient and II is sufficient, but III is not.
- C. II is sufficient and III is sufficient, but I is not.
- D. III is sufficient, but I is not and II is not.

ID: d27973f9 Answer

Correct Answer: B

Rationale

Choice B is correct. It's given that in triangles ABC and FGH , angles B and G each measure 51° , the length of $\overline{AB}$ is 6 , and the length of $\overline{FG}$ is 18 . For statement I, if the measure of angle A is equal to the measure of angle F , then triangles ABC and FGH have two pairs of congruent angles, angles A and F and angles B and G . When 2 pairs of angles in 2 triangles are congruent, the triangles are similar, so statement I is sufficient. For statement II, if the measure of angle C is equal to the measure of angle H , then triangles ABC and FGH have 2 pairs of congruent angles, angles C and H and angles B and G . When 2 pairs of angles in 2 triangles are congruent, the triangles are similar, so statement II is sufficient. For statement III, if the length of $\overline{FH}$ is 3 times the length of $\overline{AC}$, then the ratio of the length of $\overline{FH}$ to the length of $\overline{AC}$ is 3 . It's given that the length of $\overline{FG}$ is 18 and the length of $\overline{AB}$ is 6 . Therefore, the ratio of the length of $\overline{FG}$ to the length of $\overline{AB}$ is also 3 , so there are two pairs of sides in proportion. However, angle B isn't between sides AB and AC , and angle G isn't between sides FG and FH . Since the given angle measures aren't between the proportional sides, this information isn't sufficient to prove that the triangles are similar. Therefore, statements I and II are each sufficient, but statement III isn't sufficient.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID bd0354a4

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	

ID: bd0354a4

Acute angle P has a measure of p radians, and acute angle T has a measure of t radians. The function g is defined by $g(x) = p(8)^x + 6t$. If $\sin p = \cos t$, which of the following represents the value $g(0)$ in terms of t ?

- A. $6t + \pi$
- B. $6t$
- C. $5t + \pi$
- D. $5t + \frac{\pi}{2}$

ID: bd0354a4 Answer

Correct Answer: D

Rationale

Choice D is correct. It's given that acute angle P has a measure of p radians and that acute angle T has a measure of t radians. If $\sin p = \cos t$, it follows that $p + t = \frac{\pi}{2}$, or $p = \frac{\pi}{2} - t$. It's also given that function g is defined by $g(x) = p(8)^x + 6t$. To determine the value of $g(0)$ in terms of t , substituting 0 for x and $\frac{\pi}{2} - t$ for p in the equation $g(x) = p(8)^x + 6t$ yields $g(0) = \left(\frac{\pi}{2} - t\right)(8)^0 + 6t$, or $g(0) = \left(\frac{\pi}{2} - t\right)(1) + 6t$. This is equivalent to $g(0) = \frac{\pi}{2} - t + 6t$, or $g(0) = 5t + \frac{\pi}{2}$. Therefore, $5t + \frac{\pi}{2}$ represents the value of $g(0)$ in terms of t .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID d6e1f157

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	

ID: d6e1f157

A circle has a diameter of **12.6** centimeters. What is the radius of the circle, in centimeters?

ID: d6e1f157 Answer

Correct Answer: 6.3, 63/10

Rationale

The correct answer is **6.3**. The diameter of a circle is equal to **$2r$** , where **r** represents the radius of the circle. It’s given that a circle has a diameter of **12.6** centimeters. Therefore, **$12.6 = 2r$** . Dividing both sides of this equation by **2** yields **$6.3 = r$** . It follows that the radius of the circle, in centimeters, is **6.3**. Note that 6.3 and 63/10 are examples of ways to enter a correct answer.

Question Difficulty: Easy

Question ID ff8f68e6

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	

ID: ff8f68e6

The measure of angle A is $\frac{\pi}{216}$ radians. The measure of angle B is **36** times the measure of angle A . What is the value of **cos B** ?

- A. **0**
- B. $\frac{1}{2}$
- C. $\frac{\sqrt{2}}{2}$
- D. $\frac{\sqrt{3}}{2}$

ID: ff8f68e6 Answer

Correct Answer: D

Rationale

Choice D is correct. It’s given that the measure of angle A is $\frac{\pi}{216}$ radians and the measure of angle B is **36** times the measure of angle A . Therefore, the measure of angle B is **36** $\left(\frac{\pi}{216}\right)$ radians, or $\frac{\pi}{6}$ radians. A trigonometric ratio can be found using the unit circle, or a circle with radius **1** unit. For a unit circle centered at the origin in the xy-plane, the cosine of $\frac{\pi}{6}$ radians is the x-coordinate of the point resulting from traveling a distance of $\frac{\pi}{6}$ counterclockwise from the point **(1, 0)**. This results in the point $\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$. Therefore, the value of **cos B** is $\frac{\sqrt{3}}{2}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the value of **sin B** , not **cos B** .

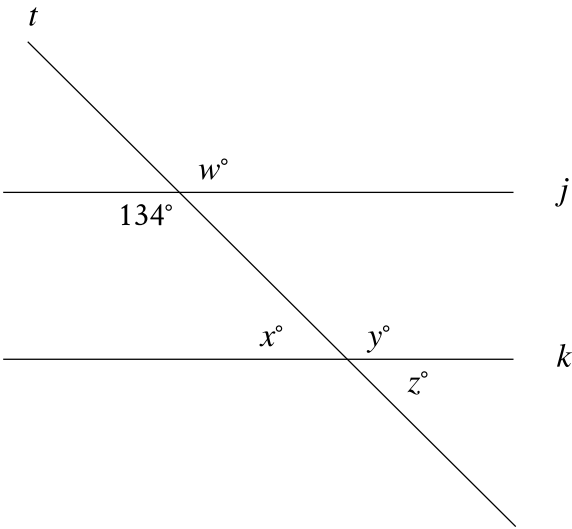
Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Hard

Question ID 9096db02

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 9096db02



Note: . igure not drawn to scale.

In the figure, line t intersects lines j and k . Which additional piece of information is sufficient to prove that lines j and k are parallel?

- A. $x = 46$
- B. $y = 46$
- C. $w = 134$
- D. $z = 134$

ID: 9096db02 Answer

Correct Answer: A

Rationale

Choice A is correct. In the figure shown, line t , which is a transversal, intersects lines j and k . If two lines are intersected by a transversal and the measures of a pair of same-side interior angles sum to 180° , then the lines are parallel. In the figure, the angle measuring 134° and the angle measuring x° are a pair of same-side interior angles. If $x = 46$, $(134 + x)^\circ = (134 + 46)^\circ$, or $(134 + x)^\circ = 180^\circ$. Therefore, $x = 46$ is sufficient to prove that lines j and k are parallel.

Choice B is incorrect. The angle measuring y° and the angle measuring 134° are alternate interior angles. If $y = 46$ rather than $y = 134$, then lines j and k would not be parallel.

Choice C is incorrect. The angle measuring w° and the angle measuring 134° are vertical angles. Thus, $w = 134$ whether lines j and k are parallel or not.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID 95fc265c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	

ID: 95fc265c

$$x^2 + y^2 - 10x - 8y - 14n = 0$$

The given equation represents circle A in the xy -plane, where n is a constant. Point $(11, 8)$ lies on circle B, which has the same center but twice the diameter as circle A. What is the value of n ?

ID: 95fc265c Answer

Correct Answer: -2

Rationale

The correct answer is -2 . It's given that $x^2 + y^2 - 10x - 8y - 14n = 0$ represents circle A in the xy -plane, where n is a constant. The equation of a circle in the xy -plane can be written in the form $(x - h)^2 + (y - k)^2 = r^2$, where (h, k) is the center of the circle and r is its radius. By completing the square, the equation $x^2 + y^2 - 10x - 8y - 14n = 0$ can be rewritten as $\left(x^2 - 10x + \left(\frac{-10}{2}\right)^2\right) + \left(y^2 - 8y + \left(\frac{-8}{2}\right)^2\right) = 14n + \left(\frac{-10}{2}\right)^2 + \left(\frac{-8}{2}\right)^2$. This is equivalent to $(x^2 - 10x + 25) + (y^2 - 8y + 16) = 14n + 25 + 16$, which can be written as $(x - 5)^2 + (y - 4)^2 = 14n + 41$. It follows that $(5, 4)$ is the center of circle A and $\sqrt{14n + 41}$ is its radius. It's given that circle B has the same center as circle A. Therefore, the center of circle B is $(5, 4)$. It's also given that circle B has twice the diameter as circle A. Therefore, circle B has twice the radius as circle A. Since the radius of circle A is $\sqrt{14n + 41}$, it follows that the radius of circle B is $2\sqrt{14n + 41}$. Substituting 5 for h , 4 for k , and $2\sqrt{14n + 41}$ for r in the equation $(x - h)^2 + (y - k)^2 = r^2$ yields $(x - 5)^2 + (y - 4)^2 = \left(2\sqrt{14n + 41}\right)^2$, or $(x - 5)^2 + (y - 4)^2 = 4(14n + 41)$, which represents an equation for circle B. It's given that the point $(11, 8)$ lies on circle B. Substituting 11 for x and 8 for y in the equation $(x - 5)^2 + (y - 4)^2 = 4(14n + 41)$ yields $(11 - 5)^2 + (8 - 4)^2 = 4(14n + 41)$, which gives $6^2 + 4^2 = 4(14n + 41)$, or $52 = 4(14n + 41)$. Dividing both sides of this equation by 4 yields $13 = 14n + 41$. Subtracting 41 from both sides of this equation yields $-28 = 14n$. Dividing both sides of this equation by 14 yields $-2 = n$. Therefore, the value of n is -2 .

Question Difficulty: Hard

Question ID 5b866237

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	

ID: 5b866237

$$(x - 3)^2 + (y + 16)^2 = 289$$

The graph of the given equation is a circle in the xy -plane. The point (q, s) lies on the circle. Which of the following is NOT a possible value of q ?

- A. -17
- B. -3
- C. 3
- D. 17

ID: 5b866237 Answer

Correct Answer: A

Rationale

Choice A is correct. It's given that the graph of the equation $(x - 3)^2 + (y + 16)^2 = 289$ is a circle in the xy -plane. For a circle in the form $(x - h)^2 + (y - k)^2 = r^2$, its center is (h, k) and its radius is r . It follows that $r^2 = 289$. Taking the positive square root of both sides of this equation yields $r = 17$. Therefore, the center of this circle is $(3, -16)$ and the radius is 17 . It follows that the x -coordinates of the points that lie on the circle must be within 17 units of the x -coordinate of the center of the circle. Therefore, the least possible x -coordinate of a point that lies on the circle is $3 - 17$, or -14 , and the greatest possible x -coordinate of a point that lies on the circle is $3 + 17$, or 20 . So for any point (q, s) to lie on the circle, it must be true that $-14 \leq q \leq 20$. Since -17 does not satisfy this inequality, it is NOT a possible value of q .

Choice B is incorrect. Since -3 satisfies the inequality $-14 \leq q \leq 20$, it is a possible value of q .

Choice C is incorrect. Since 3 satisfies the inequality $-14 \leq q \leq 20$, it is a possible value of q .

Choice D is incorrect. Since 17 satisfies the inequality $-14 \leq q \leq 20$, it is a possible value of q .

Question Difficulty: Medium

Question ID a141c7be

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: a141c7be

The table gives the areas and perimeters of two similar rectangles, where n is a constant.

	Area (square inches)	Perimeter (inches)
Rectangle A	630	210
Rectangle B	2,520	n

What is the value of n ?

- A. 2,100
- B. 1,680
- C. 840
- D. 420

ID: a141c7be Answer

Correct Answer: D

Rationale

Choice D is correct. It’s given that rectangle A and rectangle B are similar. Therefore, there is a positive constant k such that the length of each side of rectangle B is k times the length of the corresponding side of rectangle A and the area of rectangle B is k^2 times the area of rectangle A. It’s also given that the area of rectangle A is 630 square inches and that the area of rectangle B is 2,520 square inches. It follows that $2,520 = 630k^2$. Dividing both sides of this equation by 630 yields $4 = k^2$, or $k^2 = 4$. Taking the positive square root of each side of this equation yields $k = 2$. Therefore, the length of each side of rectangle B is 2 times the length of the corresponding side of rectangle A. It follows that the perimeter of rectangle B is 2 times the perimeter of rectangle A. Since the perimeter of rectangle A is 210 inches, the perimeter of rectangle B is $2(210)$, or 420, inches. Therefore, $n = 420$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from using the area scale factor instead of the side length scale factor.

Question Difficulty: Hard

Question ID 318bbe9a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	

ID: 318bbe9a

A circle has center P , and points A and B lie on the circle. The measure of arc AB is 45° and the length of arc AB is 4π units. What is the length, in units, of the radius of the circle?

ID: 318bbe9a Answer

Correct Answer: 16

Rationale

The correct answer is **16**. It's given that the measure of arc AB is 45° and the length of arc AB is 4π units. In a circle, the circumference corresponds to an arc measure of 360° . Therefore, the ratio of the measure of arc AB to 360° is equal to the ratio of the length of arc AB to the circumference. Since the circumference of a circle is $2\pi r$, it follows that $\frac{45}{360} = \frac{4\pi}{2\pi r}$. This equation can be rewritten as $\frac{1}{8} = \frac{2}{r}$. Multiplying both sides of this equation by $8r$ yields $1(r) = 2(8)$, or $r = 16$. Therefore, the length, in units, of the radius of the circle is **16**.

Question Difficulty: Hard

Question ID 8030c51e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: 8030c51e

The length of one side of an equilateral triangle is **28** inches. What is the perimeter, in inches, of this triangle?

- A. **28**
- B. **31**
- C. **84**
- D. **112**

ID: 8030c51e Answer

Correct Answer: C

Rationale

Choice C is correct. The perimeter of a triangle is the sum of the lengths of the three sides of the triangle. It's given that the length of one side of an equilateral triangle is **28** inches. Since the lengths of the three sides of an equilateral triangle are equal, it follows that the perimeter, in inches, of this equilateral triangle is **$28 + 28 + 28$** , or **84**.

Choice A is incorrect. This is the length of one side, not the perimeter, of this equilateral triangle.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the perimeter of a square, not an equilateral triangle, with side lengths of **28** inches.

Question Difficulty: Easy

Question ID b38e68e6

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	

ID: b38e68e6

In right triangle QRS , the length of hypotenuse SQ is $4\sqrt{82}$ and the length of side QR is 4 . What is the length of side RS ?

ID: b38e68e6 Answer

Correct Answer: 36

Rationale

The correct answer is **36**. It's given that side SQ is the hypotenuse of right triangle QRS . It's also given that the length of side SQ is $4\sqrt{82}$ and the length of side QR is 4 . It follows that sides QR and RS are the legs of triangle QRS . The Pythagorean theorem states that for a right triangle, the sum of the squares of the lengths of the two legs is equal to the square of the length of the hypotenuse. Let b represent the length of side RS . By the Pythagorean theorem, $4^2 + b^2 = \left(4\sqrt{82}\right)^2$, or $16 + b^2 = 1,312$. Subtracting **16** from both sides of this equation yields $b^2 = 1,296$. Taking the positive square root of both sides of this equation yields $b = \mathbf{36}$. Therefore, the length of side RS is **36**.

Question Difficulty: Hard

Question ID acb06db7

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: acb06db7

In triangle ABC , the measure of angle A is 52° and $AC = 30$. In triangle PQR , the measure of angle P is 52° and $PR = 120$. Which additional piece of information is sufficient to prove that triangle ABC is similar to triangle PQR ?

- A. $AB = 50$ and $PQ = 50$.
- B. $AB = 50$ and $QR = 200$.
- C. The measures of angle B and angle R are 32° and 96° , respectively.
- D. The measures of angle B and angle Q are 52° and 32° , respectively.

ID: acb06db7 Answer

Correct Answer: C

Rationale

Choice C is correct. It's given that the measure of angle A is 52° and the measure of angle P is 52° . Therefore, angle A is congruent to angle P . Choice C states that the measures of angle B and angle R are 32° and 96° , respectively. The triangle sum theorem states that the measures of the angles in a triangle sum to 180° . In triangle ABC , if the measure of angle B is 32° , it follows that the measure of angle C is $(180 - (52 + 32))^\circ$, or 96° . In triangle PQR , if the measure of angle R is 96° , it follows that the measure of angle Q is $(180 - (52 + 96))^\circ$, or 32° . Therefore, in triangles ABC and PQR , angle B is congruent to angle Q , and angle C is congruent to angle R . If two triangles have three pairs of congruent angles, then the triangles are similar. Therefore, the additional piece of information that is sufficient to prove that triangle ABC is similar to triangle PQR is that the measures of angle B and angle R are 32° and 96° , respectively.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

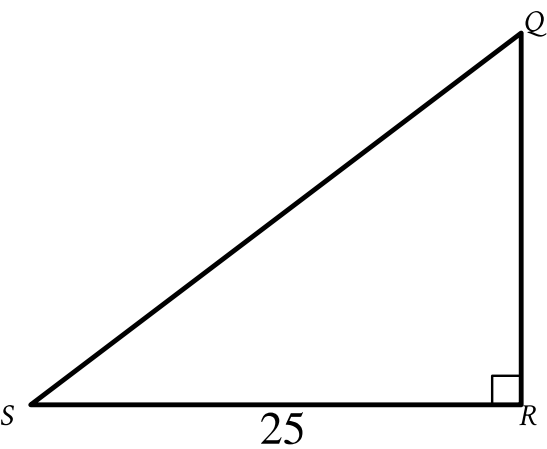
Choice D is incorrect. If the measure of angle B is 52° , it follows that the measure of angle C is $(180 - (52 + 52))^\circ$, or 76° . If the measure of angle Q is 32° , which isn't equal to any angle measure in triangle ABC , the triangles can't be similar.

Question Difficulty: Hard

Question ID 835e170f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	

ID: 835e170f



Note: Figure not drawn to scale.

In triangle QRS shown, $QR < RS$. Which expression represents the length of QS ?

- A. $25 \cos Q$
- B. $25 \sin Q$
- C. $\frac{25}{\cos Q}$
- D. $\frac{25}{\sin Q}$

ID: 835e170f Answer

Correct Answer: D

Rationale

Choice D is correct. The figure shows that triangle QRS is a right triangle. Each of the given choices is an expression containing $\sin Q$ or $\cos Q$. For an acute angle in a right triangle, the sine of the angle is the length of the opposite leg divided by the length of the hypotenuse, and the cosine of the angle is the length of the adjacent leg divided by the length of the hypotenuse. It follows that $\sin Q = \frac{RS}{QS}$ and $\cos Q = \frac{QR}{QS}$, where $QR < RS$. It's given in the figure that the length of side RS is 25 . Since only the equation $\sin Q = \frac{RS}{QS}$ contains RS , an expression representing QS can be found by substituting 25 for RS in this equation, which yields $\sin Q = \frac{25}{QS}$. Multiplying both sides of this equation by QS yields $QS \cdot \sin Q = 25$. Dividing both sides of this equation by $\sin Q$ yields $QS = \frac{25}{\sin Q}$. Therefore, the expression $\frac{25}{\sin Q}$ represents the length of QS .

Choice A is incorrect and may result from conceptual or calculation errors.

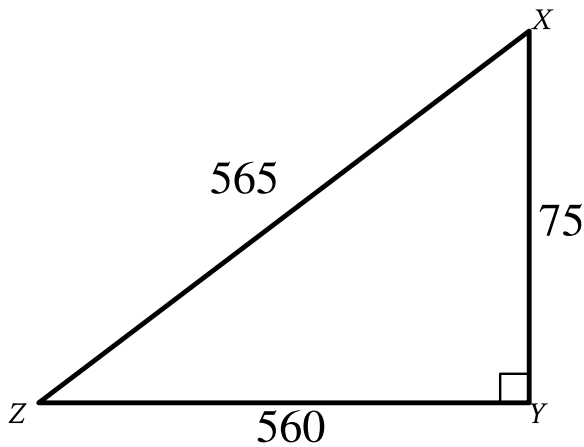
Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question ID 9b378af8

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	

ID: 9b378af8



Note: Figure not drawn to scale.

In triangle XYZ , $\cos Z = \frac{560}{c}$. What is the value of c ?

ID: 9b378af8 Answer

Correct Answer: 565

Rationale

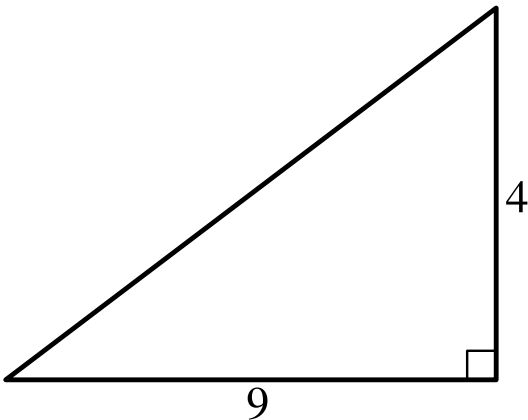
The correct answer is **565**. In triangle XYZ , angle Y is a right angle. Therefore, triangle XYZ is a right triangle. The cosine of an acute angle in a right triangle is the ratio of the length of the side adjacent to that angle to the length of the hypotenuse. In right triangle XYZ , side YZ is the side adjacent to angle Z and side XZ is the hypotenuse. It's given that the length of side YZ is **560** and the length of side XZ is **565**. Therefore, $\cos Z = \frac{560}{565}$. It's also given that $\cos Z = \frac{560}{c}$. It follows that $\frac{560}{c} = \frac{560}{565}$. Therefore, $c = \mathbf{565}$.

Question Difficulty: Medium

Question ID 71c5aff7

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: 71c5aff7



Note: Figure not drawn to scale.

The figure shows the lengths, in units, of two sides of a triangle. What is the area, in square units, of the triangle?

- A. 18
- B. 13
- C. 9
- D. 4

ID: 71c5aff7 Answer

Correct Answer: A

Rationale

Choice A is correct. The area, A , of a triangle is given by the formula $A = \frac{1}{2}bh$, where b is the length of the base of the triangle and h is the height of the triangle. In the right triangle shown, the length of the base of the triangle is 9 units, and its height is 4 units. Substituting 9 for b and 4 for h in the formula $A = \frac{1}{2}bh$ yields $A = \frac{1}{2}(9)(4)$, or $A = 18$. Therefore, the area of the triangle, in square units, is 18 .

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the length of the base of the triangle, in units, not the area of the triangle, in square units.

Choice D is incorrect. This is the height of the triangle, in units, not the area of the triangle, in square units.

Question Difficulty: Easy

Question ID e7dd3b96

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	

ID: e7dd3b96

What is the radius of the circle in the xy -plane defined by $(x + 3)^2 + (y + 5)^2 = 289$?

ID: e7dd3b96 Answer

Correct Answer: 17

Rationale

The correct answer is **17**. The equation of a circle in the xy -plane can be written in the form $(x - h)^2 + (y - k)^2 = r^2$, where (h, k) is the center of the circle and r is the radius of the circle. It's given that the circle in the xy -plane is defined by $(x + 3)^2 + (y + 5)^2 = 289$. Therefore, $r^2 = 289$. Taking the positive square root of both sides of this equation yields $r = 17$. Thus, the radius of the circle is **17**.

Question Difficulty: Medium

Question ID 8f1570b6

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	

ID: 8f1570b6

In the xy -plane, circle M is the graph of the equation $(x - 4)^2 + (y - 5)^2 = 9$. Circle P has the same center as circle M but has a radius that is twice the radius of circle M . Which equation represents circle P ?

- A. $(x - 4)^2 + (y - 5)^2 = 18$
- B. $(x - 4)^2 + (y - 5)^2 = 36$
- C. $(x - 8)^2 + (y - 10)^2 = 9$
- D. $(x - 8)^2 + (y - 10)^2 = 18$

ID: 8f1570b6 Answer

Correct Answer: B

Rationale

Choice B is correct. An equation of a circle in the xy -plane can be written as $(x - h)^2 + (y - k)^2 = r^2$, where (h, k) is the center and r is the radius. It's given that in the xy -plane, circle M is the graph of the equation $(x - 4)^2 + (y - 5)^2 = 9$, which is equivalent to $(x - 4)^2 + (y - 5)^2 = 3^2$. Therefore, the center of circle M is $(4, 5)$ and the radius of circle M is 3 . It's given that circle P has the same center as circle M but has a radius that is twice the radius of circle M . Therefore, the center of circle P is $(4, 5)$ and the radius of circle P is $2(3)$, or 6 . Substituting 4 for h , 5 for k , and 6 for r in the equation $(x - h)^2 + (y - k)^2 = r^2$ yields $(x - 4)^2 + (y - 5)^2 = 6^2$, which can be rewritten as $(x - 4)^2 + (y - 5)^2 = 36$.

Choice A is incorrect. The graph of this equation has a radius of $\sqrt{2}(3)$, not $2(3)$.

Choice C is incorrect. The graph of this equation has center $(8, 10)$, not $(4, 5)$, and a radius of 3 , not $2(3)$.

Choice D is incorrect. The graph of this equation has center $(8, 10)$, not $(4, 5)$, and a radius of $\sqrt{2}(3)$, not $2(3)$.

Question Difficulty: Hard

Question ID e00517ba

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	

ID: e00517ba

A circle in the xy -plane has the equation $(x - 19)^2 + (y - 17)^2 = 64k^2$, where k is a positive constant. Which of the following gives the center of the circle and its radius?

- A. The center is at $(19, 17)$, and the radius is 8 .
- B. The center is at $(19, 17)$, and the radius is $8k$.
- C. The center is at $(19, 17)$, and the radius is 64 .
- D. The center is at $(19, 17)$, and the radius is $64k$.

ID: e00517ba Answer

Correct Answer: B

Rationale

Choice B is correct. The equation of a circle in the xy -plane can be written as $(x - h)^2 + (y - p)^2 = r^2$, where the center of the circle is (h, p) and the radius of the circle is r . It's given that the circle in the xy -plane has the equation $(x - 19)^2 + (y - 17)^2 = 64k^2$, where k is a positive constant. It follows that $h = 19$, $p = 17$, and $r^2 = 64k^2$. Taking the positive square root of each side of $r^2 = 64k^2$ yields $r = 8k$. Therefore, the center of the circle is at $(19, 17)$ and the radius is $8k$.

Choice A is incorrect. This circle has the equation $(x - 19)^2 + (y - 17)^2 = 64$, not $(x - 19)^2 + (y - 17)^2 = 64k^2$.

Choice C is incorrect. This circle has the equation $(x - 19)^2 + (y - 17)^2 = 4,096$, not $(x - 19)^2 + (y - 17)^2 = 64k^2$.

Choice D is incorrect. This circle has the equation $(x - 19)^2 + (y - 17)^2 = 4,096k^2$, not $(x - 19)^2 + (y - 17)^2 = 64k^2$.

Question Difficulty: Medium

Question ID cb3792bb

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: cb3792bb

The area of one of the bases of a right circular cylinder is **81** square centimeters, and the volume of the cylinder is **2,187** cubic centimeters. What is the height, in centimeters, of the cylinder?

- A. **27**
- B. **2,025**
- C. **2,106**
- D. **177,147**

ID: cb3792bb Answer

Correct Answer: A

Rationale

Choice A is correct. The volume, V , of a right circular cylinder is given by the formula $V = \pi r^2 h$, where πr^2 is the area of one of the bases of the cylinder and h is the height. It's given that the area of one of the bases of the cylinder is **81** square centimeters and that the volume of the cylinder is **2,187** cubic centimeters. Substituting **2,187** for V and **81** for πr^2 in the formula $V = \pi r^2 h$ yields **2,187** = **81** h . Dividing both sides of this equation by **81** yields **27** = h . Therefore, the height, in centimeters, of the cylinder is **27**.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

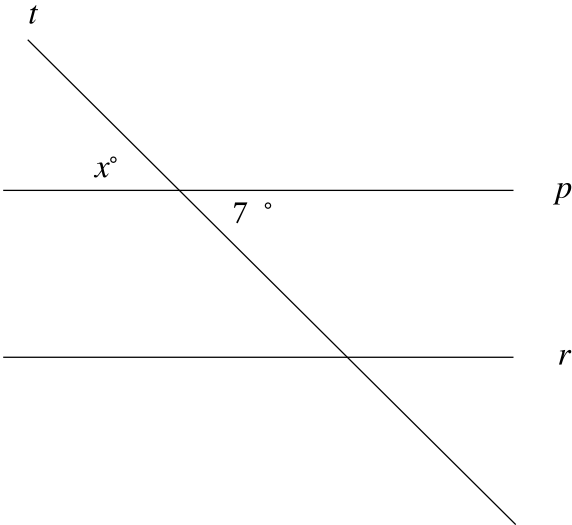
Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Medium

Question ID c0ba1169

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: c0ba1169



Note: . igure not drawn to scale.

In the figure, line *p* is parallel to line *r*, and line *t* intersects both lines. What is the value of *x*?

- A. 36
- B. 72
- C. 180
- D. 252

ID: c0ba1169 Answer

Correct Answer: B

Rationale

Choice B is correct. In the figure shown, the angles with measures *x*° and 72° are vertical angles. Vertical angles have equal measure, so the two marked angles have equal measure. Therefore, the value of *x* is 72.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty: Easy

Question ID 69d9d256

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	

ID: 69d9d256

A circle has center O , and points A and U lie on the circle. Line segments AE and UE are tangent to the circle at points A and U , respectively. The distance between point O and point E is 510 centimeters and the distance between point O and point U is 78 centimeters. What is the area, in square centimeters, of quadrilateral $OAEU$?

ID: 69d9d256 Answer

Correct Answer: 39312

Rationale

The correct answer is **39,312**. It's given that the circle has center O and points A and U lie on the circle. It follows that segments OA and OU are radii of the circle and therefore congruent. It's also given that line segments AE and UE are tangent to the circle at points A and U , respectively. Since line segments AE and UE are tangent to the circle from a common external point, point E , line segments AE and UE are congruent and form right angles with radii OA and OU , respectively. Therefore, quadrilateral $OAEU$ consists of two congruent right triangles: triangle OAE and triangle OUE . It's given that the distance between point O and point E is 510 centimeters and the distance between point O and point U is 78 centimeters. It follows that the area of triangle OUE can be found by first using the Pythagorean theorem to calculate the distance between point U and point E . For right triangle OUE , where angle U is a right angle, the Pythagorean theorem states that $(OU)^2 + (UE)^2 = (OE)^2$. Substituting 78 for OU and 510 for OE in this equation yields $78^2 + (UE)^2 = 510^2$, or $6,084 + (UE)^2 = 260,100$. Subtracting 6,084 from both sides of this equation yields $(UE)^2 = 254,016$. Taking the positive square root of both sides of this equation yields $UE = 504$. Thus, triangle OUE is a right triangle with legs of length 78 centimeters and 504 centimeters. It follows that the area of triangle OUE is $(\frac{1}{2})(78)(504)$ square centimeters, or 19,656 square centimeters. Since triangle OAE is congruent to triangle OUE , the area of triangle OAE is also 19,656 square centimeters. Therefore, the area of quadrilateral $OAEU$ is $19,656 + 19,656$ square centimeters, or 39,312 square centimeters.

Question Difficulty: Hard